

2009

Introduction to PP(M)SB for Interns by Interns

This handbook provides an introduction to PETRONAS Penapisan (Melaka) Sdn. Bhd., and a brief description of the process and utility units available at the refinery.



Revision History

No.	Date	Prepared by	Amendments	Reviewed & Approved by
1	21/08/'07	Sinthi Laya Thillaivarna	Overview PSR-1 : Unit 11 Unit 12 Unit 13 PSR-2 : Unit 21	Prakash K.K. Principal Engineer APC
2	11/02/'08	Nurafira Abdul Aziz Muhammad Fauwaz Hassan	PSR-1 : Unit 14 Unit 15 Unit 16 PSR-2 : Unit 22 Unit 23 Unit 24 Unit 25 Unit 26 Unit 27 Unit 28 Unit 29 Unit 30 Unit 31 Unit 32 Unit 39	Prakash K.K. Principal Engineer APC
3	10/07/'08	Muhammad Fauwaz Hassan	Correction made on previous issue	Prakash K.K. Principal Engineer APC
4	20/02/'09	Shobana D/O Balakrishnan	Utilities: Unit 51 Unit 52 Unit 53 Unit 54 Unit 55 Unit 56 Unit 58 Unit 64 Unit 65 Overview of COGEN Plant	Prakash K.K. Principal Engineer APC

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1.0 Introduction

The objective of this handbook is to develop a common reference for interns entering Melaka Refinery. This handbook provides an initial set of topics covering the history of Melaka refinery, the importance of safety and the refinery organization. This is then followed by a description of all the different process and utility units available in PP (M) SB. Each of this process and utility unit has a general description which is then followed by a detailed description.

2.0 Melaka Refinery History

Petronas Penapisan (Melaka) Sdn. Bhd. [PP (M) SB], located in Sungai Udang, Melaka, is a wholly-owned subsidiary of Petroliaam Nasional Berhad or PETRONAS, the national oil corporation. PP (M) SB manages and operates the refinery complex in Malacca. PP (M) SB is the second refinery complex in the country after the first refinery which was constructed in Kerteh, Terengganu. Petronas also has a third refinery in Engen, South Africa. PP (M) SB consists of Petronas Second Refinery Phase 1 (PSR 1) and Petronas Second Refinery Phase 2 (PSR 2). The Malacca refinery complex is now within an area of 926 acres and is still expanding.

PSR 1, wholly owned by Petronas, was built-in in September 1987. Plant operations were commissioned in 1994 processing local sweet feedstock (sulfur<0.5wt %) for the design capacity of 100,000 barrels per day (BPSD) utilizing hydroskimming configuration (sweet train). The main process units at PSR1 include:

- Crude Distillation Unit
- Naphtha Hydrotreating Unit
- Catalytic Reforming Unit with Continuous Catalytic Regenerator
- Saturated Gas Concentration Unit
- Sour Water Stripping Unit
- Kerosene & Heavy Naphtha Treating Unit
- Mercury Removal Unit

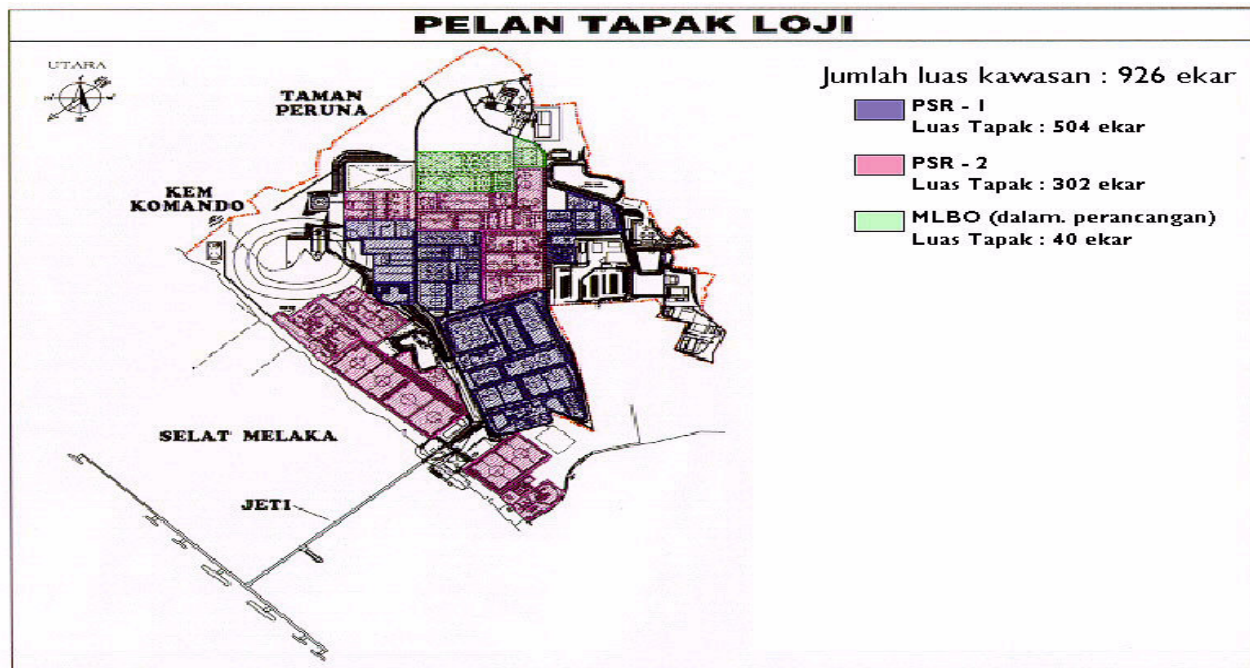
PSR 2, incorporated in May 1991, is operated by the Malaysian Refining Company (MRC). MRC consists of Petronas (53%) and Conoco-Phillips (47%). Production for PSR 2 with deep conversion facilities, started in 1998 with a capacity of 100,000 BPSD processing up to 70% sour feedstock (sulfur>2.5wt%), mainly middle east crudes, utilizing complex conversion configuration (sour train). The main process units at PSR 2 include:

- Crude and Vacuum Distillation Units

-
- Naphtha Hydrotreating Unit
 - Catalytic Reforming Unit with Continuous Catalyst Regenerator
 - Saturated Gas Recovery Unit
 - C₆ Isomerization Unit
 - LPG and Naphtha Treating Units
 - Kerosene Treating Unit/Caustic Neutralization Unit
 - Distillate Hydrotreating Unit
 - Hydrocracker Unit
 - Delayed Coker Unit
 - Hydrogen Production Unit
 - Hydrogen Collection and Distribution System

The two refineries, PSR1 and PSR2 are designed to operate as an integrated complex with common utility, offsite and marine facilities.

▪ *PP(M)SB complex layout*



▪ *PP(M)SB complex location*



3.0 Safety

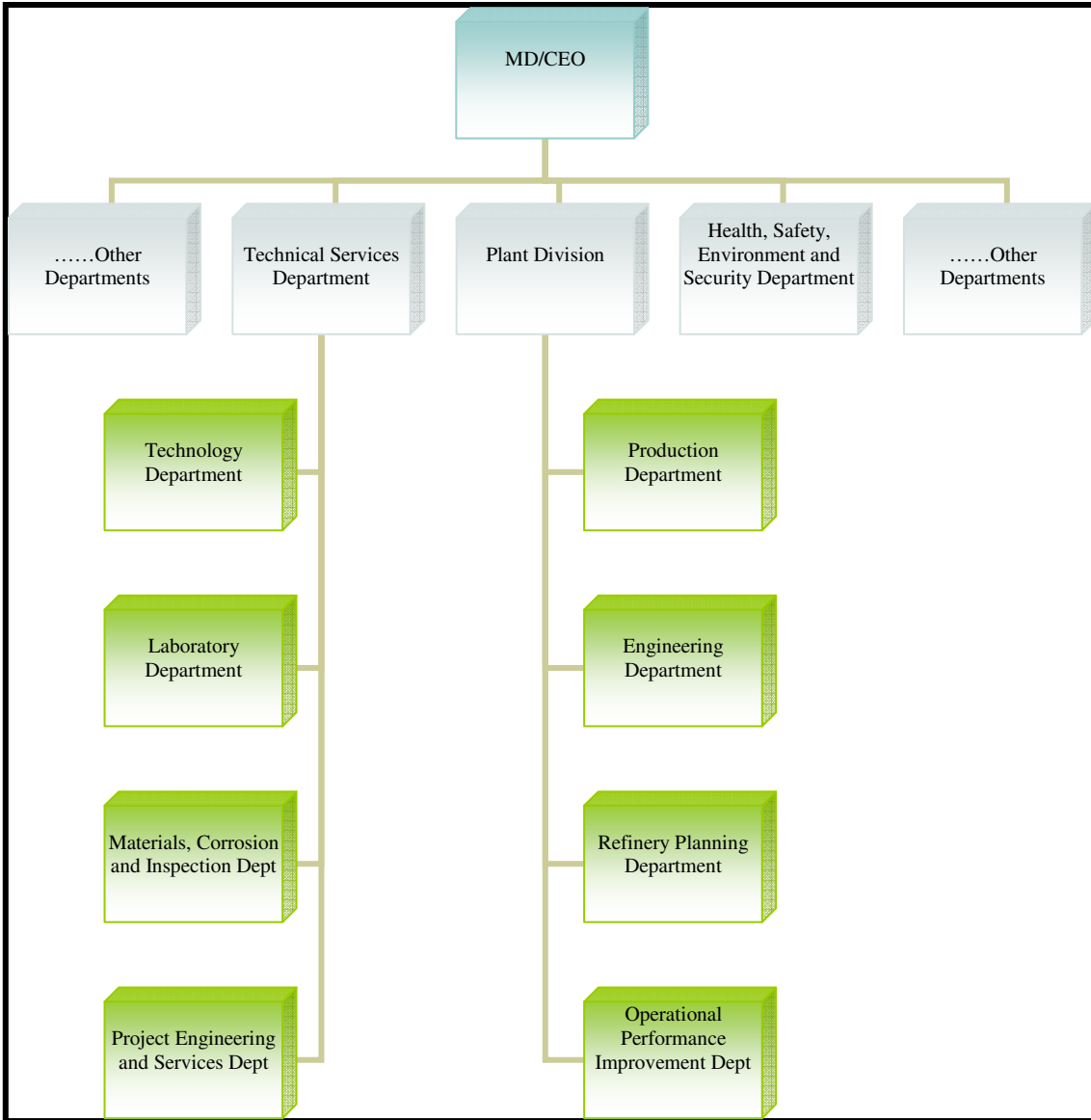
In the refinery, our operation requires us to deal with hydrocarbons. Thus, safety is taken very seriously in PP (M) SB. The Health, Safety, Environment and Security Department, HSE, plays the important role of managing all issues involving health, safety and environment in this refinery. HSE acts as both a preventative measure and a reactive measure to all safety hazards in PP (M) SB. It is very important to run the refinery safely as it is the objective of the management to run the plant within its safe operation limit while protecting all staffs of the refinery from work hazards.

Safety in PP (M) SB is managed via the HSE objectives/targets. HSE objectives/targets are established by the management each fiscal year, and a 5-year plan is maintained in order to continuously improve HSE performance and move the organization towards achieving HSE excellence. Management also provides resources essential to the implementation, maintenance and continuous improvement of safety in PP (M) SB. Resources include; personnel, specialized skills, facilities, equipment, technology and finance. During the engineering design phase, every effort is made to reduce the risks associated with a process, system or facility utilizing a variety of Standards to a level that is very low.

PP (M) SB also maintains procedures to identify the potential for, and the response to emergency situations. Purchase of materials such as chemicals, catalysts, activated carbons and etc are also carried out based on procedure HSEMS-20160, Chemical Approval System to ensure that the HSE impact posed by these materials are properly evaluated before purchase and use. In order to ensure that the company and the employees fully understand the hazards presented by the materials and processes in use at the refinery, information pertaining to the hazards of the highly hazardous chemicals in the process and information concerning the technology of the process is made necessary. PP (M) SB also manages safety by providing Personal Protection Equipment (PPE) such as safety shoes, safety hat, one piece NOMEX coverall, safety glasses and etc to all their staffs.

4.0 Organization

Organizational Chart of PP (M) SB



Basic Function of Each Department

1. Health, Safety, Environment and Security Department, HSE

Manages all issues involving health, safety and environmental in the refinery. HSE acts as both a preventative measure and a reactive measure to all safety hazards in PP (M) SB.

2. Technical Services Department, TSD

Mainly caters to the needs of plant operations.

▪ Technology Department, TECH

Ensures that the Production Department meets the expectations of the planner by monitoring and troubleshooting shooting.

i. Distillation Section

Responsible for unit 11, 14, 15, 16, 21, 24, 26, 27, 28. Provides technical support within refinery in order to assist in optimizing production and minimizing plant upsets through plant performance monitoring.

ii. Cracking Section

In-charge for unit 30, 31, 32, 39. Offers technical support by monitoring, troubleshooting and proposing for improvement modifications for optimizing production.

iii. Reforming and Treating Section

Responsible for unit 12, 13, 17, 22, 23, 25, 29. Ensures that production operate to meet the required expectation by providing technical support and assist in optimizing production.

iv. Utility Treating Energy and Environment Section

Provides technical support within refinery utility system and respective units in order to assist in optimizing production and minimizing plant upsets through plant performance monitoring.

v. Advanced Process Control and Optimization Section

Supports the Production Department by implementing and maintaining the plant advanced control system in order to stabilize and optimize the plant.

vi. Process Safety Management Section

Process Safety Management is the application of management principles and systems to the identification, understanding, and control of process hazards to prevent process-related injuries and incidents

▪ Laboratory Department, LAB

Play an important role as they verify feed/product quality and provide information on the conditions of the unit performance via sample testing.

▪ Materials, Corrosion and Inspection Department, MCI

Responsible for the procurement and management of all materials ranging from office supplies to plant materials and spare parts of equipment.

▪ Project Engineering and Services Department, PESD

Carry out projects concerning equipment modifications or process changes.

3. Plant Division

All matters regarding the operational aspect of the refinery is handled by this division.

- **Production Department**

All process operations including utilities, storage and shipping of feed and products are taken care by this department.

- **Engineering Department**

Plant assets management, major maintenance, maintenance engineering, reliability management and turnarounds are taken care by this department.

- **Refinery Planning Department**

Feed and product shipping schedules and plant productions are taken care by this department.

- **Operational Performance Improvement Department**

This department is divided further to two sections which are the Performance Improvement Management, which takes care of benchmarking, knowledge management in the refinery and is the custodian of RMS , and Six Sigma which improves performance in the refinery by carrying out projects.

5.0 The Refining Process

Crude/condensate oil is fossil fuel made naturally from decaying plants and animals living in ancient seas million of years ago. Crude oil consists of a mixture of chemical compounds (not a single chemical compound) with each compound having its own boiling point. These chemical compounds in crude oil are call hydrocarbons since they are a combination of hydrogen and carbon atoms only. Some compounds are as simple as CH_4 (methane) while others are as complex as $\text{C}_{85}\text{H}_{60}$. Impurities such as sulphur, nitrogen, oxygen, metal, basic sediment and water are also present in crude oil but in small amounts.

Since crude oil consists of a mixture of chemical compounds with each compound having its own boiling temperature, distillation process is carried out to refine petroleum crude oil and condensates into valuable and marketable petroleum products.

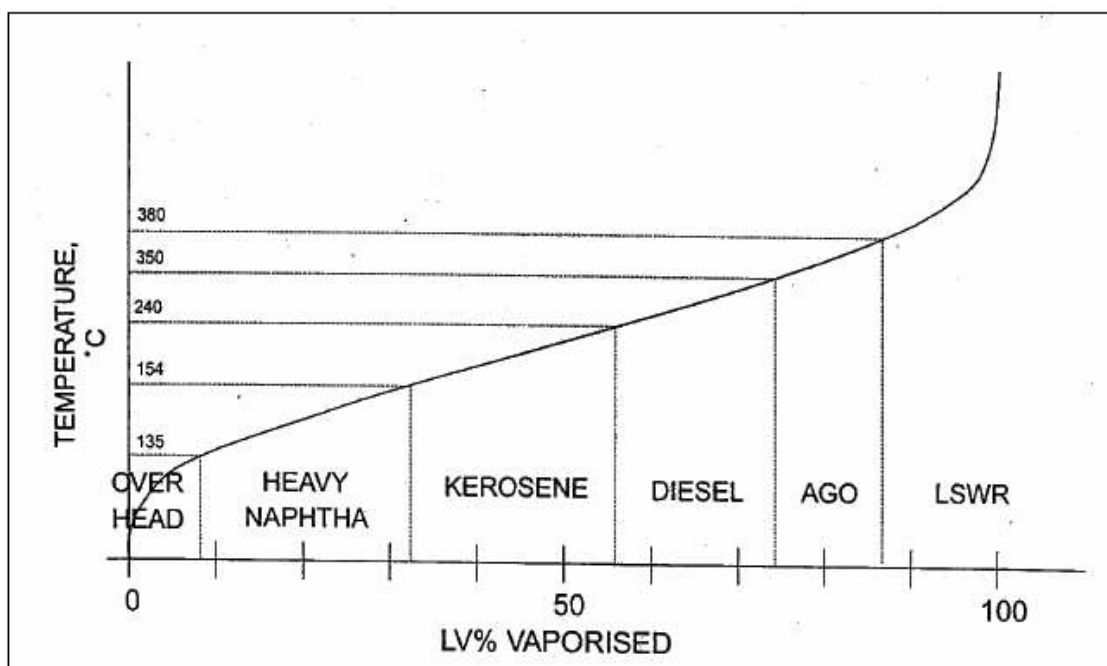


Figure 5.1 :: Separating crude into products according to their boiling point range

The crude oil and condensates from the Terengganu and Bintulu oil fields are transported by ocean tankers to the Sungai Udang Port. These feedstocks are then sent to the refinery storage tanks via the associated network of pipelines.

About 75,000BPSD of crude oil and 25,000BPSD of condensates are pumped to the process units and gradually heated in various heat exchangers provided in the preheat train. Heating of the feedstock are by means of heat exchanging using heat from the product streams and also in separate furnaces for the feedstock to be heated to their respective distillation temperatures. The heated crude oil and condensates are then sent to their respective distillation columns where the hydrocarbons are separated according to their boiling temperature.

All finished products are stored in their respective tanks and are subjected to quality control before distribution. The products for export are loaded via dedicated pipelines onto ships at the Sungai Udang Port. For the domestic market, products are shipped via coastal vessels, road tankers and product pipelines for inland distribution. The products to the Klang Valley and to the Kuala Lumpur International Airport (KLIA) in Sepang are transported via pipelines.

Product Description

Final Products from PSR1	Final Products from PSR2	Product Description
Fuel Gas	Fuel Gas	reused in the refinery as fuel in heaters and etc.
LPG	LPG	used in cooking gas cylinders
Isopentane		used as a blending component in producing motor gasoline
Petrochemical Naphtha	Petrochemical Naphtha	is an important feedstock in the petrochemical industry
Reformate		used as a blending component in producing motor gasoline
Medium Naphtha		- used as a blending component in producing motor gasoline - is an important feedstock in the petrochemical industry
Heavy Naphtha		used as a blending component in producing Dual Purpose Kerosene
Kerosene	Kerosene	used as jet fuel and illuminating kerosene
Diesel		used as a blending component in producing automotive diesel
AGO		used as a blending component in producing automotive diesel
LSWR	Gasoline	- used as fuel oil - feedstock for further processing (such as in the catalytic cracking unit) - used as fuel in vehicles

Table 5.1: Final products from PSR 1 and PSR 2 and their respective product description

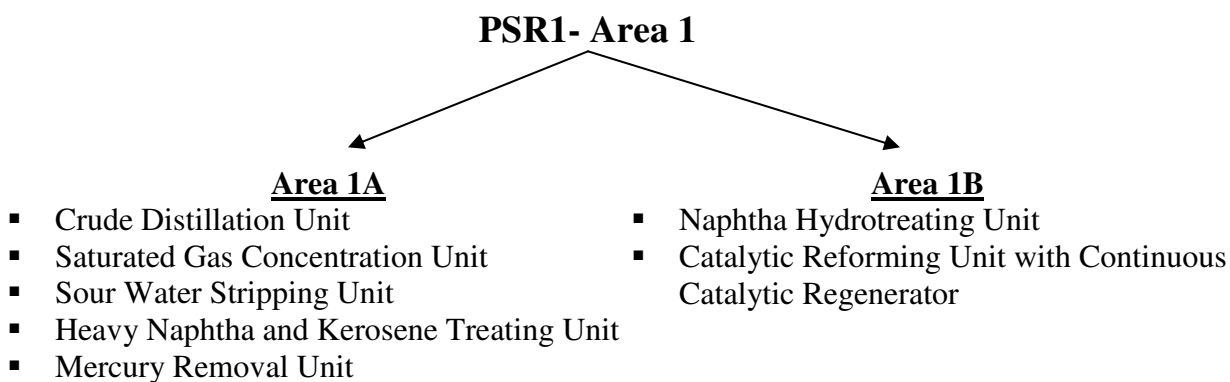
6.0 PETRONAS Second Refinery – Phase 1 (PSR-1)

PSR 1 processes local sweet feedstock (sulfur<0.5wt %) for the design capacity of 100,000 barrels per day (BPSD) utilizing hydroskimming configuration (sweet train). Feedstocks used are Tapis/Miri Light Crude, Terengganu Condensates and Bintulu Condensates.

The main process units at PSR1 include:

- Crude Distillation Unit
- Naphtha Hydrotreating Unit
- Catalytic Reforming Unit with Continuous Catalytic Regenerator
- Saturated Gas Concentration Unit
- Sour Water Stripping Unit
- Kerosene & Heavy Naphtha Treating Unit
- Mercury Removal Unit

PSR 1 is divided into two areas for operating convenience. PSR1 includes area 1A and area 1B.



UNIT	ABBREVIATION	UNIT NO	CAPACITY
Crude Distillation Unit	CDU	11	100,000 BPSD
Naphtha Hydrotreating Unit	NHT	12	20,000 BPSD
Catalytic Reforming Unit with Continuous Catalytic Regenerator	CRU and CCR	13	20,000 BPSD
Saturated Gas Concentration Unit	SGCU	14	13,600 kg/hr
Sour Water Stripping Unit	SWSU	15	16,000 kg/hr
Heavy Naphtha and Kerosene Treating Unit	HN/KTU	16	<u>High Tapis:</u> 3,240 BPSD HN 19,033 BPSD Kero <u>Low Tapis:</u> 3,955 BPSD HN 17,770 BPSD Kero

Table 6.1: Summary of all process units in PSR 1

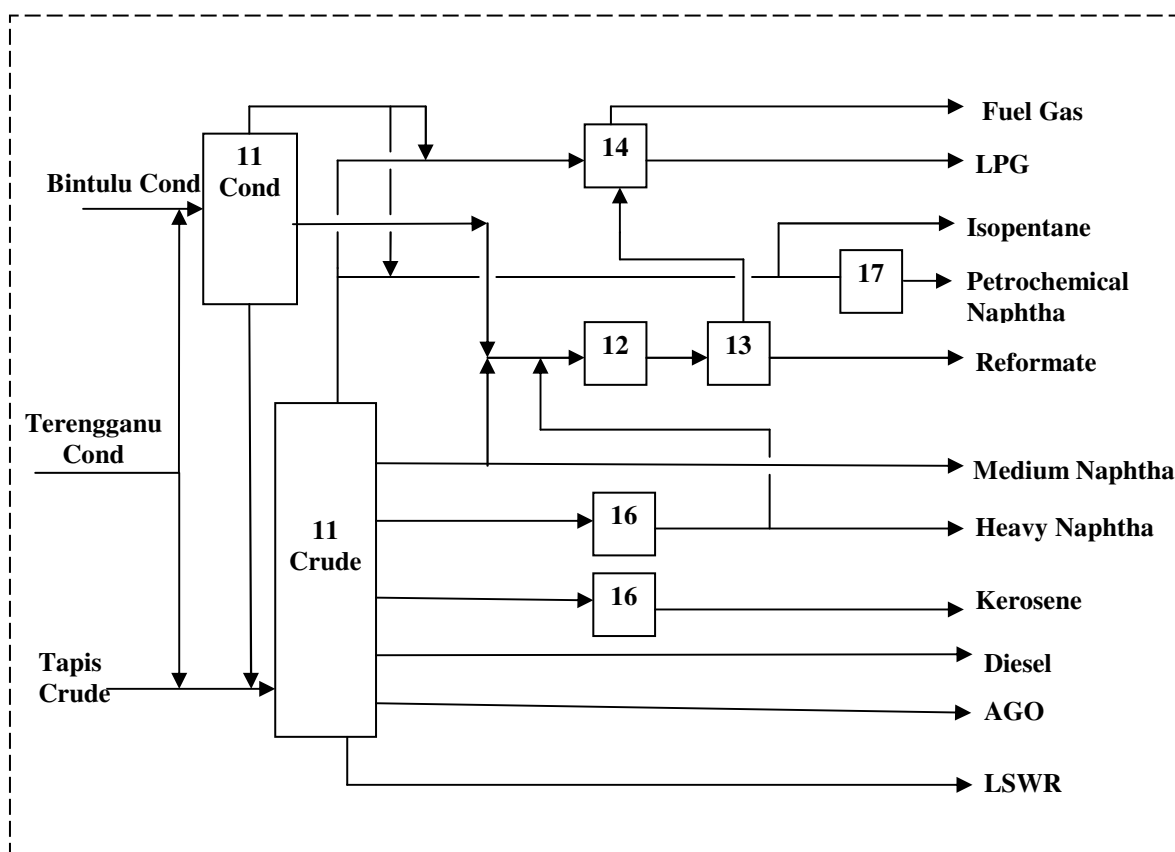


Figure 6.1 : PSR 1 Block Flow Diagram

Crude Distillation Unit (Unit 11)

CDU consists of a crude section (~75,000barrels/day) and a condensate section (25,000barrels/day). It includes a condensate tower, a condensate stabilizer, a crude tower, a crude stabilizer, a naphtha splitter, and a deisopentanizer. CDU distills Tapis crudes, Bintulu condensates and Terengganu condensates. The crude section processes Tapis, Miri Light and Masa crude with a mixture of condensates (condensate tower bottom) while the condensate section processes Bintulu and Terengganu condensates. Ratio of the crude and condensate tower's feedstock depends on the production planner.

The products formed are off gas, liquefied petroleum gas (LPG), Isopentane, Light Naphtha, Condensate Naphtha, Medium naphtha, Heavy Naphtha, Kerosene, Diesel, Atmospheric Gas Oil and Low Sulphur Waxy Residue (LSWR).

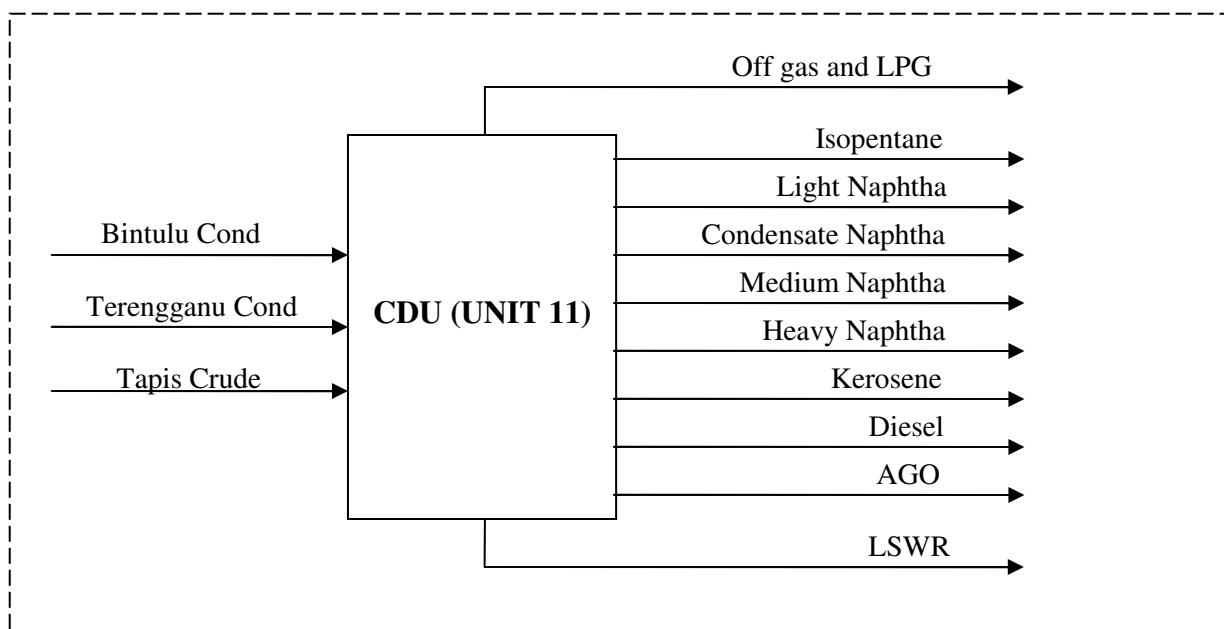


Figure 6.2 : Unit 11 Block Flow Diagram

Product stream from CDU unit	Downstream unit
Off gas	Fuel Gas System
LPG	Saturated Gas Concentration Unit
Isopentane	Gasoline Blending via Storage
Light Naphtha	Petrochemical Chemical Naphtha(PCN)/Gasoline Blending via Storage
Condensate Naphtha	Naphtha Hydrotreating Unit
Medium Naphtha	Naphtha Hydrotreating Unit/PCN/Gasoline Blending via Storage
Heavy Naphtha	Heavy Naphtha and Kerosene Treating Unit
Kerosene	Heavy Naphtha and Kerosene Treating Unit
Diesel	Diesel Blending via Storage/LSWR In-line Blending
Atmospheric Gas Oil	Diesel Blending via Storage/LSWR In-line Blending
Atmospheric Residue	LSWR In-line Blending

Table 6.2 : CDU products with their respective downstream units

The CDU unit's performance depends on the adjustment of the cut point between light naphtha and medium naphtha. The nominal cut points between these two products can be adjusted between 70°C and 92°C. As such, the CDU is designed to perform under two different cases which are the Base Case (70°C Cut) and the Alternate Case (92°C).

After the crude in-line mixing project in the year 2000, the crude section has been running at 120% feed throughput (90 000 BPSD). The capacity of the condensate section and the crude section depends on its operating cases. Two typical cases are as follows:

High Tapis Case

Feed Component	Crude Section(BPSD)	Condensate Section(BPSD)
Tapis Crude	75,000	-
Bintulu Condensate	-	12,000
Terengganu Condensate	-	13,000
Condensate Tower Bottom	4,080	-
Total Charge	79,080	25,000

Low Tapis Case

Feed Component	Crude Section(BPSD)	Condensate Section(BPSD)
Tapis Crude	51,000	-
Bintulu Condensate	-	17,000
Terengganu Condensate	24,000	8,000
Condensate Tower Bottom	3,802	-
Total Charge	78,802	25,000

Figure 6.3 : The crude section

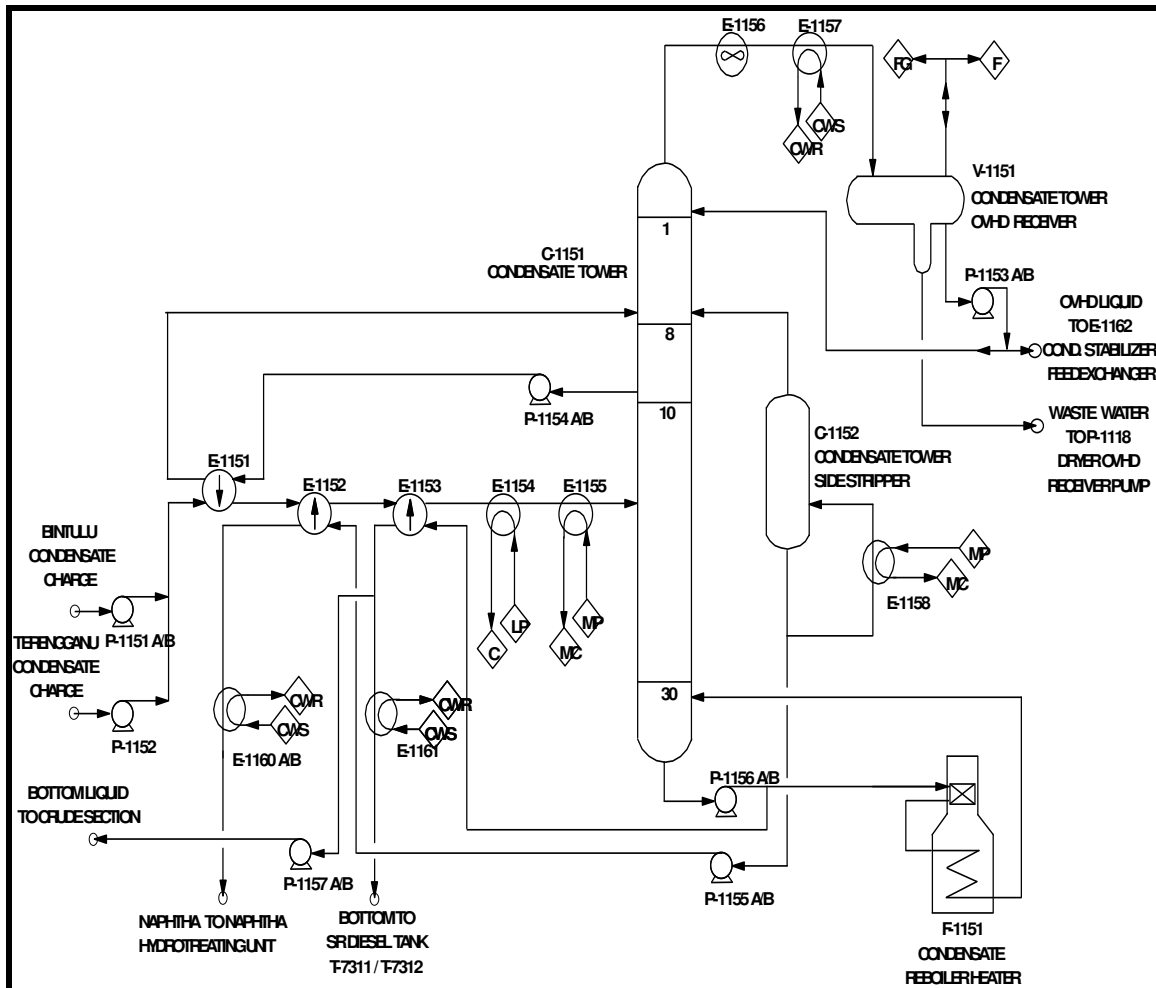


Figure 6.4 : The condensate section

Preheat train

Preheat train's main task is to maximize heat recovery from other streams, such as crude and condenser tower pumparounds, to heat the feed to 180-200 °C before entering the furnace heater, F-1101. This allows for savings in fuel gas at the furnace and utility consumption for cooling. There are 4 pre-desalter heat exchangers and 7 post-desalter heat exchangers.

Desalter

Crude contains salt and base sediments that can cause corrosion, catalyst poisoning, and heat exchanger fouling. The typical salt content in crude is 1 pound salt for every 1000 barrels of crude. Desalters are used to remove these salt and base sediments.

Pre-Flash Drum

In the pre-flash drum, light hydrocarbons in crude vaporize due to the sudden drop in pressure (from 18.5 kg/cm²_g to 6.5 kg/cm²_g). This is the most economical way to flash charge since heat is not required.

Furnace

The furnace is used to heat the feed up to 340 °C. Heating mediums used are fuel gas and combustion air from the atmosphere.

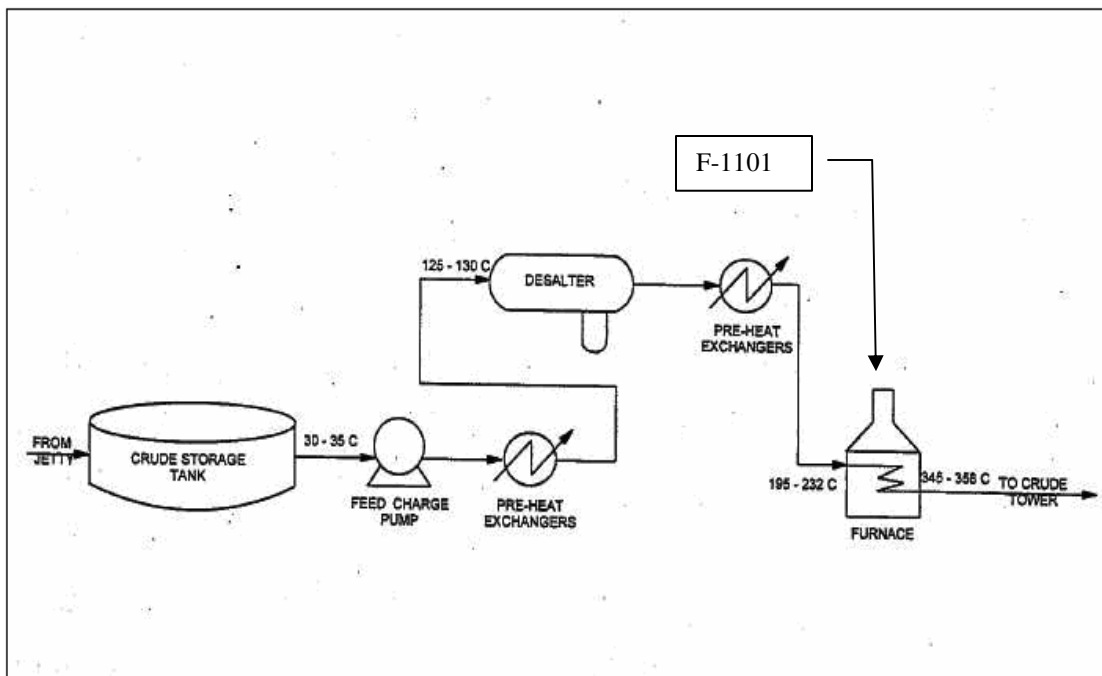


Figure 6.5 : Crude flow to crude tower

Crude Tower

Two phase crude feed from the pre-flash drum (in the form of vapour and liquid) enters the crude tower and is separated into different products according to their boiling points. Reflux and reboiling are used to enhance separation efficiency. Side strippers and dryers are used to recover products at their specific cut-points.

Component	Cutpoints (°C)
Overhead gas	< 135
Heavy naphtha	135 - 154
Kerosene	154 - 240
Diesel	240 - 350
Atmospheric Gas Oil (AGO)	350 - 380
Low Sulphur Waxy Residue (LSWR)	> 380

Overhead systems

Overhead vapour leaves the crude tower at 106°C and is then condensed to liquid at 40°C. This liquid stream is then split into the reflux feed and stabilizer feed.

Non-condensables are compressed in a compressor and is send to the stablilizer as well as to the crude tower condenser as spillback. This is to control the overhead receiver pressure.

Stabilizer

Stabilizer is essential as it is used to produce LPG with $< 1\text{mol}\%$ of C_5 and to produce naphtha.

Splitter

Splitter separates naphtha from the stabilizer bottom into light naphtha and medium naphtha.

De-isopentanizer

De-isopentanizer reduces the vapour pressure of light naphtha from the splitter by removing isopentane. Isopentane is used in gasoline blending.

Pumparounds

Pumparounds are product streams, from the crude and condenser towers, recycled back as feed into these towers to condense vapour that would otherwise have gone to the lighter cuts thus regulating internal reflux. Since the vapour load in the tower is reduced, a smaller tower diameter is required. As a result, smaller trays are required in the tower. Pumparounds are also used to recover heat from the product stream to the feed stream.

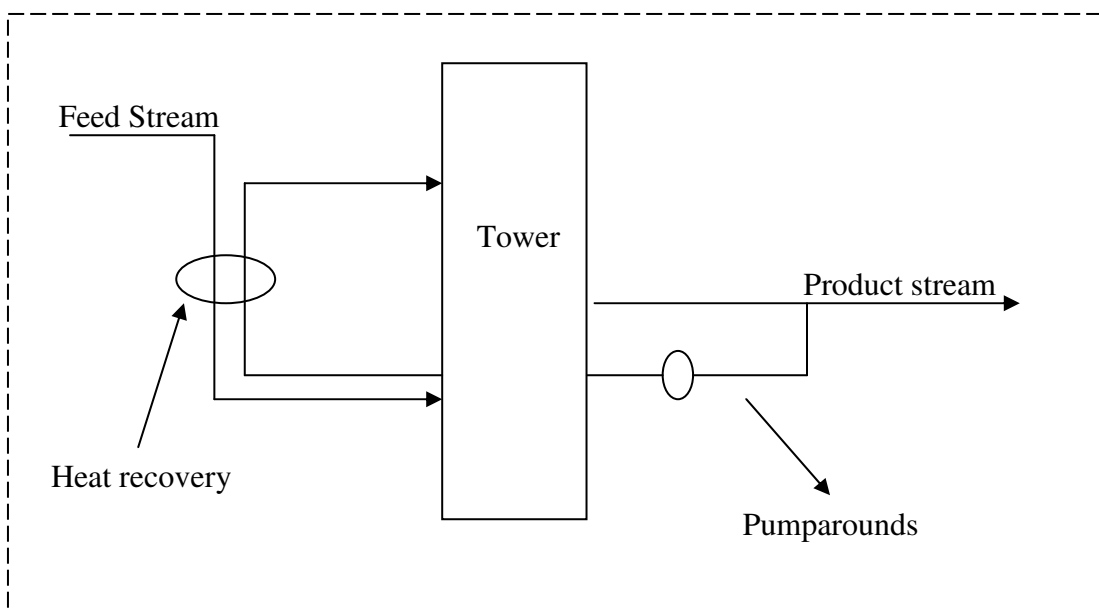


Figure 6.6 : Pumparounds

Strippers (Heavy Naphtha, Kerosene, Diesel and AGO)

Strippers are used to remove light components from each product to adjust their flash point. Light components will vapourize due to the heat absorbed if the stripper is used with a reboiler.

AGO and Diesel dryers

Water contained in diesel and AGO is eliminated by flashing under vacuum conditions (0.2kg/cm^2) since water will flash at a lower temperature if compared to either diesel or AGO.

Condensate Tower

Similar to the crude tower, the condensate tower also has a pumparound system, side strippers and a pre-heat train. The condensate tower has three product streams which are the overhead gas stream, the condensate naphtha stream and the condensate tower bottom.

Condensate stabilizer : Condensate stabilizer removes light ends (LPG) in the overhead products from the condensate tower before sending it to the de-isopentanizer.

Naphtha Hydrotreating Unit (Unit 12)

Naphtha consists of light naphtha, medium naphtha and heavy naphtha. Naphtha is C_5 to C_{11} hydrocarbons. The boiling range of naphtha is generally from 35°C to 205°C although the naphtha in PSR 1 has a boiling range from 80 - 165°C while PSR 2 naphtha has a boiling range from 35 - 165°C . The naphtha in the refinery is from the delayed coker unit (PSR 2), the hydrocracker unit (PSR 2) and straight run naphtha from the crude unit (PSR 1 and 2). Naphtha from these units are sold as low octane gasoline without further processing, Petrochemical Naphtha (PCN) while the rest are generally upgraded to be sold as high or low octane gasoline. Naphtha is processed in the NHT unit and the CRU in PSR 1 but in the NHT, CRU and ISOM unit in PSR 2.

The function of the NHT unit is to use hydrogen gas to treat naphtha in order to prepare the feed stream to the CRU with the right and required specifications. The NHT unit removes sulphur, nitrogen, oxygen and metals from the untreated naphtha, to protect the downstream catalysts (especially the catalyst in CRU) and prevent corrosion problems in downstream units. The NHT unit also saturates olefins. The main process in this unit is the hydrogenation process occurring at 336°C and at $39\text{ kg/cm}^2_{\text{g}}$. The reactions in the NHT unit are catalytic reactions. The catalyst used is alumina based, with Cobalt-Molybdenum or Nickel-Molybdenum as the active metals. The reactions in this unit require hydrogen thus makeup hydrogen originating from the CRU is used. The reactions (especially olefin saturation) are highly exothermic reactions. The recycle gas and off gas in this unit contains $0.5 - 5\text{ mol\% H}_2\text{S}$. H_2S gas is extremely poisonous. One breath of this gas at this concentration can cause unconsciousness and death.

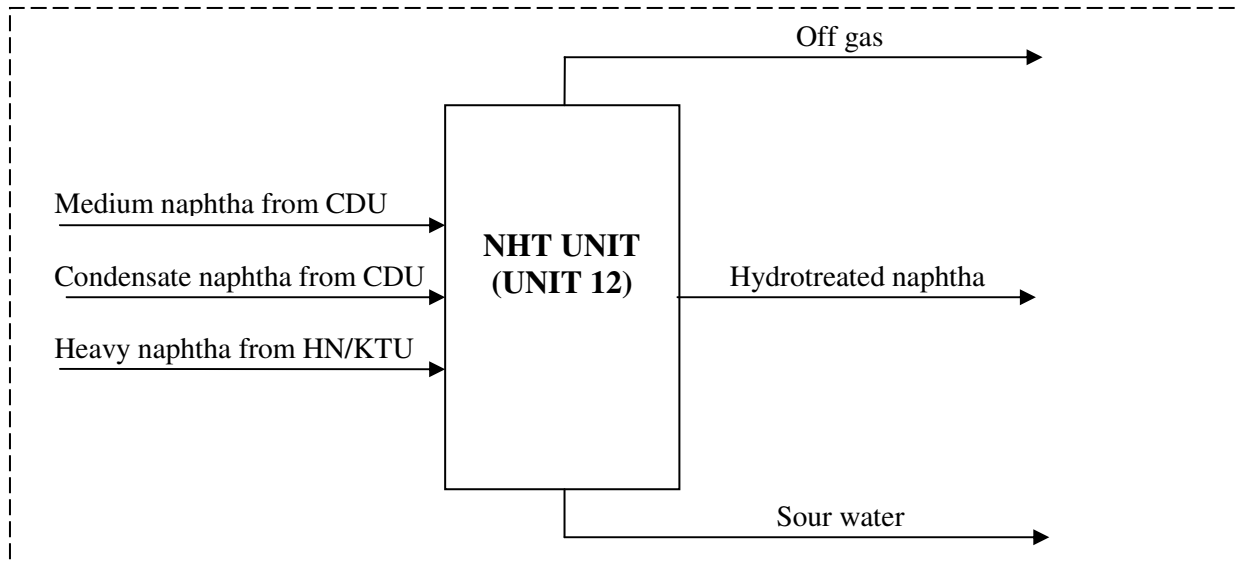


Figure 6.7: Unit 12 Block Flow Diagram

The reactor section

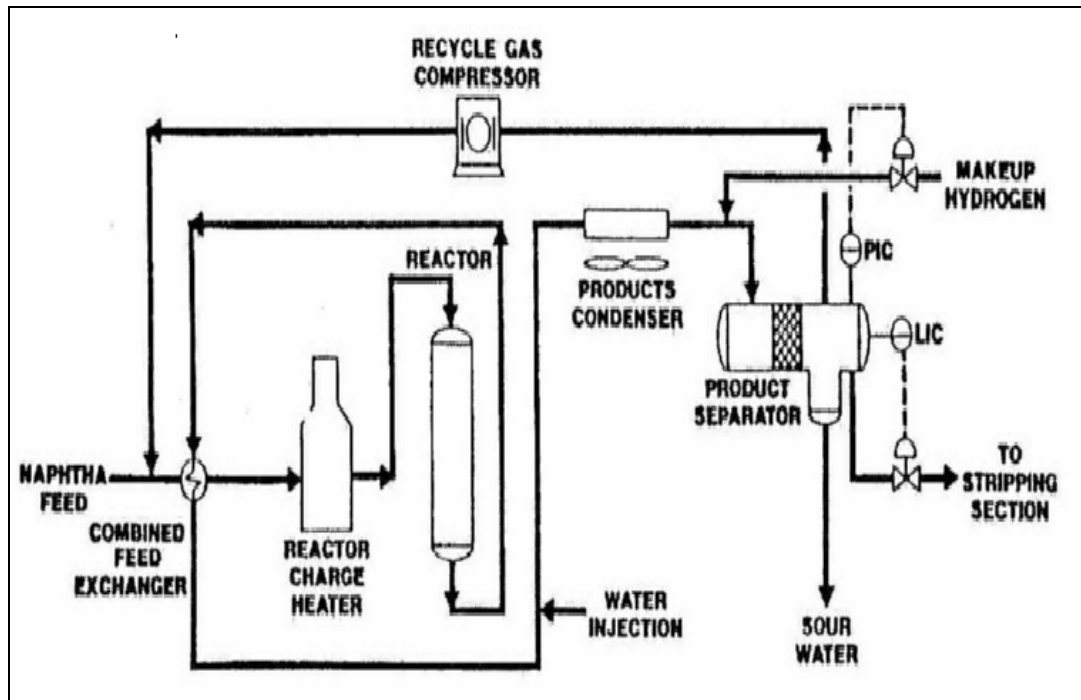


Figure 6.8 : The reactor section with recycle gas

Combined feed exchanger (CFE)

The mixture of exchanger feed and recycle hydrogen is in the shell side of this feed exchanger and the feed is 100% vapour when leaving the feed exchanger. The effluent from the reactor, which contains high concentrations of H_2S , passes through the CFE on the tube side allowing heat transfer to occur with the feed stream (which is in the shell side).

Fire Heater

Provides heat to increase the reactor feed temperature that required in the reactor.

Reactor

The reactor in the NHT unit is a packed bed catalytic reactor. This is where all the reactions occur. Reactor feed and all of the materials in the reactor must remain in vapor state at all times to avoid preferential flow and channeling. The hydrogen and hydrocarbon feed will enter at the top of the reactor and leave from the bottom of the reactor as downflow (in a mixture of vapor and liquid). The reactions that take place in the reactor are as follows:

Removal of sulphur

For good performance of the bimetallic catalyst used in the CRU reactor, the naphtha feed must contain less than 0.5wt ppm of sulphur. For optimum selectivity and stability of the catalyst, the hydrotreated naphtha should contain sulphur well below the 0.5wt ppm limit, i.e. 0.2 wt ppm or less.

Removal of nitrogen

Removal of nitrogen from naphtha is much more difficult if compared to the removal of sulphur. Rate of removal of sulphur from naphtha is five times faster if compared to the rate of removal of nitrogen. Naphtha feed to the CRU must have less than 0.5 wt ppm nitrogen. Nitrogen if fed to the CRU will form ammonia and will then react with chloride to form ammonium chloride (NH_4Cl_2). Ammonium chloride salts will then deposit in the CRU recycle gas circuit and/or stabilizer

overhead system leading to corrosion or plugging. It will also disturb the chloride-water balance that is crucial for the CRU catalyst's performance.

Removal of oxygen

Generally only small amounts of oxygenate compounds are present in the untreated naphtha. The carbon-hydroxyl bond is hydrogenated to form water and the corresponding hydrocarbon. Thus removing oxygen.

Olefin saturation

Presence of olefins in the CRU enhances the amount of coke produced on the CRU catalyst. Since the presence of coke reduces the activity of the catalyst, olefins are saturated. Olefin saturation is very rapid, about 80% the rate of sulphur removal. Olefin saturation is a very exothermic process.

Removal of metals

Most metals are a permanent poison for the CRU catalyst and can pose environmental problems if they are contained in fuel products. Even in ppm levels these metals can reduce the CRU catalyst's life. As such, these metals are removed before the hydrotreated naphtha is fed to the CRU.

Product condenser

The reactor effluent passes through the condenser to condense all naphtha present.

Water injection

Water injection is used to control salt deposits and fouling/pressure drop (caused by salt buildup due to the presence of nitrogen).

Product separator

The mixture of vapour and liquid from the bottoms of the reactor is separated in the product separator. The vapour from the tops of the separator is recompressed, combined

with fresh H_2 and mixed with the reactor feed. The liquid from the bottoms of the separator is fed into a stripper.

The stripping section

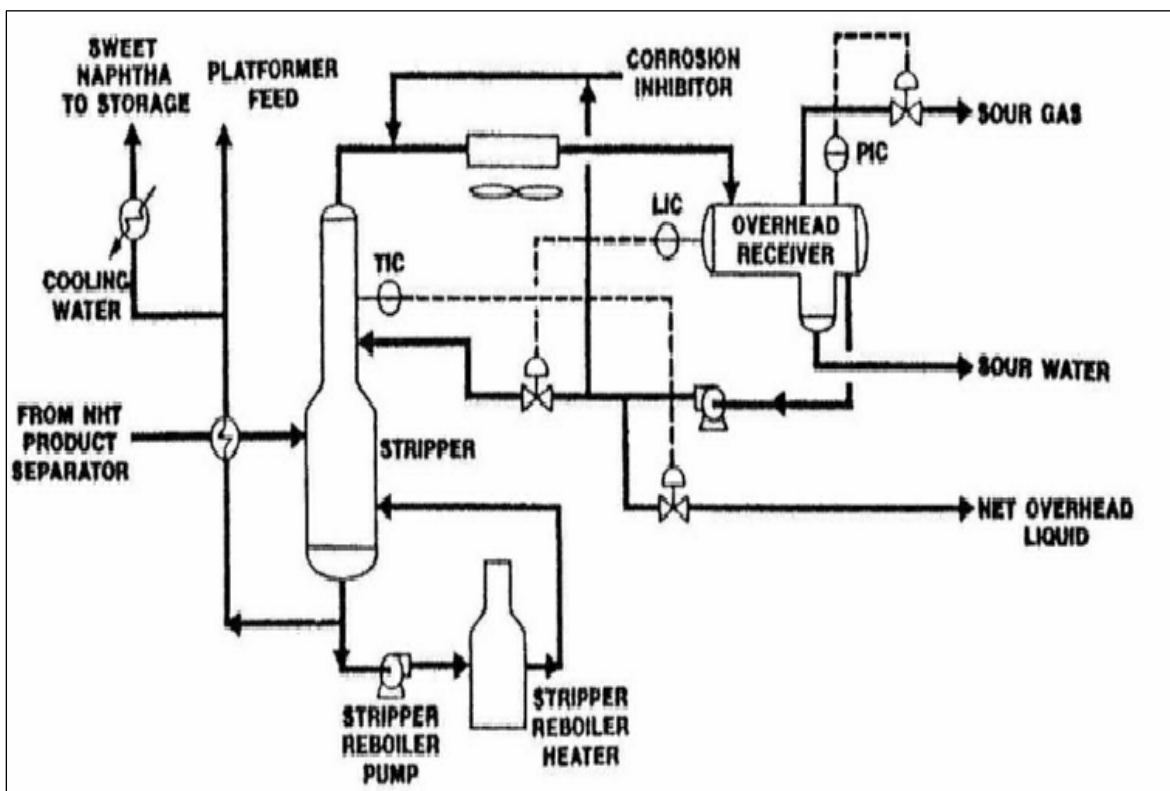


Figure 6.9 : The stripping section

Stripper

The stripper removes H_2S , H_2O , hydrogen and other non-condensables, together with the resultant H_2S and NH_3 from the reactions in the reactor. The stripper only has an overhead gas product but not an overhead liquid product. The reflux to feed ratio is approximately 0.25. Stripper bottoms are the main product (treated naphtha/distillate) of the NHT unit. Stripped naphtha from the NHT unit should meet the following specifications:

Component	Specification
Sulphur	< 0.5 wt ppm
Nitrogen	< 0.5 wt ppm
Chlorides	< 0.5 wt ppm
Endpoint temperature	400 F max
Lead	< 20 wt ppb max
Arsenic	1 wt ppb max
Iron with chloride	1 wt ppm max
Copper with heavy metals	< 25 wt ppb max

Reboiler

A fired reboiler is used in the NHT unit to supply heat.

Catalytic Reforming Unit with Continuous Catalytic Regenerator (Unit 13)

Unit 13 contains mainly of two sections which are the catalytic reforming unit, CRU and the continuous catalytic regenerator, CCR. CRU and CCR exist together but have different functions. This unit depends on the NHT unit to ensure that it gets 'clean' enough feed to be processed. This is to avoid contaminants or poisons to come in contact with the catalyst in this unit and destroy its activity.

Catalytic Reforming Unit

CRU can also be divided into two sections which are the reactor section and the debutanizer section. The CRU operates at a temperature of 530°C and a pressure of 6 kg/cm² g. The catalyst used in this unit is the alumina based catalyst with platinum (0.29 wt% Pt). The primary function of this catalyst is to promote the dehydrogenation reaction in the reactors.

CRU's main function is to produce reformate from heavy naphtha. This is to increase the value of naphtha by upgrading it to a component that can contribute in increasing the octane number of motor gasoline. CRU increases the RON of naphtha up to 101 by converting it to aromatics

The most desired product from the CRU unit is the reformate but the process also produces LPG, and hydrogen from the dehydrogenation reaction that occurs in the reactors in this unit. The significant quantity of hydrogen produced is utilized by other units such as the hydrocracker and hydrotreaters.

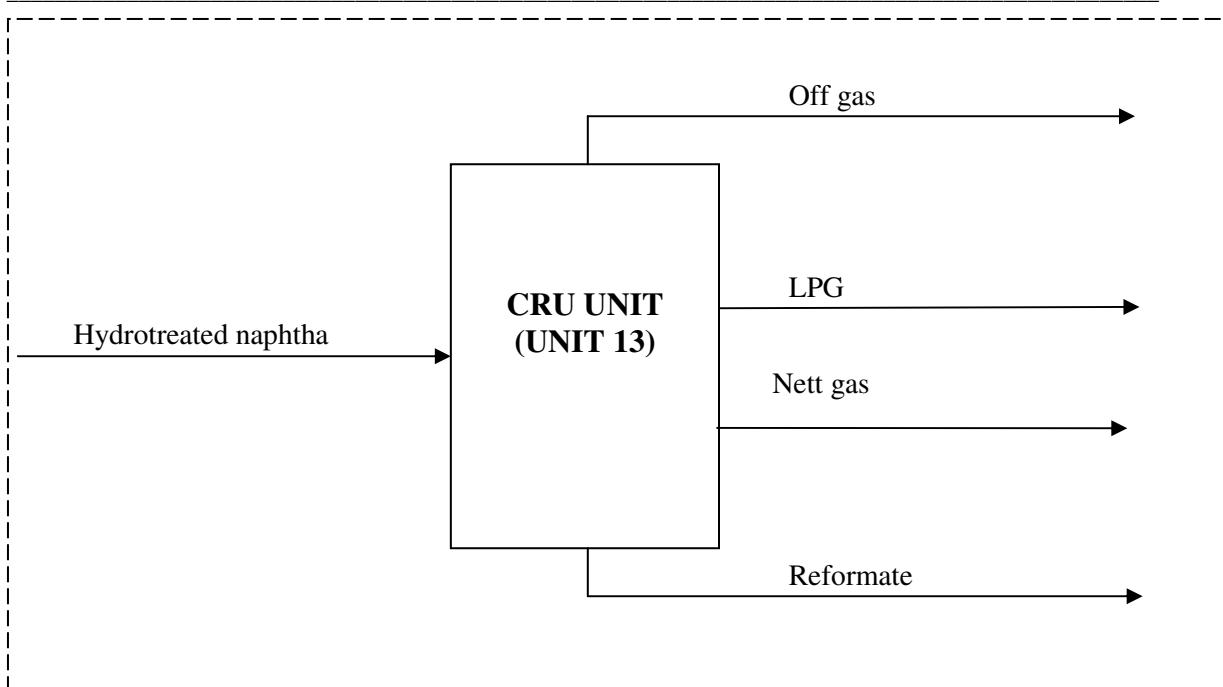


Figure 6.10 : Unit 13 Block Flow Diagram

Product streams from the CRU unit	Downstream unit
Off gas	Fuel Gas System
LPG	Saturated Gas Concentration Unit
Nett Gas	Naphtha Hydrotreating Unit
Reformate	Blend Stock Storage and Blending System

Table 6.1 : CRU products with their respective downstream units

Product	Product Specification		
	RVP	RON	C5+ specification
Reformate	100.0	25.0 KPa	-
LPG	-	-	< 1.5 vol%

Table 6.2 : CRU products with their product specifications

RON

Research octane number (RON) measures the anti-knock performance of gasoline. The higher the RON number, the better the anti-knock performance of gasoline.

RVP

Reid Vapour Pressure (RVP) measures volatility. The higher the value of RVP, the more volatile gasoline.

Summary of the sub-units within the CRU unit

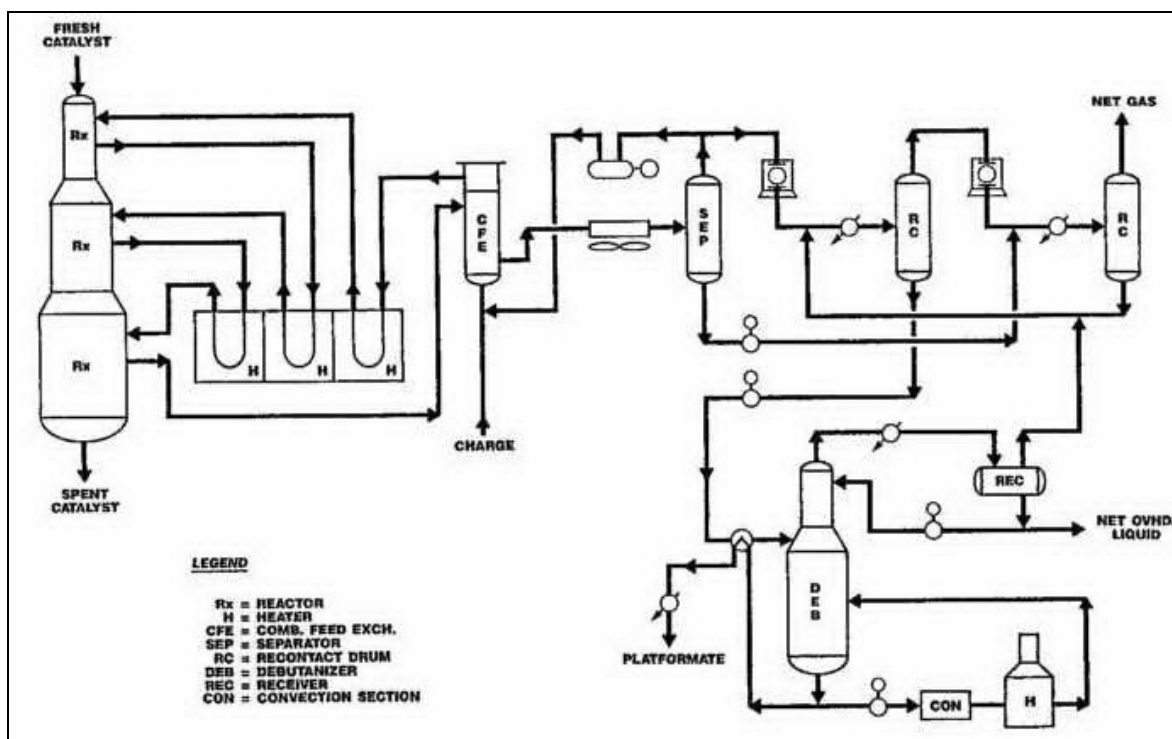


Figure 6.11: The simplified flow diagram of the CRU unit

CRU Reactor Section

Reactors

In the reactors, a combination of high temperature and low pressure promotes the dehydrogenation reaction that converts naphtha into high octane components such as the aromatics. Alumina based platinum catalyst is used to accelerate the reaction. In order to maximize conversion, four reactors are utilized in series. The four reactors are stacked together in a downward sequence with the first one at the top.

Heaters

To provide the heat required for the endothermic reaction in this unit.

Product Separator

The reactor effluent from the reactor goes into the product separator to produce hydrogen rich gas and hydrocarbon rich liquid. Produced mainly to satisfy the demand for hydrogen from hydrogen consuming units, the hydrogen rich gas is distributed throughout the plant according to the daily plan generated by the Production Planners. The hydrocarbon rich liquid is sent to the debutanizer to undergo further separation.

CRU Debutanizer Section

Debutanizer Column

The debutanizer column basically works as a distillation column. Its main function is to distill the hydrocarbon rich liquid from the reactor section to produce two valuable products namely LPG and reformat. Product specifications can be met by adjusting the temperature controller at the column or the heat duty of the bottom reboiler. LPG from the column is sent to the treating units while the reformat is sent to the intermediate tank before being blended into the final product. This column is also called a stabilizer because it removes unstabilized light ends from the reformat.

Continuous Catalytic Regenerator

Without catalyst regeneration, catalyst will be useless after 1-2 weeks. The CCR section is designed especially to regenerate the used R-164 catalyst, by burning off coke from the catalyst in a continuous process, so that fresh catalyst can be supplied continuously to the reactors. The catalyst used in the reactor flows downward through all four reactors and due to the high temperature condition, coke tends to deposit on the catalyst, hence deactivating the catalyst.

The used catalyst is then lifted from the bottom of the reactor to the top of the regeneration tower. In the regeneration tower, the catalyst flows down due to the gravitational force in the tower. The tower operates at a temperature between 490-570°C and at atmospheric pressure with the presence of air. It then goes through three major steps which are burning, chlorination and drying/cooling.

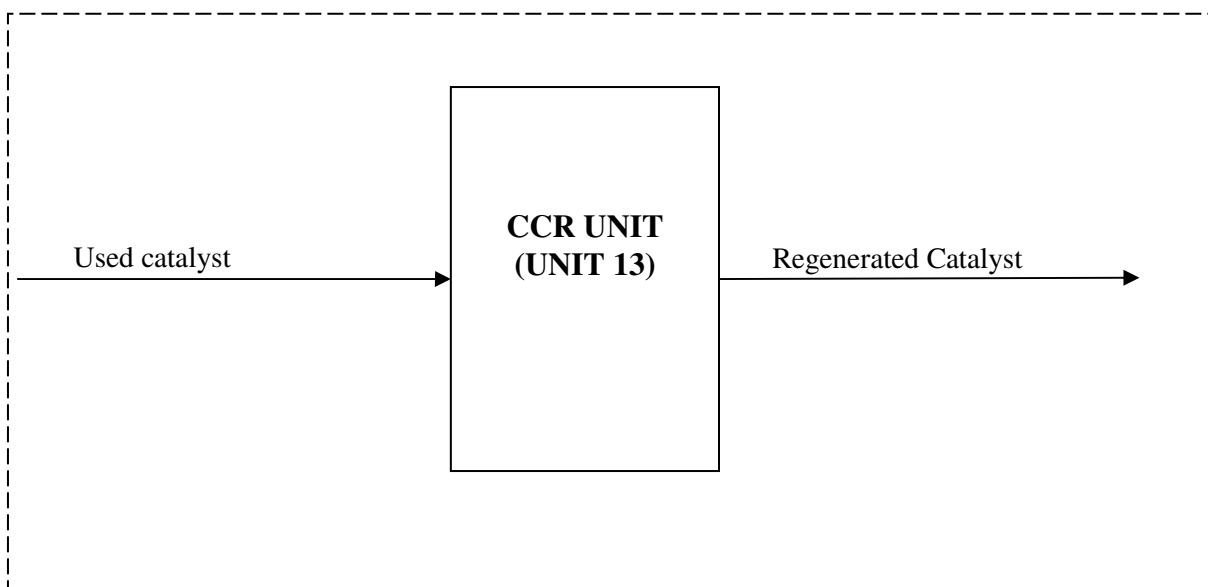


Figure 6.12 : Unit 13 Block Flow Diagram

The Regeneration Tower

Burning Zone

The catalyst from the bottom of the reactor first enters the burning zone in the regeneration tower where coke is burnt off the catalyst in the presence of heat and air. The amount of oxygen used is vital (0.6 mol% - 0.8 mol %) as too much oxygen will lead to high temperatures that could destroy the catalyst.

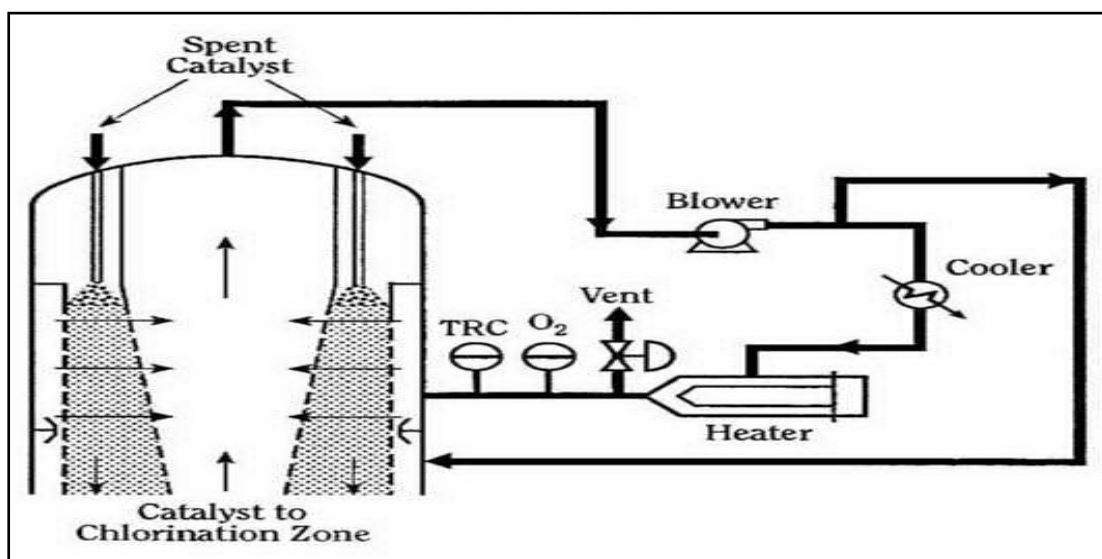


Figure 6.13 : Burning zone in the regeneration tower

Chlorination Zone

After coke burning, the catalyst gets chlorinated in the chlorination zone. A chemical compound containing chlorine is injected into the zone and with the presence of air, the chlorine is able to redisperse platinum on the catalyst surface. Chlorination is also important as a certain amount of chlorine is needed on the catalyst to boost catalyst activity. It is also important to ensure that all coke is completely burned in the burning zone so that no coke is present in this oxygen rich environment.

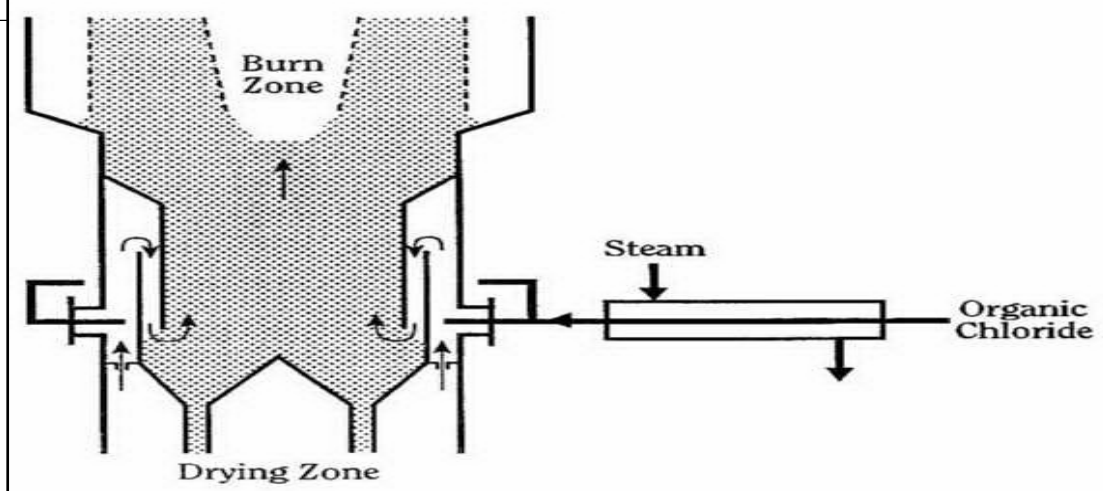


Figure 6.14: Chlorination zone in the regeneration tower

Drying/Cooling Zone

Water accumulated on the catalyst in the burning zone and the chlorination zone is removed in this section. Too much water will cause the chloride to be stripped off from the catalyst in the reactor. Thus the catalyst is first dried using hot air and then cooled using cold air before leaving the regenerator and sent into the reactor.

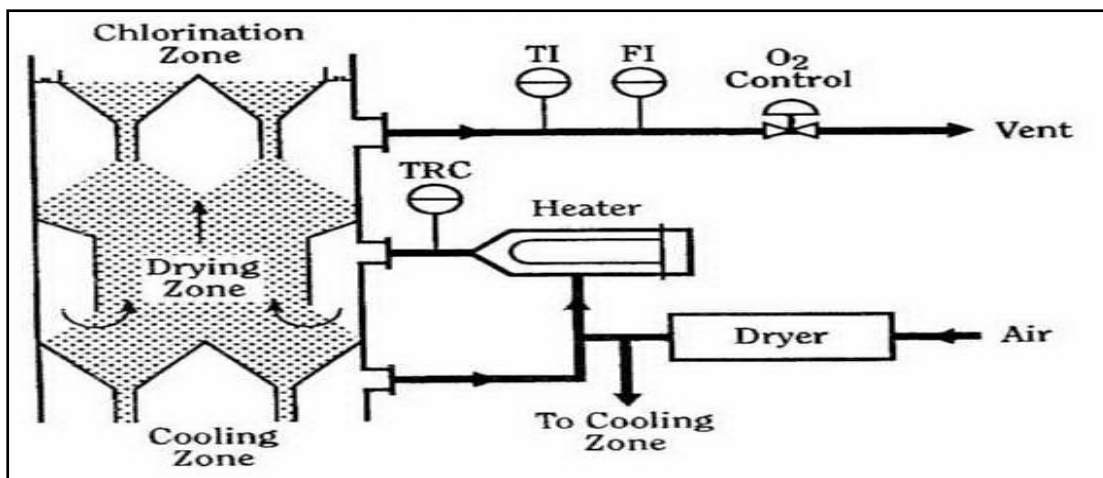


Figure 6.15 : Drying/cooling zone in the regeneration tower

Saturated Gas Concentration Unit (Unit 14)

Saturated Gas Concentration Unit (SGCU) plays a role to process feed from unit 11 ; crude distillation unit and feed from unit 13 ; catalytic reforming unit to produce fuel gas and LPG. This unit also removes hydrogen sulfide from the feed through caustic wash. SGCU has a design capacity of 13,600 kg/hr feed.

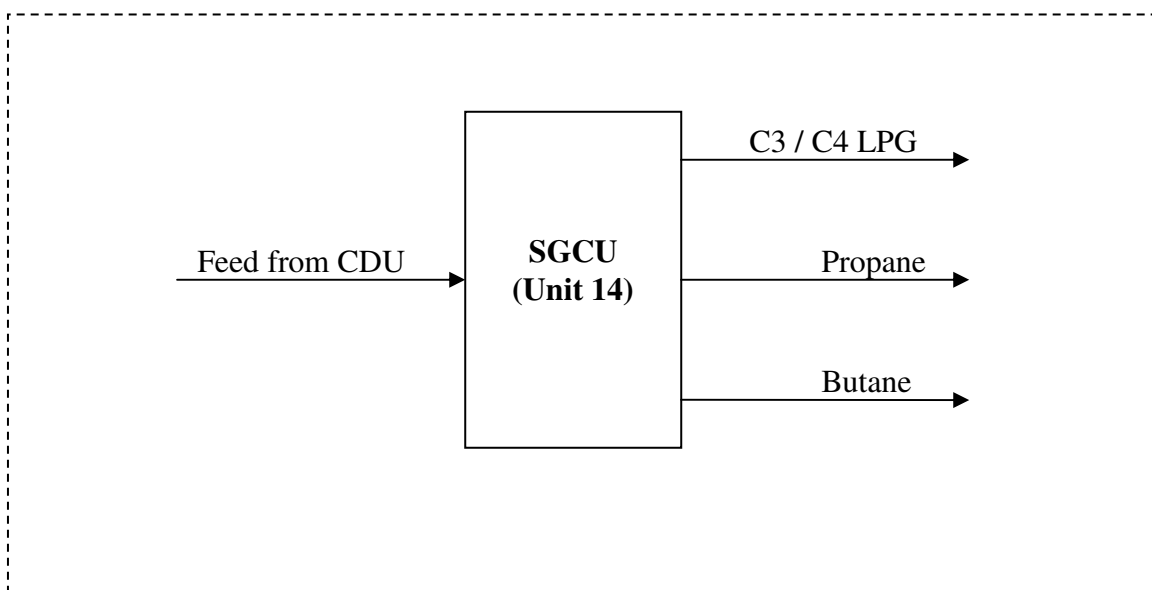


Figure 6.16 : Unit 14 Block Diagram

Summary of sub-units within the SGCU unit

This unit is divided to three sections which are the Fresh Feed Section, Caustic Treating Section and Fractionation Section.

Fresh Feed Section

The feed consists of four kinds of gases. These gases are fed to Compressor Suction Drum (V-1401) from Stabilizer Overhead Receiver (V-1109) in the Crude Distillation

[illegible]

Crude Stabilizer Overhead Liquid & Condensate Stabilizer Overhead Liquid

After separation of hydrocarbon from water, the hydrocarbon liquid in the high pressure receiver (V-1402) is sent to the caustic treating unit.

This liquid is fed to the caustic treater (V-1403) directly to remove hydrogen sulfide from the liquid.

Caustic Treating Section

High pressure liquid and the platforming debutanizer overhead liquid are fed to this section. This section consists of a caustic treater (V-1403) and a sand filter (V-1404). Hydrogen sulfide in feed is removed to meet LPG quality specifications. When liquid passes through the caustic treater, it will be filled by caustic solution.

When caustic have reached 70% spent, the caustic solution is sent to the Caustic Degassing Drum (V-1409) and is replaced by fresh caustic. The caustic solution is carried in the light liquid from the caustic treater (V-1403) and is filtered at the sand filter (V-1404) and treated light liquid is sent to the next section.

Spent caustic solution drawn off from the caustic treater (V-1403) and sand filter (V-1404) is sent to the spent Caustic Degassing Drum (V-1409).

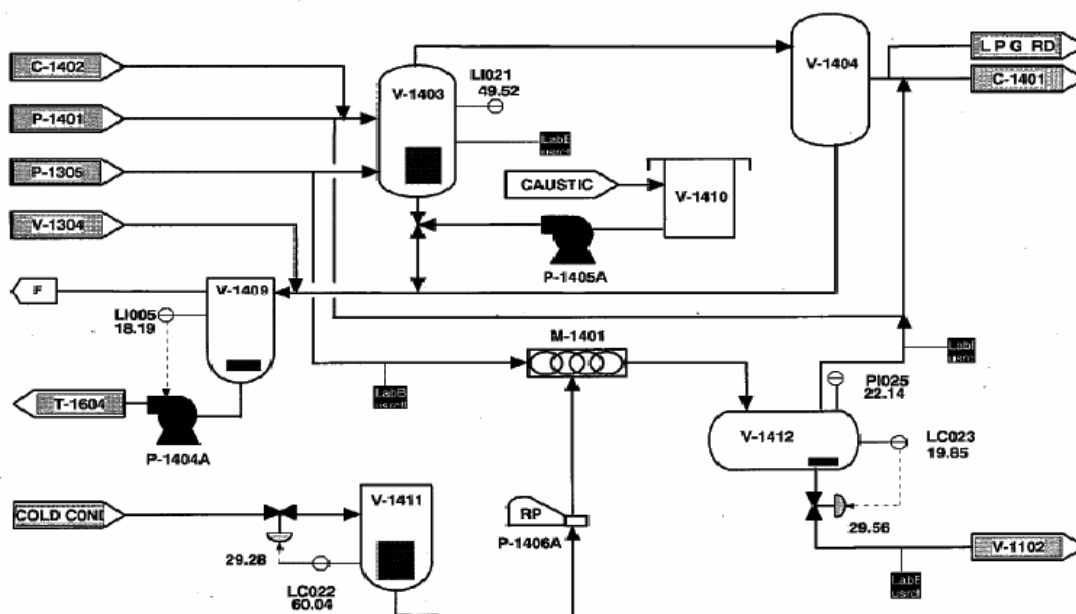


Figure 6.17 : Caustic Treating Section

Fractionation Section

Deethanizer

Treated light liquid from the sand filter (V-1404) is sent to the Deethanizer (C-1401) to strip C2 and lighter components after being heated up in the Deethanizer Feed Exchanger (E-1403). Deethanizer overhead vapor goes to the Deethanizer Overhead Receiver (V-1406) via Deethanizer Overhead Condenser (E-1404). C2 and lighter components are sent to the fuel gas system and the overhead receiver liquid is pumped to the top of the deethanizer as total reflux. The deethanizer bottom liquid is sent to the C3/C4 splitter (C-1402).

C3/C4 Splitter

C3/C4 splitter overhead vapor goes to the C3/C4 splitter overhead receiver (V-1408) via the C3/C4 splitter overhead condenser (E-1405).

The receiver liquid (C3) is pumped to the top tower as reflux and run down to the LPG Storage (T-7501 A/B/C/D) via the splitter net overhead cooler (E-1406) as part of the LPG product. The bottom liquid (C4) runs down via deethanizer feed exchanger (E-1403) and splitter bottom cooler (E-1407) and is separated into two. One will be drawn off to the butane bullet (V-7301) and another one, after mixing with C3 which comes from the C3/C4 splitter overhead receiver is sent to LPG storage (V-7501 A/B/C/D).

Also, C3 liquid is used for refrigerant of the refrigeration equipment (A-1308) in the Platforming Unit.

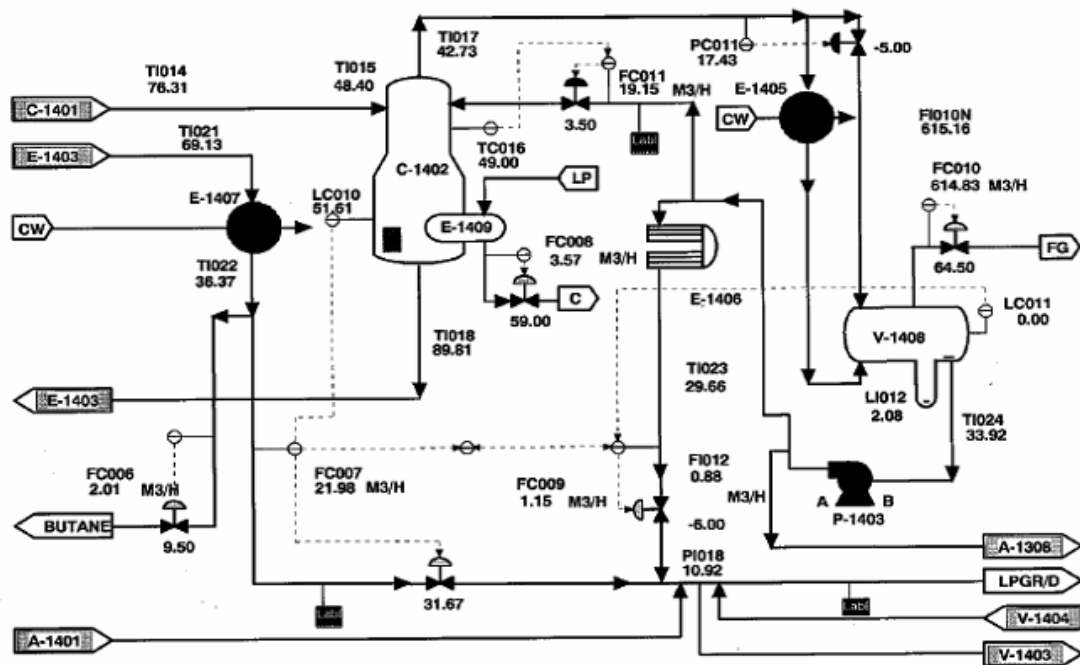


Figure 6.18 : C3/C4 Splitter Section

Sour Water Stripping Unit (Unit 15)

The Sour Water Stripping Unit has the function to treat sour water from process unit to make it suitable for use in the Crude Unit desalter. It has a design capacity of 16 ton/hr of feed.

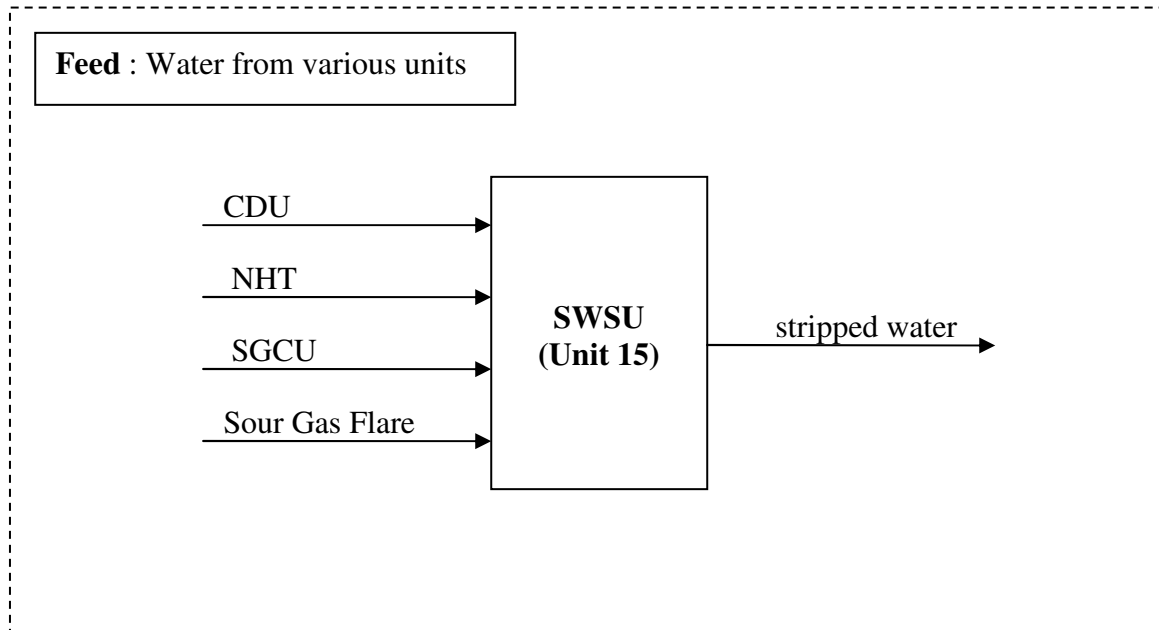


Figure 6.19 : Unit 15 Block Diagram

Summary of sub-units within the SWSU unit

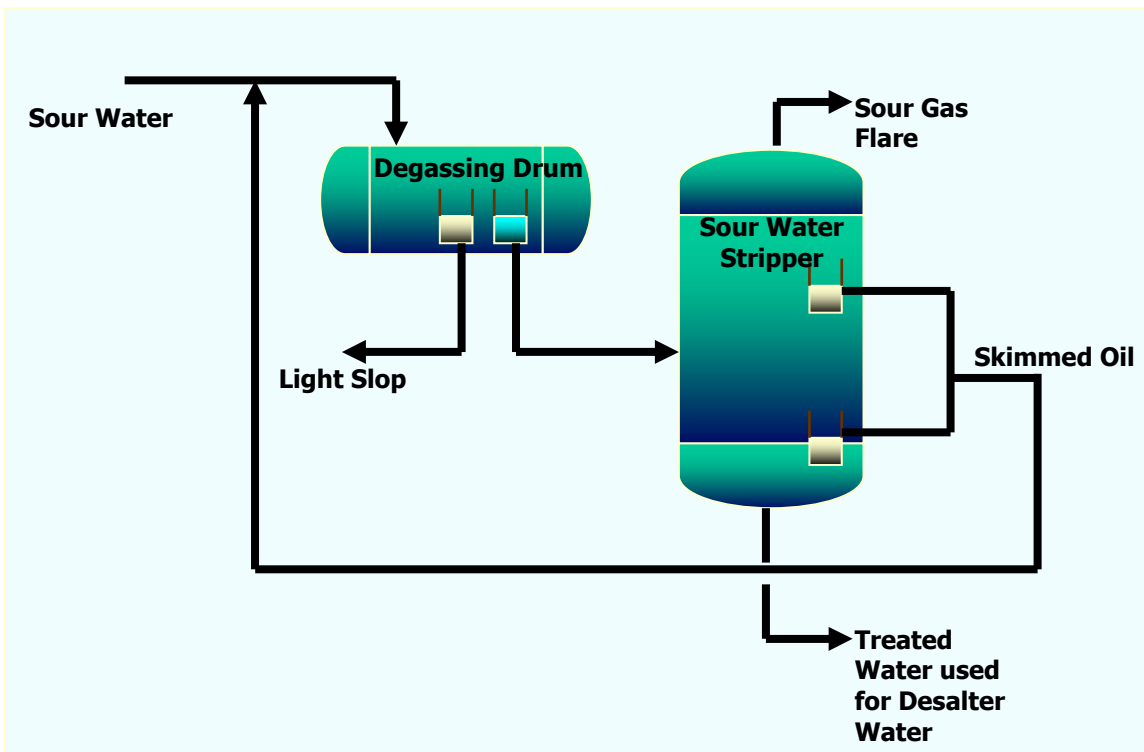


Figure 6.20 : SWSU Process

Fresh Feed and Pretreatment Section

Fresh feed of sour water comes from Crude Distillation Unit, the Naphtha Hydrotreating Unit and the Sour Gas Flare System.

All sour water will be fed in to the Degassing Drum (V-1501) for degassing and de-oiling. An inner baffle will divide the Degassing Drum into two sections ; a gas-oil-water separation section and a water holding section. The gas-oil-water separation section keeps the feed sour water for a sufficient period of time to allow separation of gas and oil from water. Separated oil will be gathered in an oil container, then pumped out by Slop Oil Pump (P-1502) and transferred to Light Slop Oil Tank (T-7701). Degassed and de-oiled sour water is overflowed via the vertical baffle and sent to the water holding section. The water will then be pumped up to the Sour Water Stripper (C-1501). The sour water is fed

to the Sour Water Stripper (C-1501) after it is heated up by heat exchanger with treated water from the bottom of the Sour Water Stripper.

Sour gas from the Degassing Drum (V-1501) is sent to the Sour Gas Flare System to be safely disposed.

Post-treatment Section

The feed sour water is supplied to the no.10 tray of the Sour Water Stripper (C-1501). Hydrogen Sulfide, H_2S and Ammonia, NH_3 are stripped by the Sour Water Stripper Reboiler (E-1502) located at the bottom section of the column.

Sour Water Stripper Top Section

Vapor from the feed stream is combined with the vapor from no.10 tray and passes through baffle trays (1-6). Overhead reflux is drawn off from the accumulator above tray no.7 and is pumped (by P-1504A/B), cooled (by E-1503) and returned to the top section of the Stripper.

Non-condensable vapor such as Hydrogen Sulfide, H_2S and Ammonia, NH_3 is discharged from the column stripper through the no.1 tray and is sent to the Sour Gas Flare System.

Sour Water Stripper Bottom Section

The stripped water which is to be drawn off from the stripper bottom will be heated and vaporized by Stripper Reboiler (E-1502) and returned to the striper bottom. The reboiler is heated by LP steam. The remainder of the stripped water is cooled by the Sour Water Stripper Feed/Bottom Exchanger (E-1501), pumped through the Sour Water Stripper Bottom Pump (P-1503 A/B) and cooled by the Stripper Bottom Cooler (E-1504).

The fresh caustic solution is supplied to the Sour Water Stripping System via the Caustic Break Tank (V-1502).

Figure 6.21 : SWSU Process Flow Diagram

Heavy Naphtha and Kerosene Treating Unit (Unit 16)

The heavy naphtha and kerosene treating unit is designed to process heavy naphtha and kerosene products derived from the blends of Tapis Crude, Terengganu and Bintulu Condensates. This unit is divided into several sections which each sections having specific purpose.

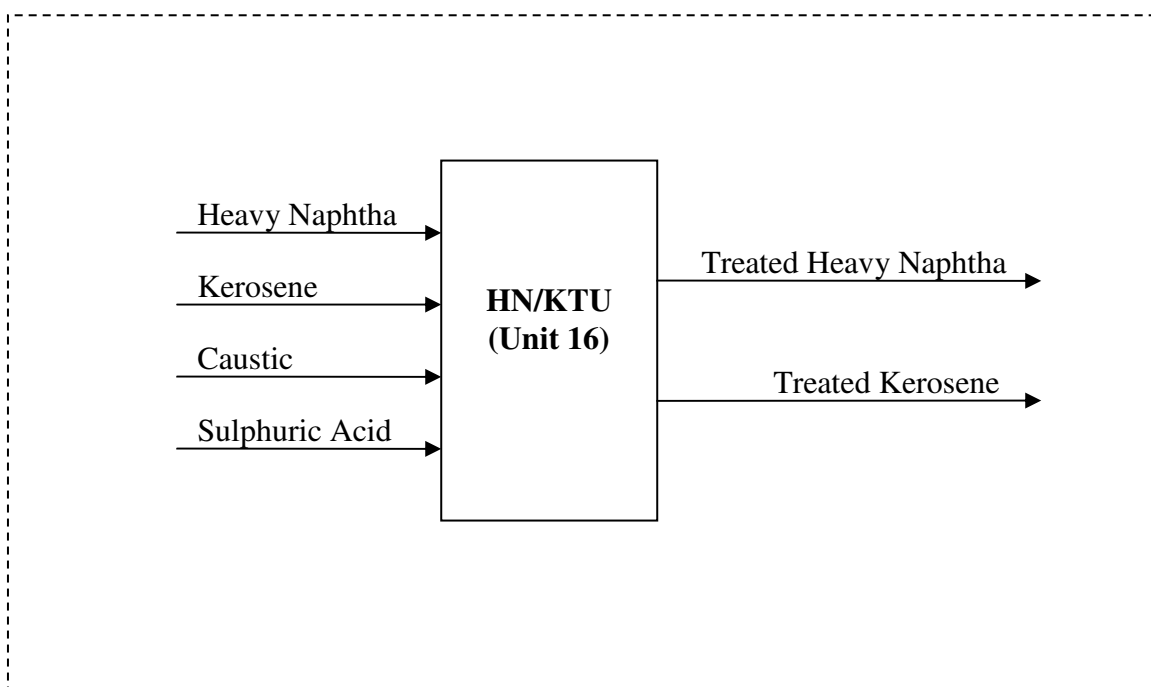


Figure 6.22: Unit 16 Block Diagram

Summary of sub-units within the HN/KTU

Caustic Prewash Section

Caustic soda (NaOH) is used in refining process to remove sulfur compounds and other acidic impurities. These impurities are present as hydrogen sulfides while the remainder in the form of organic derivatives of sulfur. It is necessary in any refinery to remove completely hydrogen sulfide in order to prevent the formation of element sulfur.

Kerosene Prewash

The kerosene from the Crude Distillation Unit is fed into the Kerosene Caustic Prewash. before entering the prewash, untreated kerosene passes through two parallel 150 micron basket strainers to remove any solid particles larger than 150 microns. In the prewash, the kerosene is mixed with caustic solution.

Heavy Naphtha Prewash

The heavy naphtha from the Crude Distillation Unit is brought directly into the Heavy Naphtha Caustic Prewash. The heavy naphtha contains acidic compounds in mixed with caustic solution in the caustic prewash. The process is similar to the Kerosene Prewash Section above.

Water Wash Section

Water wash is to remove entrained treating solution and also the last trace of water-soluble surfactants that otherwise would cause failure of water separation tests.

Kerosene Water Wash

To remove the caustic entrained with the treated kerosene from the Kerosene Caustic Prewash, the prewashed kerosene is fed to the Kerosene Water Wash. The water wash kerosene is drawn off from the top of the Kerosene Water Wash and transferred to the next section.

Heavy Naphtha Water Wash

To remove the caustic entrained with the treated heavy naphtha from the Heavy Naphtha Caustic Prewash, the prewashed heavy naphtha is fed to the Heavy Naphtha Water Wash. The water washed heavy naphtha is run down to the Heavy Naphtha Tank or transferred to the NHT unit.

Salt-Filter Section

The product from the Kerosene Water Wash will be saturated with water and may also contain an additional small amount of very fine free water droplets.

The water washed kerosene is fed to the Kerosene Salt-Filter and withdrawn from the top of it. During this time, water droplets entrained with the kerosene are separated by coagulating among droplets and increasing of water gravity from salt dissolving. The separated water is withdrawn to the Waste Water Treating Unit.

Clay Filter Section

To assure product specification will be met, the Kerosene Clay Filter is installed as part of the treating unit. It serves as a final cleanup to obtain the jet fuel which is free of solids, color bodies, oil-soluble surfactants, metallic compound and water.

Kerosene from the Kerosene Salt Filter is fed to the Kerosene Clay Filter. The filtered kerosene is drawn off from the bottom of the filter and run down to the treated Kerosene Tank.

Chemical Handling Section

Receiving of the 10° Be Caustic

50° Be caustic soda will be unloaded into the tank from the tank truck. 10° Be caustic soda is prepared by dilution of 50° Be caustic in the chemical supply unit (Unit 56). 10° Be caustic transfer pumps (P-5602 A/B) issued for transfer to each user's.

Making 3° Be Caustic Solution

This solution is made in the 3° Be Caustic Storage Tank by dilution from 10° Be caustic with cold condensate. It is necessary to calculate the volume ration of 10° Be caustic solution and the cold condensate.

3° Be caustic is sent to the Kerosene Caustic Prewash and the Heavy Naphtha Caustic Prewash by Kerosene Caustic Wash Pumps (P-1605 A/B) and Heavy Naphtha Caustic Wash Pumps (P-1606 A/B) respectively.

Receiving and Supply of Sulfuric Acid

The sulfuric acid is used to neutralize the spent caustic. The sulfuric acid is transferred to the Sulfuric Acid Drum (V-1608) from tank truck. The sulfuric acid tank is sealed by the pressure regulator with dry air to prevent entering of humid air from atmosphere and is provided with a silica gel absorbing drum in the vent line.

The sulfuric acid is transferred to the Neutralizing Drum by the Sulfuric Acid Transfer Pumps (P-1611 A/B)

Neutralization Section

This section covers the neutralization of the spent caustic from the Heavy Naphtha, Kerosene, LPG Caustic Treaters. The spent caustic solution in the Alkaline Waste Water Tank (T-1604) is withdrawn off by the Alkaline Waste Water Pump (P-1609 A/B) and fed to the Neutralizing Drum.

In the Neutralizing Drum, the spent caustic solution is neutralized by the mixed solution of sulfuric acid.

The neutralized waste water is blow down to the Vent Tank (V-1611) via the Neutralized Alkaline Waste Water Exchanger. The sour gas such as hydrogen sulfide generated by neutralizing is released from the top of the Neutralizing Drum to the Sour Flare System.

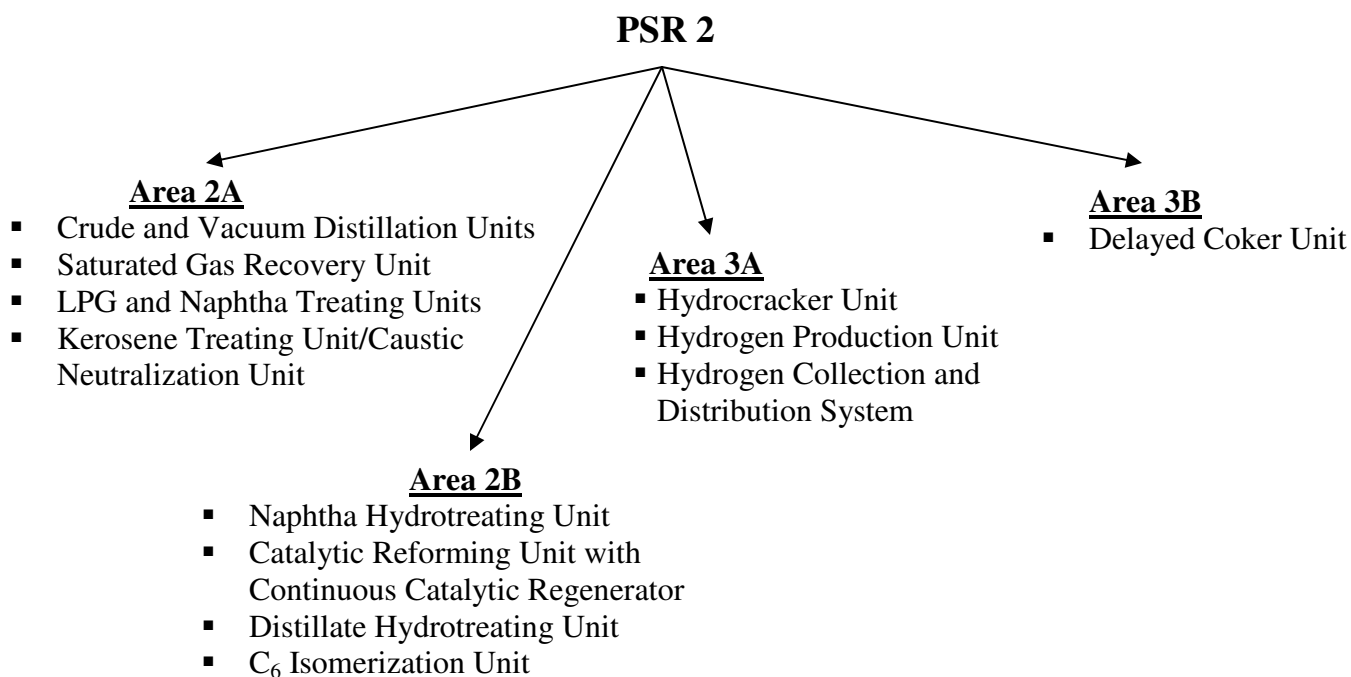
7.0 PETRONAS Second Refinery – Phase 2 (PSR-2)

PSR 2 processes up to 70% sour feedstock (sulfur>2.5wt %), mainly Middle East crude, for the design capacity of 100,000 barrels per day (BPSD) utilizing complex conversion configuration (sour train). Feedstocks used are Arabian Heavy Crude and Basrah Light Crude.

The main process units at PSR 2 include:

- Crude and Vacuum Distillation Units
- Naphtha Hydrotreating Unit
- Catalytic Reforming Unit with Continuous Catalyst Regenerator
- Saturated Gas Recovery Unit
- C₆ Isomerization Unit
- LPG and Naphtha Treating Units
- Kerosene Treating Unit/Caustic Neutralization Unit
- Distillate Hydrotreating Unit
- Hydrocracker Unit
- Delayed Coker Unit
- Hydrogen Production Unit
- Hydrogen Collection and Distribution System

PSR 2 is divided into four areas for operating convenience. PSR 2 includes area 2A, area 2B, area 3A and area 3B.



UNIT	ABBREVIATION	UNIT NO	CAPACITY
Crude Distillation Unit	CDU	21	100,000 BPSD
Vacuum Distillation Unit	VDU	21	42,000 BPSD
Naphtha Hydrotreating Unit	NHT	22	35,000 BPSD
Catalytic Reforming Unit with Continuous Catalyst Regenerator	CRU and CCR	23	26,000 BPSD
Saturated Gas Recovery Unit	SGRU	24	Not applicable
C ₆ Isomerization Unit	C6 ISOM	25	9,000 BPSD
LPG Treating Unit	LTU	26	8,500 BPSD
Naphtha Treating Unit	NTU	27	15,250 BPSD
Kerosene Treating Unit	KTU	28	13,800 BPSD
Distillate Hydrotreating Unit	DHT	29	35,000 BPSD
Hydrocracker Unit	HCK	30	26,500 BPSD
Delayed Coker Unit	DCU	32	21,000 BPSD
Hydrogen Production Unit	HPU	39	13,400 Nm ³ /hr

Table 7.1 : Summary of all process units in PSR 2

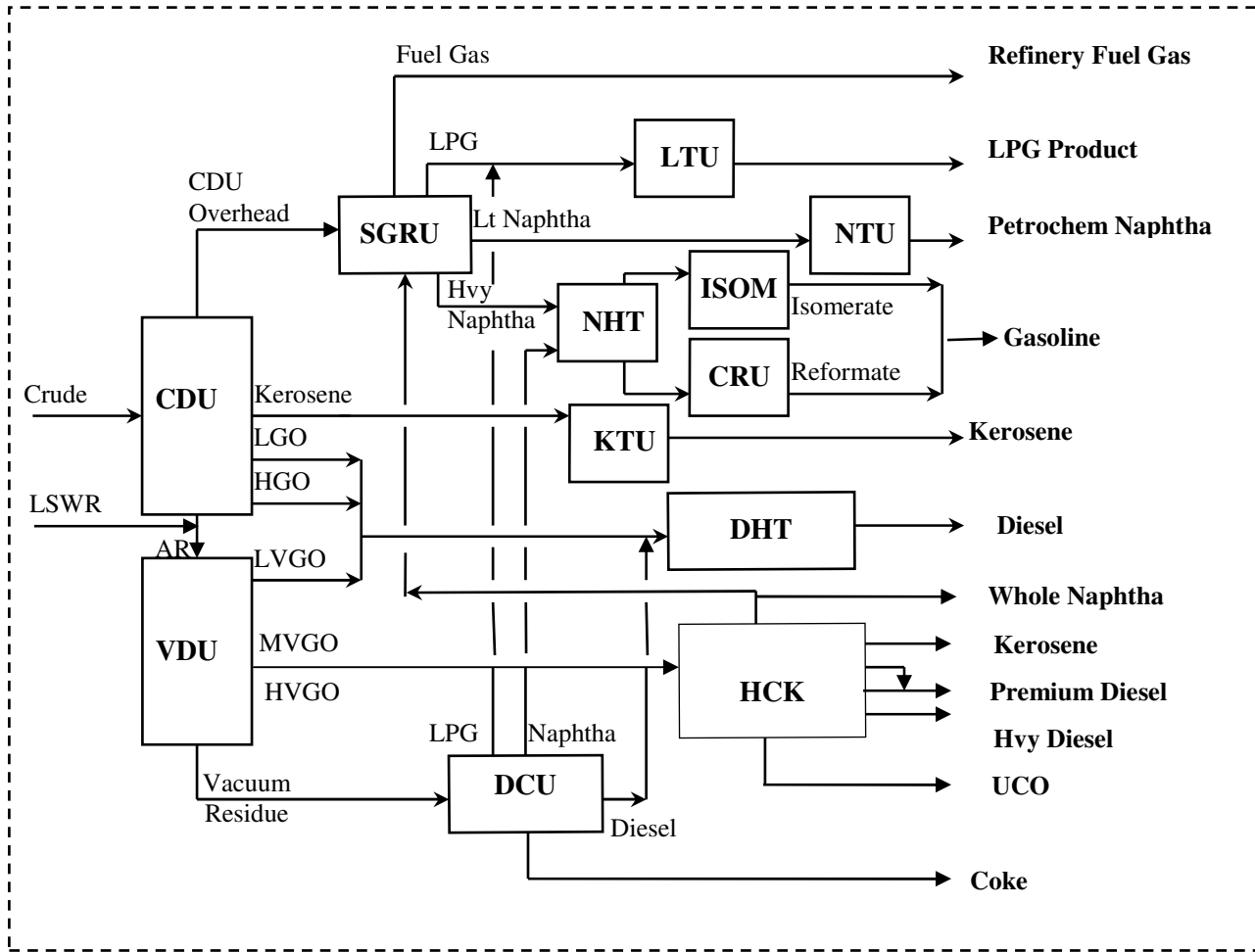


Figure 7.1 : PSR 2 Block Flow Diagram

Crude and Vacuum Distillation Units (Unit 21)

The crude and vacuum distillation unit fractionates raw crude to produce components which are used as feed in downstream units.

Crude Distillation Unit (CDU)

The objective of the CDU is to provide a **primary fractionation of crudes** to produce **straight run blendstocks** of distillate products (after suitable treating processes) and feedstock for other downstream process units. CDU has a design capacity of 100 000 BPSD. CDU is designed to distill **Arabian Heavy** and **Basrah Light Crudes**. The products formed are the unit's overhead gasses which consist of LPG and the full range of naphtha, kerosene, light gas oil, heavy gas oil and atmospheric residue.

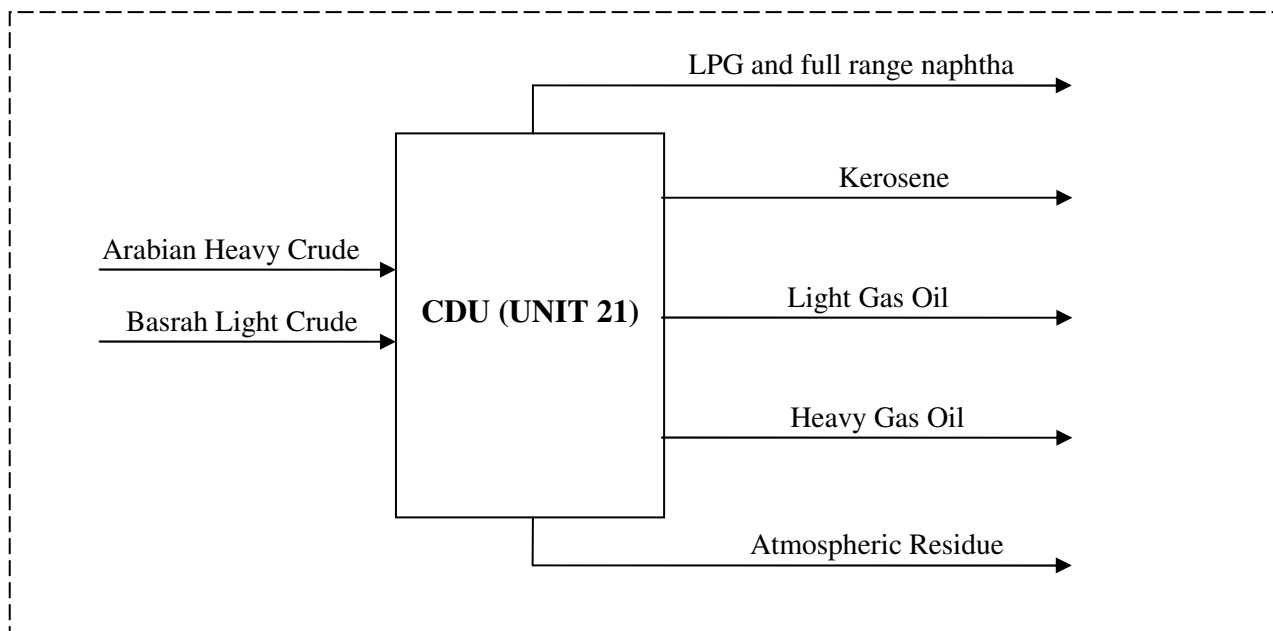


Figure 7.2 : Unit 21 Block Flow Diagram

Product stream from the CDU unit	Downstream unit
Off gas	Saturated Gas Recovery Unit
Naphtha	Saturated Gas Recovery Unit
Kerosene	Kerosene Treating Unit
Light Gas Oil	Distillate Hydrotreating Unit
Heavy Gas Oil	Distillate Hydrotreating Unit
Atmospheric Residue	Vacuum Distillation Unit

Table 7.2 : CDU products with their respective downstream units

The CDU processes either of the following design feedstock:

- Design feed 1
Processing of Basrah Light Crude. This sets the design of the top section of the unit.
- Design feed 2
Processing of Arabian Heavy Crude. This sets the design of the bottom section of the unit.

In addition to the design crudes, the CDU processes a number of alternative crudes. The ranges of alternative crudes to be processed are Tapis, Duri, Iranian Light, Iranian Heavy, Basrah and FAO Blend. The unit is also designed to process a maximum of 2000 BPSD of wild naphtha from the distillate hydrotreating unit and 100 BPSD of wild naphtha from the vacuum gas oil hydrotreating unit.

The diagram illustrates the process flow of a chemical plant, divided into several main sections:

- DESALTER:** Located at the top left, it shows a desalter unit with associated pumps (P-21201, P-21202, P-21203, P-21204, P-21205, P-21206, P-21207, P-21208, P-21209, P-21210, P-21211, P-21212, P-21213, P-21214, P-21215, P-21216, P-21217, P-21218, P-21219, P-21220, P-21221, P-21222, P-21223, P-21224, P-21225, P-21226, P-21227, P-21228, P-21229, P-21230, P-21231, P-21232, P-21233, P-21234, P-21235, P-21236, P-21237, P-21238, P-21239, P-21240, P-21241, P-21242, P-21243, P-21244, P-21245, P-21246, P-21247, P-21248, P-21249, P-21250, P-21251, P-21252, P-21253, P-21254, P-21255, P-21256, P-21257, P-21258, P-21259, P-21260, P-21261, P-21262, P-21263, P-21264, P-21265, P-21266, P-21267, P-21268, P-21269, P-21270, P-21271, P-21272, P-21273, P-21274, P-21275, P-21276, P-21277, P-21278, P-21279, P-21280, P-21281, P-21282, P-21283, P-21284, P-21285, P-21286, P-21287, P-21288, P-21289, P-21290, P-21291, P-21292, P-21293, P-21294, P-21295, P-21296, P-21297, P-21298, P-21299, P-21300, P-21301, P-21302, P-21303, P-21304, P-21305, P-21306, P-21307, P-21308, P-21309, P-21310, P-21311, P-21312, P-21313, P-21314, P-21315, P-21316, P-21317, P-21318, P-21319, P-21320, P-21321, P-21322, P-21323, P-21324, P-21325, P-21326, P-21327, P-21328, P-21329, P-21330, P-21331, P-21332, P-21333, P-21334, P-21335, P-21336, P-21337, P-21338, P-21339, P-21340, P-21341, P-21342, P-21343, P-21344, P-21345, P-21346, P-21347, P-21348, P-21349, P-21350, P-21351, P-21352, P-21353, P-21354, P-21355, P-21356, P-21357, P-21358, P-21359, P-21360, P-21361, P-21362, P-21363, P-21364, P-21365, P-21366, P-21367, P-21368, P-21369, P-21370, P-21371, P-21372, P-21373, P-21374, P-21375, P-21376, P-21377, P-21378, P-21379, P-21380, P-21381, P-21382, P-21383, P-21384, P-21385, P-21386, P-21387, P-21388, P-21389, P-21390, P-21391, P-21392, P-21393, P-21394, P-21395, P-21396, P-21397, P-21398, P-21399, P-21400, P-21401, P-21402, P-21403, P-21404, P-21405, P-21406, P-21407, P-21408, P-21409, P-21410, P-21411, P-21412, P-21413, P-21414, P-21415, P-21416, P-21417, P-21418, P-21419, P-21420, P-21421, P-21422, P-21423, P-21424, P-21425, P-21426, P-21427, P-21428, P-21429, P-21430, P-21431, P-21432, P-21433, P-21434, P-21435, P-21436, P-21437, P-21438, P-21439, P-21440, P-21441, P-21442, P-21443, P-21444, P-21445, P-21446, P-21447, P-21448, P-21449, P-21450, P-21451, P-21452, P-21453, P-21454, P-21455, P-21456, P-21457, P-21458, P-21459, P-21460, P-21461, P-21462, P-21463, P-21464, P-21465, P-21466, P-21467, P-21468, P-21469, P-21470, P-21471, P-21472, P-21473, P-21474, P-21475, P-21476, P-21477, P-21478, P-21479, P-21480, P-21481, P-21482, P-21483, P-21484, P-21485, P-21486, P-21487, P-21488, P-21489, P-21490, P-21491, P-21492, P-21493, P-21494, P-21495, P-21496, P-21497, P-21498, P-21499, P-21500, P-21501, P-21502, P-21503, P-21504, P-21505, P-21506, P-21507, P-21508, P-21509, P-21510, P-21511, P-21512, P-21513, P-21514, P-21515, P-21516, P-21517, P-21518, P-21519, P-21520, P-21521, P-21522, P-21523, P-21524, P-21525, P-21526, P-21527, P-21528, P-21529, P-21530, P-21531, P-21532, P-21533, P-21534, P-21535, P-21536, P-21537, P-21538, P-21539, P-21540, P-21541, P-21542, P-21543, P-21544, P-21545, P-21546, P-21547, P-21548, P-21549, P-21550, P-21551, P-21552, P-21553, P-21554, P-21555, P-21556, P-21557, P-21558, P-21559, P-21560, P-21561, P-21562, P-21563, P-21564, P-21565, P-21566, P-21567, P-21568, P-21569, P-21570, P-21571, P-21572, P-21573, P-21574, P-21575, P-21576, P-21577, P-21578, P-21579, P-21580, P-21581, P-21582, P-21583, P-21584, P-21585, P-21586, P-21587, P-21588, P-21589, P-21590, P-21591, P-21592, P-21593, P-21594, P-21595, P-21596, P-21597, P-21598, P-21599, P-21600, P-21601, P-21602, P-21603, P-21604, P-21605, P-21606, P-21607, P-21608, P-21609, P-21610, P-21611, P-21612, P-21613, P-21614, P-21615, P-21616, P-21617, P-21618, P-21619, P-21620, P-21621, P-21622, P-21623, P-21624, P-21625, P-21626, P-21627, P-21628, P-21629, P-21630, P-21631, P-21632, P-21633, P-21634, P-21635, P-21636, P-21637, P-21638, P-21639, P-21640, P-21641, P-21642, P-21643, P-21644, P-21645, P-21646, P-21647, P-21648, P-21649, P-21650, P-21651, P-21652, P-21653, P-21654, P-21655, P-21656, P-21657, P-21658, P-21659, P-21660, P-21661, P-21662, P-21663, P-21664, P-21665, P-21666, P-21667, P-21668, P-21669, P-21670, P-21671, P-21672, P-21673, P-21674, P-21675, P-21676, P-21677, P-21678, P-21679, P-21680, P-21681, P-21682, P-21683, P-21684, P-21685, P-21686, P-21687, P-21688, P-21689, P-21690, P-21691, P-21692, P-21693, P-21694, P-21695, P-21696, P-21697, P-21698, P-21699, P-21700, P-21701

Preheat train

Desalter

61

Furnace

The furnace is used to heat feed. Heating mediums used are fuel gas and combustion air from the atmosphere. The designed duty of the feed furnace is 76MW. 60% of the absorbed heat goes to the radiant section. The remaining heat is used to generate MP steam in the convection section.

Crude Charge Heater

Preheated crude is partially vaporized here. The typical vaporization is 60wt%.

Pumparounds

Pumparounds are product streams, from the crude tower, recycled back as feed into this tower to condense vapour that would otherwise have gone to the lighter cuts, thus regulating internal reflux. Since the vapour load in the tower is reduced, a smaller tower diameter is required. As a result, smaller trays are required in the tower. Pumparounds are also used to recover heat from the product stream to the feed stream.

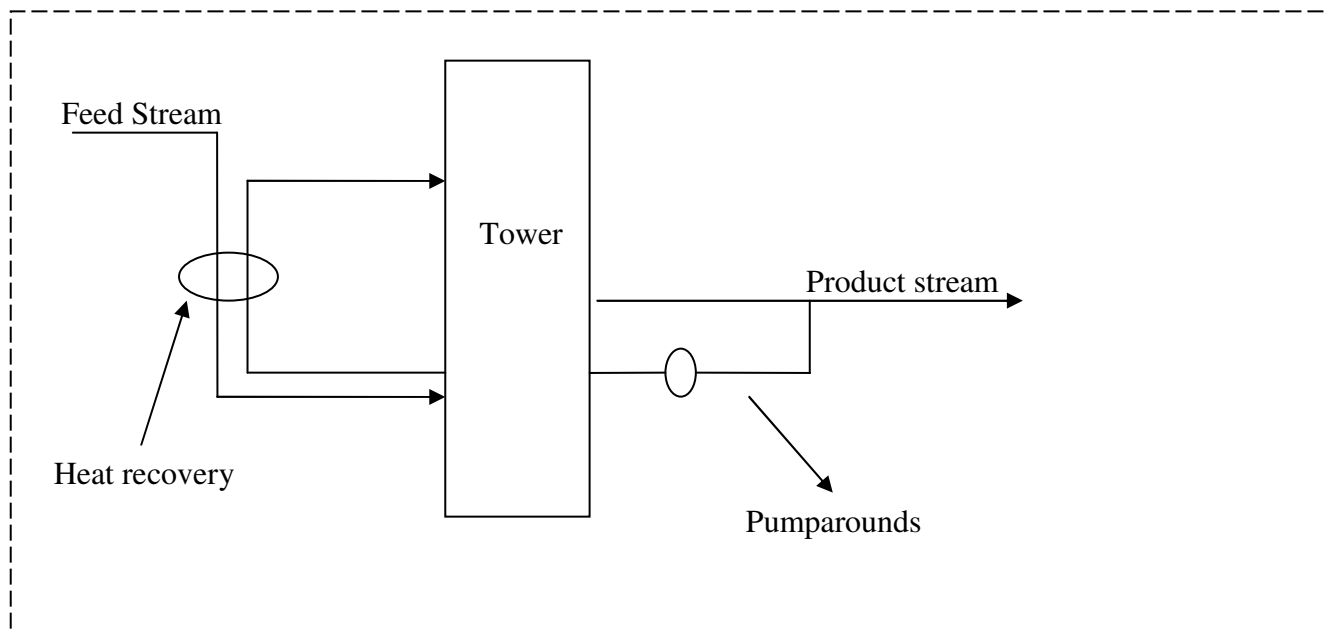


Figure 7.4: Pumparounds

Stripper

Strippers are used to remove light components from each product to adjust their flash point. Light components will vaporize due to the heat absorbed if the stripper is used with a reboiler.

Crude Distillation Column

Two phase crude feed from the pre-flash drum (in the form of vapour and liquid) enters the crude tower and is separated into different products according to their boiling points. Reflux and reboiling are used to enhance separation efficiency. Side strippers and dryers are used to recover products at their specific cut-points. Stripping steam is fed to the bottom of the column, to reduce hydrocarbon partial pressure, to aid product fractionation. Atmospheric residue is drawn at the bottom of the column and sent to the vacuum tower.

Component	TBP Cutpoints (°C)
Naphtha/Kerosene	155
Kerosene/Light Gas Oil	230
Light Gas Oil/Heavy Gas Oil	340
Heavy Gas Oil/Atmospheric Residue	370

Table 7.3 : Table of TBP Cutpoints of various components

Vacuum Distillation Unit (VDU)

The objective of the VDU is to fractionate a mixture of sour train CDU's bottom and the LSWR from the sweet train CDU (from PSR 1) under vacuum to provide feedstock for other downstream processing units including the HCK and the DCU. Vacuum required to reduce the boiling point of heavy hydrocarbon fractions. The VDU is designed to allow continuous processing of atmospheric residue, derived from 100.000 BPSD of sour Middle East crudes. The VDU is designed for a capacity of 62,000 BPSD made up of Arabian Heavy atmospheric residue, from the CDU, with a balance for LSWR from PSR 1 and/or storage. The unit is also capable of producing asphalt. The feed for asphalt operation is atmospheric residue produced from a blend of Arabian Heavy crude plus a lighter crude. The products formed are sour water, light vacuum gas oil, medium vacuum gas oil, heavy vacuum gas oil and vacuum residue.

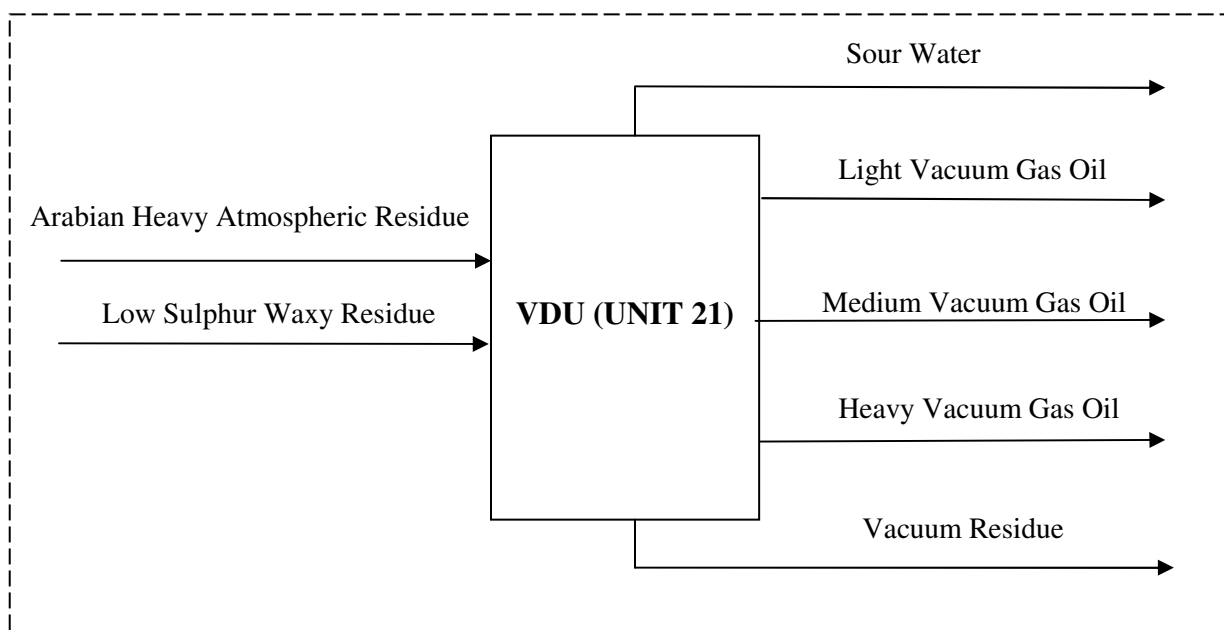


Figure 7.5 : Unit 21 Block Flow Diagram

Product stream from the VDU unit	Downstream unit
Sour Water	Sour Water Stripping Unit
Light Vacuum Gas Oil	Distillate Hydrotreating Unit
Medium Vacuum Gas Oil	Hydrocracker
Heavy Vacuum Gas Oil	Hydrocracker
Vacuum Residue	Delayed Coker

Table 7.4: VDU products with their respective downstream units

Summary of the sub-units within the VDU

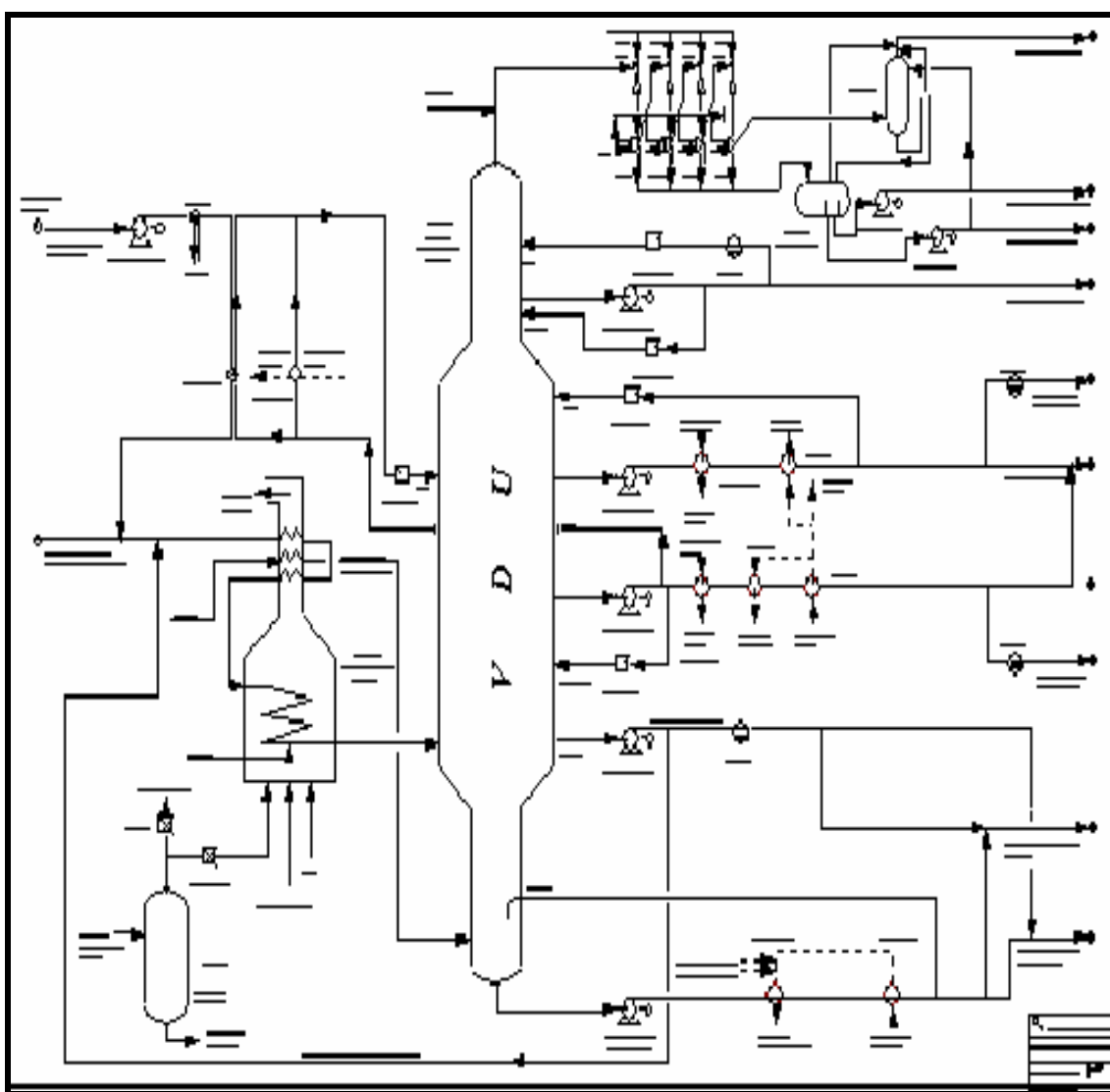


Figure 7.6: The simplified flow diagram of the VDU unit

Furnace

The furnace is used to heat feed. Heating mediums used are fuel gas and combustion air from the atmosphere. The designed duty of the feed furnace is 32MW.

Pumparounds

Pumparounds are product streams, from the crude tower, recycled back as feed into this tower to condense vapour that would otherwise have gone to the lighter cuts, thus regulating internal reflux. Since the vapour load in the tower is reduced, a smaller tower diameter is required. As a result, smaller trays are required in the tower. Pumparounds are also used to recover heat from the product stream to the feed stream.

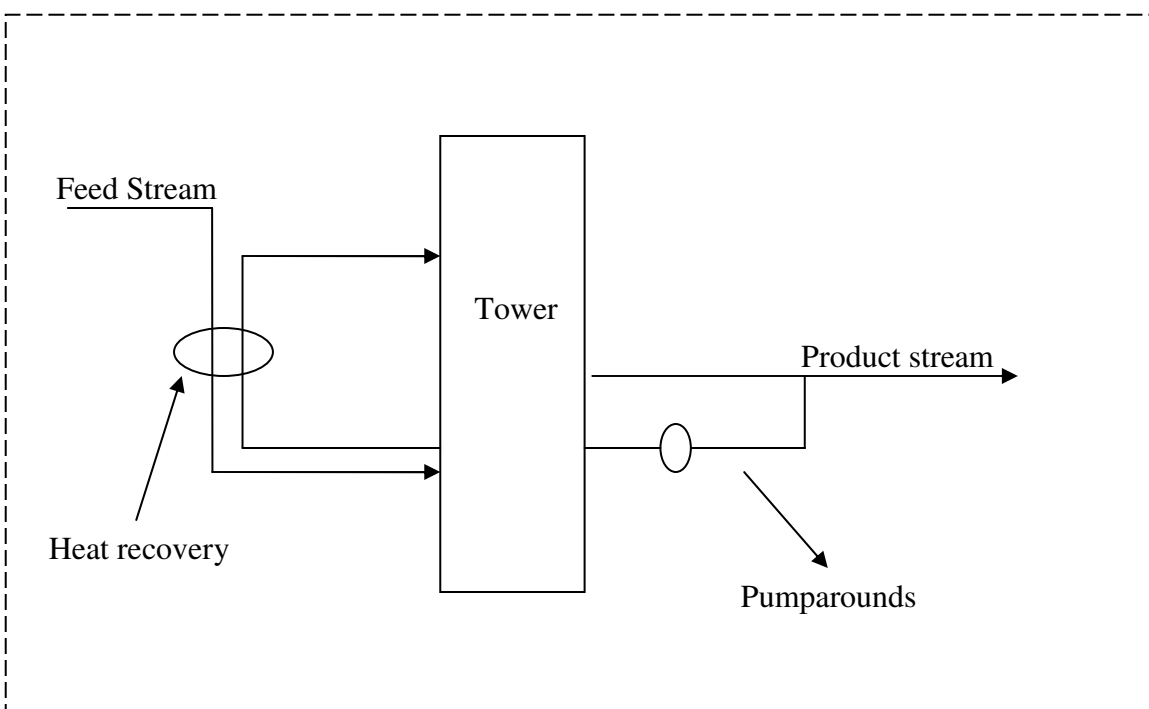


Figure 7.7 : Pumparounds

Vacuum Distillation Column

Two phase crude feed from the pre-flash drum (in the form of vapour and liquid) enters the crude tower and is separated into different products according to their boiling points under vacuum. Vacuum is maintained at 10mmHg at tower overhead by booster/ejectors. These boosters/ejectors utilize motive steam to pull vacuum through a diffuser. Steam undergoes an

isentropic expansion converting pressure to velocity (600mph) which results in a vacuum. It is analogous to a multi-stage compressor system. Reflux, reboiling and structured packing within the column are used to enhance separation efficiency. Side strippers and dryers are used to recover products at their specific cut-points.

Component	TBP Cutpoints (°C)
LVGO/MVGO	375
MVGO/HVGO	450
HVGO/XHVGO (asphalt production)	540
HVGO/Vacuum Residue (normal operation)	565

Naphtha Hydrotreating Unit (Unit 22)

The Naphtha Hydrotreating Unit, NHT is designed for a capacity of 35,000 BPSD of design feedstock. This unit plays the role to remove contaminants from the feed so that the product after fractionation will provide CCR with required sulphur and nitrogen contents.

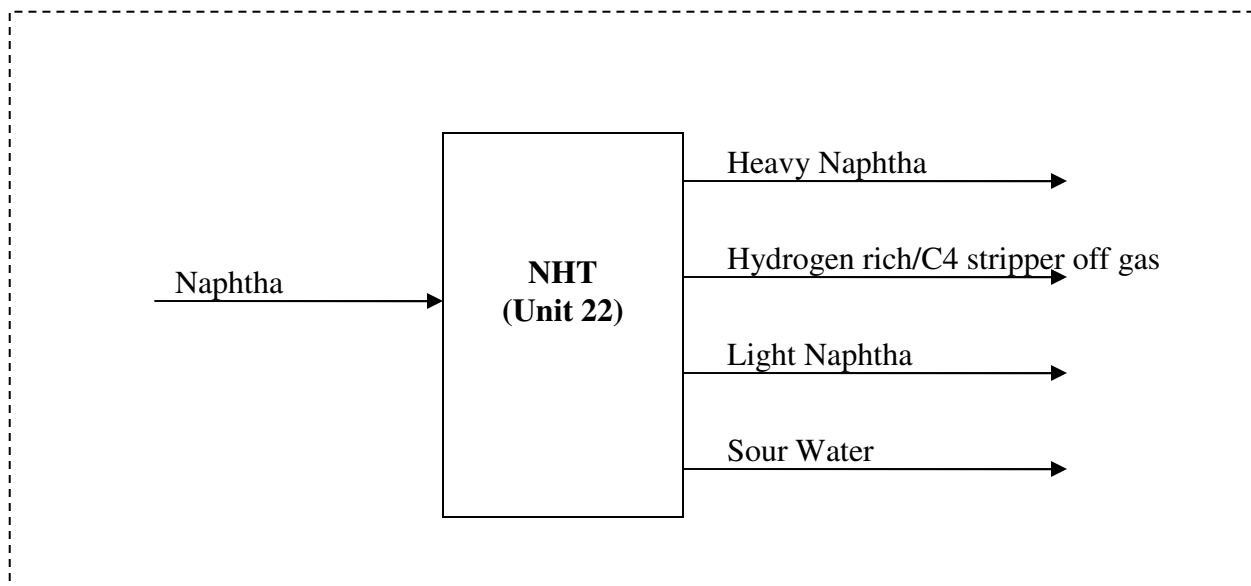


Figure 7.8 : Unit 22 Block Flow Diagram

[illegible]

Feed and Reactor Section

Naphtha from the charge pumps is combined with hydrogen rich recycle gas in the Reactor loop. The naphtha is preheated in the Combined Feed Exchanger (E-22201) with Reactor (R-22201) effluent. It is then further heated in the Charge Heater (F-22201). It is then being routed to the Reactor (R-22201), in the reactor, sulphur, nitrogen and contaminants are removed in a downflow catalyst bed.

The reactor effluent vapor exchanges with the reactor feed in the Combined Feed Exchanger (E-22201). Cold condensate is injected continuously by the Wash Water Injection pumps (P-22202 A/B) from the Water Break tank (V-22206) in order to reduce the build-up of slats.

Hydrogen rich gas from the CCR unit is introduced into the reactor effluent before it is cooled in the Product Condenser (E-22205). In here, the reactor effluent is cooled sufficiently to recover the naphtha which enters the separator (V-22202). A mesh blanket coalescer in the separator ensures separation of the naphtha, sour water and Hydrogen rich gas. The sour water is removed from the separator boot and fed to the Sour Water Treating Unit.

The hydrogen rich gas is recycled by the Recycle gas compressor (K-22201 A/B) to the front of the reactor loop, naphtha – recycle gas mixing point, upstream of the Combined Feed heat exchanger (E-22201). Naphtha from the separator and liquid from the Recycle Gas Compressor suction drum (V-22203) are combined and fed to the stripping section.

Stripper Column Section

The naphtha from the separator flows through the Stripper Feed – Dehexaniser Bottom Exchanger (E-22202) where it is preheated. Then it is routed to the Stripper Feed – Bottom Exchanger (E-22203) before entering the stripper (C-22201).

The Stripper is reboiled by reboiler heater (F-22202), a four pass furnace. The stripper bottom pumps (P-22204 A/B) re-circulate the bottom stream for reboiling.

The stripper strips Hydrogen Sulfide, water traces, light hydrocarbons and dissolved hydrogen from the feed. The vapor stream from the column passes overhead to the Stripper condenser (E-22206) and then to the Stripper receiver (V-22204). The liquid hydrocarbon is returned to the stripper by stripper reflux pumps (P-22205 A/B).

Dehexaniser Column Section

Stripped naphtha from the Stripper flows to the Dehexaniser via the Stripper Feed – Bottom Exchanger (E-22203). The Dehexaniser is reboiled by the Dehexaniser Furnace (F-22203), a six pass furnace. The dehexaniser bottom pumps (P-22206 A/B) recirculate the bottom stream for reboiling.

The Dehexaniser column produces two naphtha streams which are further processed in the CCR and C6ISOM units. The fractionation provide light naphtha overhead product which is condensed in the Dehexaniser condenser (E-22207) which is an air cooled total condenser.

The liquid hydrocarbon is returned to the Dehexaniser by its reflux pumps (P-22207 A/B). The net overhead liquid product is routed to the C6ISOM unit. The Dehexaniser bottoms liquid flows under flow control of the CCR unit through the Stripper Feed – Dehexaniser Bottom exchanger (E-22202) where it is cooled before entering the CCR.

Catalytic Reforming Unit / Continuous Catalytic Regeneration (Unit 23)

The Catalytic Reforming Unit, CRU processes treated naphtha to the required RON (adjusted for in the reactor) while the Catalytic Continuous Regeneration Unit, CCR regenerates the catalyst from the reactor.

The feedstock to this unit is heavy naphtha from the dehexaniser bottom in the Naphtha Hydrotreating Unit, NHT. The main product coming from this unit is off-gas, reformate, LPG product, net gas and makeup gas.

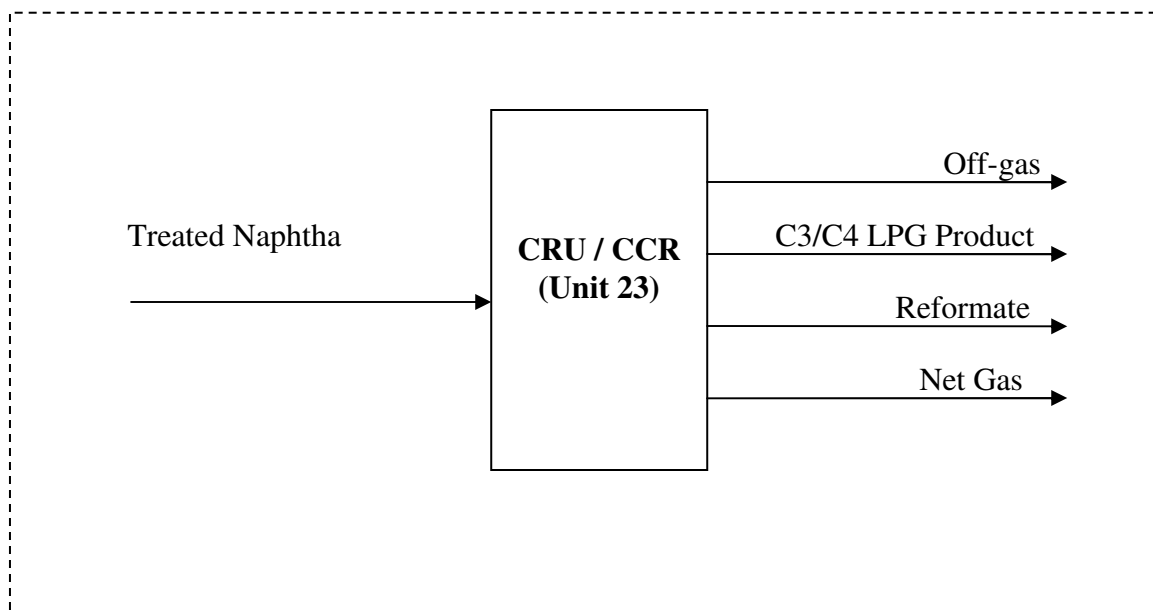


Figure 7.10 : Unit 23 Block Diagram

Summary of the sub-units within the CRU/CCR

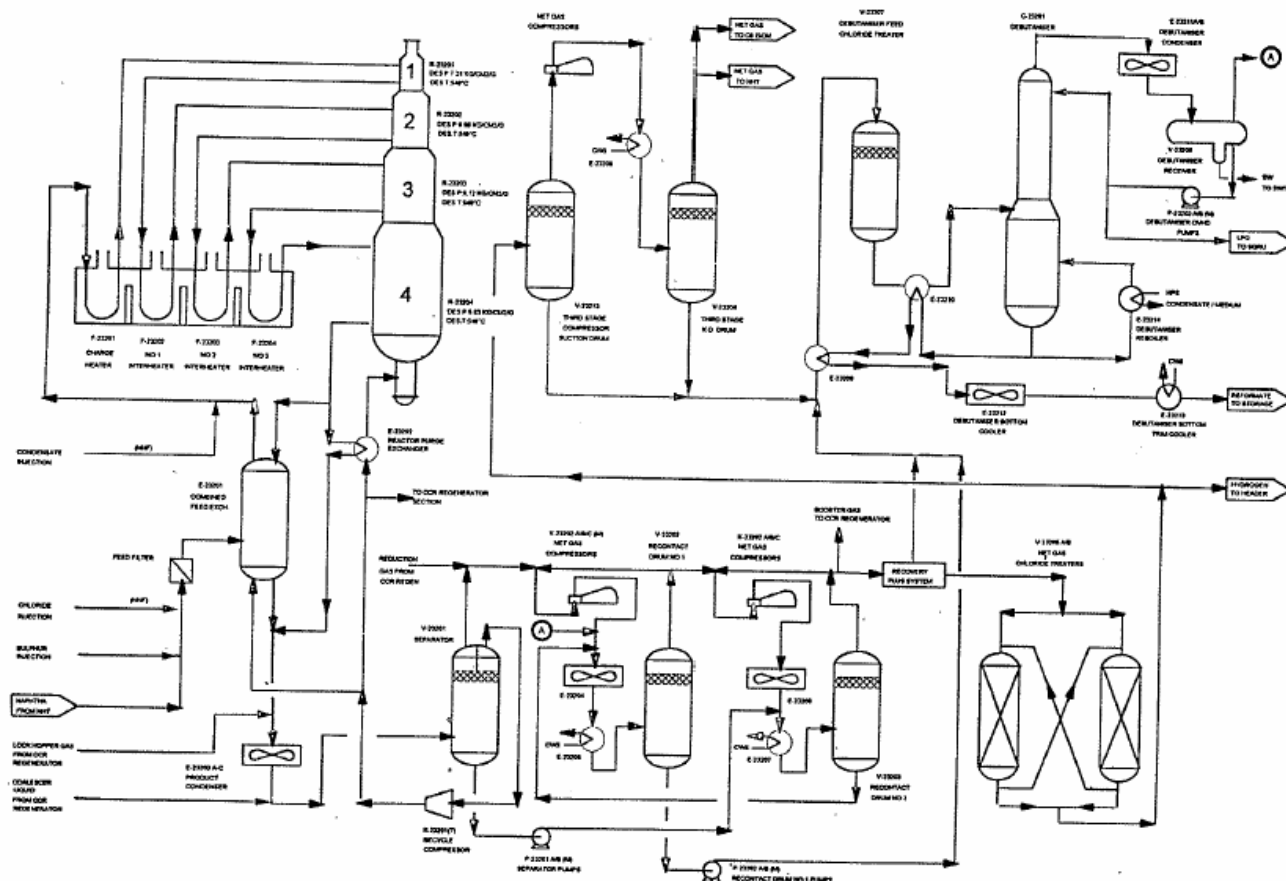


Figure 7.11 : CRU/CCR Simplified Flow Diagram

Reactor Section

The major equipments include four reactors (R-23201/2/3/4), four furnaces (F-23201/2/3/4), the combined feed heat exchanger (E-23201), separator (V-23201) and the recycle compressor (K-23201).

Heavy naphtha from the NHT dehexaniser bottom enters under flow control and is preheated by the reactor effluent in the combined feed exchanger (E-23201). In the combined feed exchanger, the feed combines with hydrogen-rich gas from the recycle compressor (K-23201). The feed is further heated in the charge heater (F-23201).

The fuel gases from all of the heaters are combined and used to generate HP steam in the convection section. The steam is used in the Naphtha Upgrading Complex and the balance is exported to the refinery.

There are four platforming reactors (R-23201/2/3/4). The heater outlet temperatures or the reactor inlet temperatures for all reactors are generally run at the same temperature. The effluent from each reactor is re-heated in the interheaters prior to entering the next reactor. The re-heating maintains the constant reactor inlet temperature since the reaction is endothermic. The reactor feed flow downwards, concurrently with the catalyst through the catalyst beds.

A small portion of the hydrogen-rich recycle gas enters the stacked reactors at the bottom after being heated in the reactor purge exchanger (E-23202), by a portion of the reactor effluent stream. This gas stream is to purge the hydrocarbon from the spent catalyst before it is regenerated in the regeneration section.

The effluent from the reactor is cooled in the combined feed exchanger against fresh feed. The combined reactor effluent from is further cooled in the product condenser (E-23203) before flowing to the separator (V-23201).

In the separator, the cooled reactor effluent is separated into a hydrocarbon liquid stream which goes forward for re-contacting and fractionation, and a hydrogen-rich gas stream. Part of the hydrogen-rich gas stream is recycled back to the reactor section via the recycle compressor (K-23201). The hydrogen rich gas is also routed to the re-contacting section and compressed by the net gas compressor (K-23202A/B/C).

Fractionation Section

This section can be divided by two, the re-contacting section and the debutanizer section. The purpose of the debutanizer is to remove light ends from the reformate product. The debutanizer receives the combined liquid streams from re-contacting section. The feed is

heated in the debutanizer feed/bottoms exchanger No.1 (E-23209). The feed then passes through the debutanizer feed chloride treater (V-23207) for removal of chlorides and is further heated in the debutanizer feed/bottoms exchanger No.2 (E-23210) before entering the debutanizer (C-23201). The debutanizer is reboiled by HP steam in the reboiler (E-23214).

The overhead product is cooled and collected in the debutanizer receiver (V-23206). The overhead pumps (P-23203A/B), return part of the liquid to the column as reflux. The net C3/C4 rich stream is routed to the saturated gas recovery unit. The off gas from the debutanizer receiver is recycled back to the re-contacting section.

The debutanizer bottoms product, reformat, is cooled by exchanging heat with the debutanizer feed in the two debutanizer feed/bottoms exchangers (E-23209/10). The reformat is then cooled to storage temperature in the debutanizer bottoms cooler (E-23212) and the debutanizer bottoms trim cooler (E-23212).

Regenerator Tower

The regenerator tower serves four functions ; coke burn, chlorination, drying and cooling.

In the burn zone, the catalyst flow is downward through an annulus between two vertical cylindrical screens. Hot, re-circulating, regeneration gas with a low concentration of oxygen is used to burn off the accumulated coke. The regeneration gas flow radically from the outside to the inside of the catalyst bed. The regeneration gas is withdrawn from the center-pipe, circulated using the regeneration blower (K-25253) and then cooled in the regeneration cooler (E-23254) before being returned to the regeneration tower.

In the chlorination zone, the catalyst flows downward through the regeneration tower. The chlorination gas flows up through the catalyst bed, counter-current to direction of catalyst flow and mixes with the circulating burn zone gases. The presence of chloride circulating in the burn zone helps with all the overall catalyst conditioning.

Instrument air for regeneration is dried in the air dryer (A-23251) and enters the bottom of the regeneration tower to cool the regenerated catalyst. The air from the cooling zone flows upward and enters the drying zone.

The flow in the drying zone is similar to that in the chlorination and cooling zone which is, hot dry air flows up, countercurrent to the flow of catalyst. The air leaving the drying zone is used again in the chlorination zone ensuring effective use of air in the regeneration tower.

Saturated Gas Recovery Unit (Unit 24)

Saturated Gas Recovery Unit (SGRU) is designed to fractionate saturated light streams and unsaturated coker material (naphtha and lighter component) to produce refinery fuel gas, LPG, butane and naphtha for Naptha Hydrotreating Unit, NHT.

The feed entering this unit is from the Crude Distillation Unit, CDU (Unit 21) of Basrah Light crude and also a mixture of C4 from PSR-1 of the High Tapis Case. On the other hand, the products produced by the SGRU are refinery fuel gas, propane, butane, light naphtha and heavy naphtha.

The major equipments involved in this unit are de-etahnizer, naphtha stabilizer, de-propanizer and de-isohexaniser.

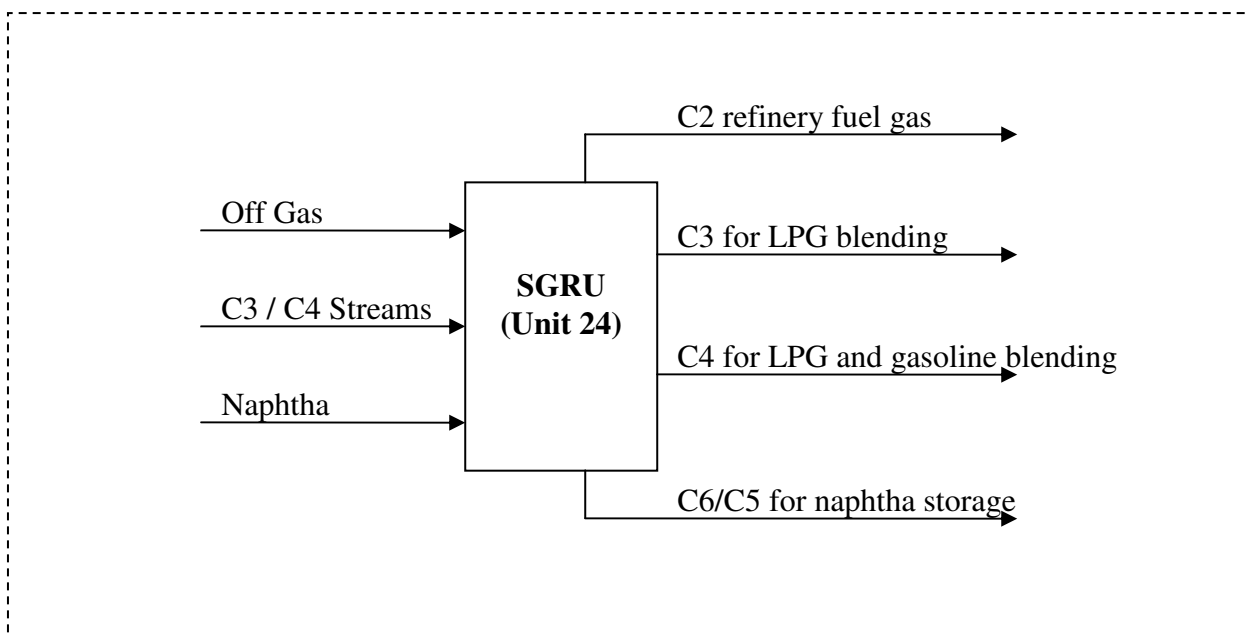


Figure 7.12 : Unit 24 Block Diagram

Summary of the sub-units within the SGRU

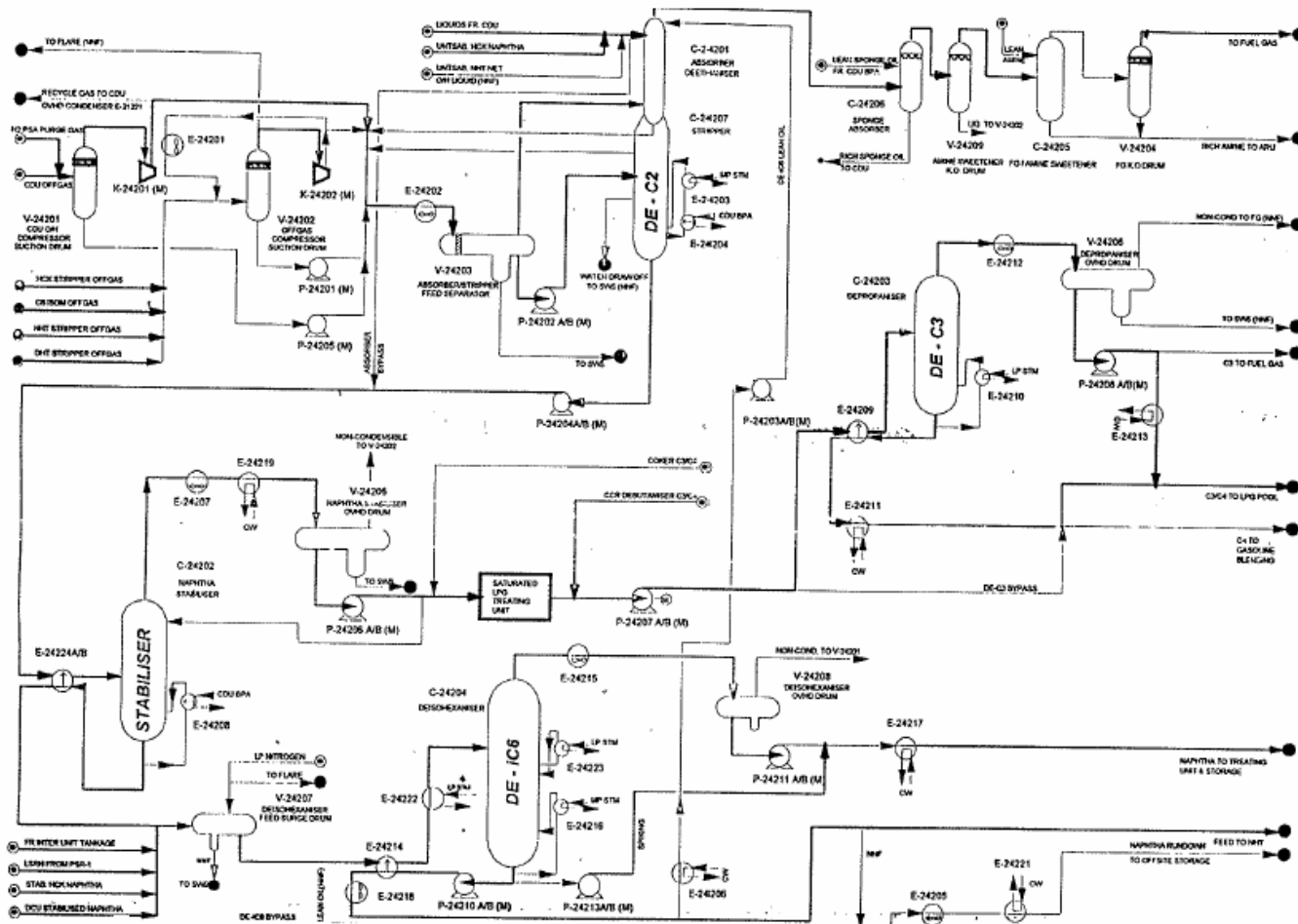


Figure 7.13 : SGRU Simplified Flow Diagram

De-ethanizer Section

Overhead gas from CDU is compressed in the CDU Overhead Compressor (K-24201) and mixed with several compressed off-gas streams before it is cooled and partially condensed by the Absorber/Stripper feed cooler (E-24202). A stream of hydrogen PSA purge gas may be fed to the CDU Overhead Compressor Suction Drum (V-24201) to fully utilize SGRU capacity to recover LPG.

The Absorber Stripper Feed Separator (V-24203) separates the incoming two-phase stream. The vapor passes out to bottom of the Absorber Deethanizer (C-24201). The

condensed liquid is pumped to the top of the Stripper (C-24207). Vapor leaving the top of the Stripper is recycled to the Absorber Stripper Feed Separator via the Absorber/Stripper Feed Cooler. Liquid from the Absorber Deethanizer is similarly recycled.

The Absorber removes C3 and C4 components from the the vapor phase into the liquid. Lean oil from the bottom of the Deisohaxanizer (C-24204) or Naphtha Stabilizer (C-24202) and liquid naphtha streams from the CDU, HCK and NHT are fed into the top of the column to absorb the C3/C4 components in vapor phase.

Light vapor from the top of the column is sent to the Sponge Absorber (C-24206) where it comes into contact with lean oil from the CDU over a packed bed. The lean gas is further processed in the Amine Sweetener (C-24205) where H₂S is removed. The sweetened gas is sent to the refinery fuel gas system.

Naphtha Stabilizer Section

Liquid from the bottom of the stripper (C-24207) is passed under flow control to the Naphtha Stabilizer (C-24202). Stabilized naphtha from the bottom is then sent under flow control to the downstream Deisohexaniser section. Overhead vapor from the Naphtha Stabilizer is condensed through the overhead condenser (E-24207) and trim condenser (E-24219), and collected in the overhead drum (V-24205)

Depropanizer Section

C3/C4 liquid from the LTU together with the liquid from the CCR is pumped under flow control to the Depropanizer column (C-24203). The liquid feed is preheated by the Depropanizer Feed/Bottoms Exchangers (E-24209A – C). The Depropanizer bottoms is reboiled using LP steam heat source in the Depropanizer Reboiler (E-24210).

C4 product from the column bottom is cooled and sent down to the C3/C4 LPG pool. Overhead vapor from the Depropanizer is totally condensed through the overhead condenser (E-24212) and collected in the overhead drum (V-24206)

Deisohexanizer Section

Naphtha from PSR-1, HCK, DCU, Naphtha Stabilizer and tankage are combined in the Deisohexanizer Feed Surge Drum (V-24207). The liquid is fed under flow control to the Deisohexanizer (C-24204) via a series of preheaters. The column has two reboilers ; Deisohexanizer Lower Reboiler (E-24216) and Deisohexanizer Side Reboiler (E-24223). The bottom outlet is sent as feed to the NHT. A branch from the bottom take-off line flows as lean oil to the Absorber Deethanizer column. The overhead naphtha is condensed and collected in the Deisohexaniser Overhead Drum (V-24208)

C6 Isomerization Unit (Unit 25)

The Naphtha Hydrotreating Unit, NHT, Catalytic Reforming Unit, CRU, and C6 Isomerization, C6ISOM units are closely coupled and have no intermediate tankage between the units. If the C6 Isomerization is shut down, the NHT and CRU will be maintained in operation with the C6 Isomerization feed processed in the CRU unit up to its capacity. If the CRU unit is shut down, the NHT and the C6 Isomerization will also be shut down.

The feedstock to the C6 Isomerization is light naphtha from the dehexaniser overhead in the NHT unit. This feedstock is primarily, n-hexane with some branched and cyclic hexanes, benzene and heptanes. The main products from this unit are off-gas and isomerate.

The purposes of C6ISOM are :

- To isomerize light naphtha isomerate (87 RON) for gasoline blend.
- Utilizes low temperature hydro-isomerization catalyst

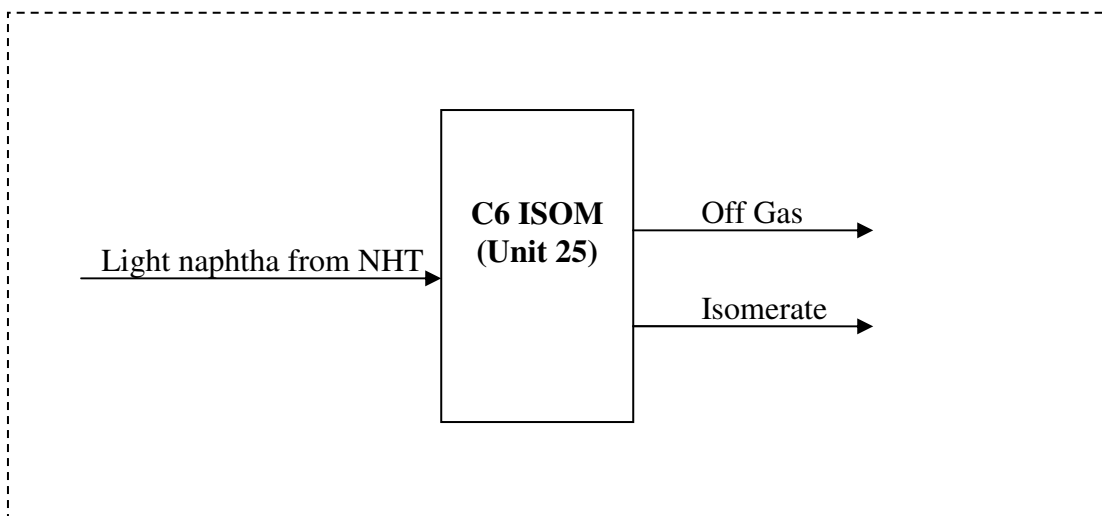


Figure 7.14: Unit 25 Block Diagram

The C6 Isomerization Unit consists of three different sections. The sections are reactor section, stabilizer section and also the deisohexaniser section.

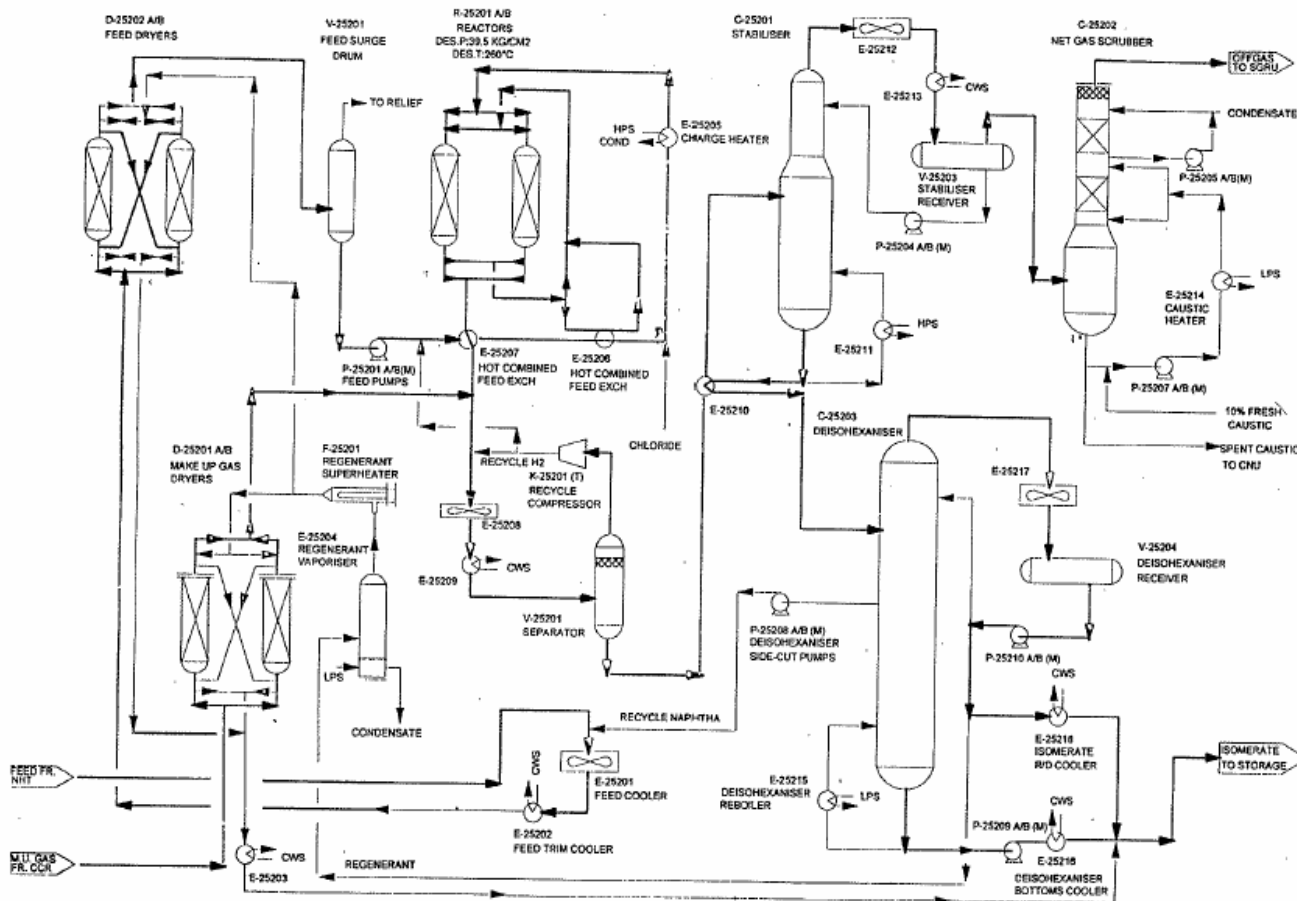


Figure 7.15 : Unit 25 Simplified Flow Diagram

Reactor Section

In the feed drying section, light naphtha fresh feed and recycled deisohexaniser are cooled and dried. From the feed dryers the dried liquid feed enters the feed surge drum (V-25201). The feed surge drum is blanketed with dry hydrogen gas from the make up gas dryers under drum pressure control. The make up gas comes from the CCR reformer unit.

From the feed surge drum, the liquid feed is pumped through the feed heat exchangers. The feed is first heated in the cold combined feed heat exchanger (E-25207). The feed is then heated in the hot combined feed heat exchanger (E-25206). After passing that heat exchanger, a small quantity of perchloroethylene is injected into the two phase combined stream as a catalyst. The feed is then heated in the charge heater (E-25205) with high pressure steam before entering the lead reactor.

The steam from the lead reactor passes to the hot combined feed exchanger (E-25206) preheating the feed and partially removing the lead reactor's exothermic heat of reaction. The stream then passes through the cold combined feed exchanger (E-25207). After that, the reactor product is further cooled in the product condensers ; E-25208 which is air cooled and E-25209 which is a cooling water exchanger. The product then enters the separator.

In the separator, the liquid hydrocarbons are separated from the hydrogen gas. The hydrogen gas leaves the separator at the top of the vessel through a mesh blanket and is routed to combine with the stabilizer gas, which is sent to the net gas scrubber section. The liquid hydrocarbons are removed from the bottom of the separator on level control and are routed to the stabilizer.

Stabilizer Section

The stabilizer, C-25201 separates the dissolved hydrogen, HCl and light ends from the deisohexaniser feed stream. The stabilizer receiver level is controlled by manipulating the reflux flow and there is no net liquid distillate. The stabilizer pressure is controlled on the non-condensed vapor outlet to the net gas scrubber.

The bottom product from the stabilizer is removed under level control and routed to the deisohexaniser.

Deisohexaniser

Stabilized liquid from the stabilizer are routed to the deisohexaniser, C-25203. It separates its feed hydrocarbons into three streams :

- Overhead isomerate, which is 2,3-dimethyl-butane and lighter boiling components
- Side draw, which is 2-methyl-pentane to n-hexane boiling components, is recycled back to combine with the light naphtha feed before entering the heater (E-25201)
- Bottom isomerate which is methyl-cyclo-pentane and heavier boiling components

The deisohexaniser overheads and bottoms are combined together and sent to the isomerate tank in the blendstock storage unit.

LPG Treating Unit (Unit 26)

The function of the LPG Treating Unit is to remove H_2S and to reduce mercaptan content plus re-entry sulphur content of various C_3 and C_4 feed stocks to produce blendstocks for LPG. The design capacity is 8500 BPSD. The specification of the product is 1ppm maximum for hydrogen sulphide, maximum of 10 ppm for mercaptan content and negative result for Doctor Test.

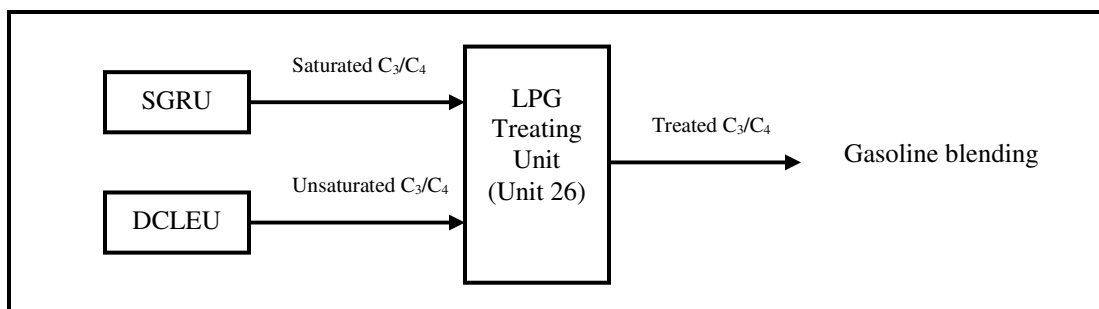


Figure 7.16: Block Diagram of Unit 26

Summary of sub-units within the LPG Treating Unit

AMINEX section

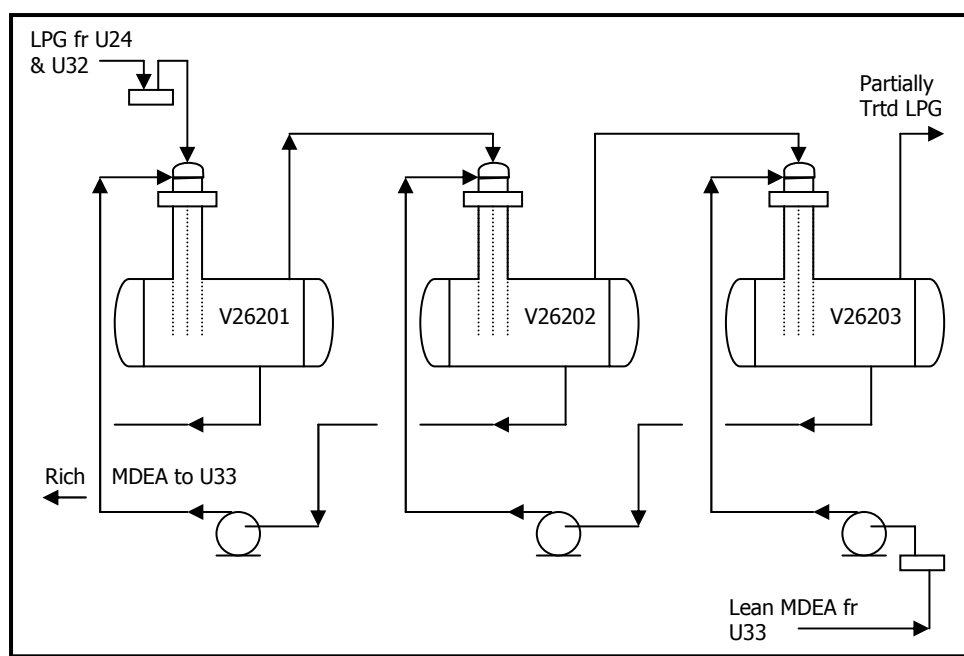


Figure 7.17 : Flow diagram of the AMINEX Section

This section is designed to remove H_2S f with lean acid that is the Methyldiethanol amine (MDEA) from the untreated LPG stream and transferred to the amine system. This section has a 3 Phase Separators equipped with fiber-film contactor to optimize the contact with fresh amine. LPG feed is introduced into top of Separator shroud and get contact with MDEA in the fiber-film contactor. MDEA flows downward adhering to the fibers until it enter aqueous phase. LPG concurrently flow and disengages upon entering hydrocarbon phase. H_2S is removed to amine stream.

THIOLEX Section

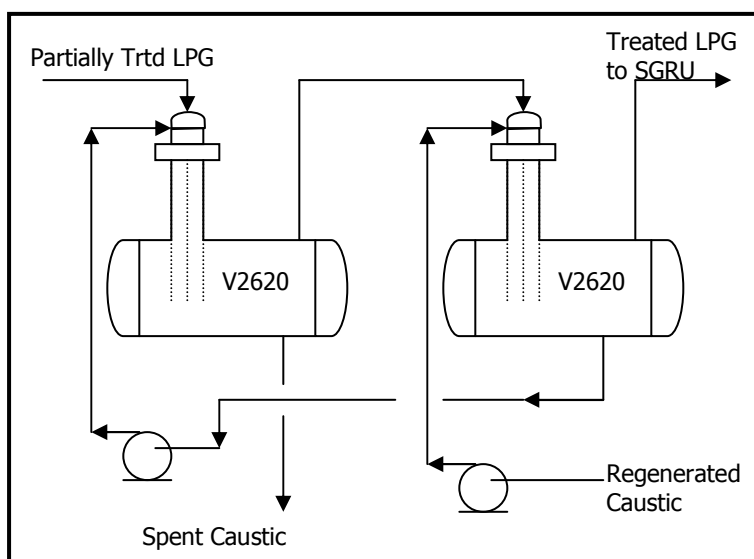


Figure 7.18: Flow diagram of the THIOLEX Section

This section is designed to extract mercaptan using caustic and convert it to sodium mercaptides and also remove residual H_2S . It consists of 2 Phase Separators equipped with fiber-film contactor. It has a similar process flow principle to AMINEX. The treated LPG exit the 2nd Separators to Unit 24 (SGRU) for further separation/direct to rundown.

REGEN section

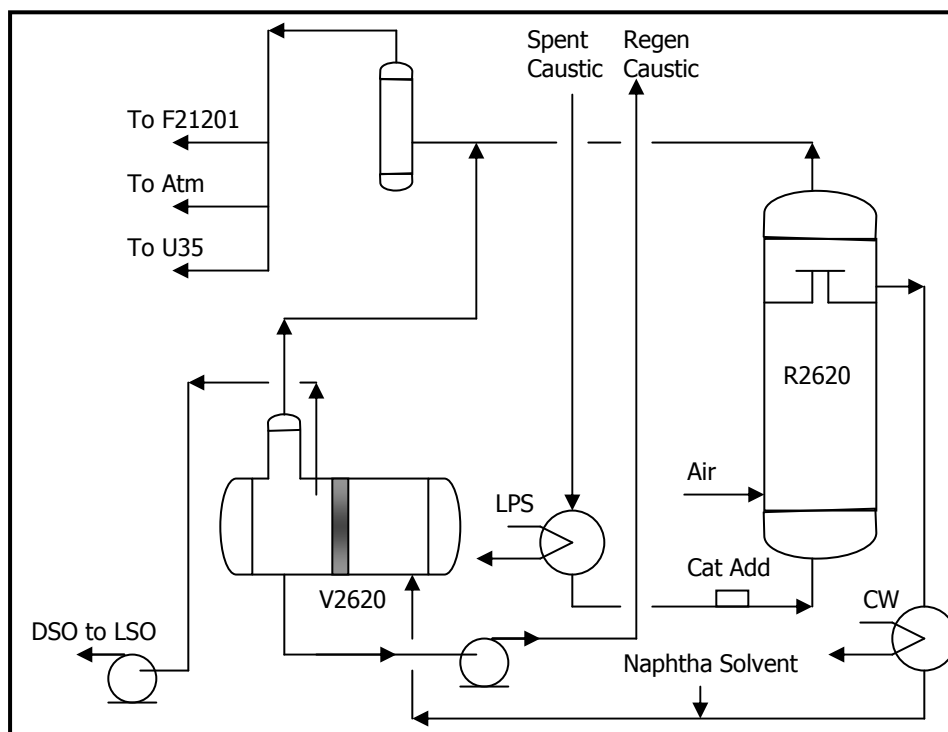


Figure 7.19: Flow diagram of the REGEN Section

This section is intended to oxidize and convert mercaptides from THIOLEX to disulphide oil (DSO). The process happens in a tower with the presence of oxidation air and liquid catalyst. THIOLEX, which in the caustic, is heated with LPS to 52°C. At the same time the caustic is regenerated to be used in THIOLEX section again. Regenerated caustic is separated from DSO Gravity Separator after being cooled with cool water to 38°C. The process releases off gas which is routed to K.O pot, where it is mixed with fuel gas which is then burnt in either CDU furnace or SGRU incinerator.

Naphtha Treating Unit (Unit 27)

Naphtha Treating Unit is designed to reduce the mercaptan content in naphtha feedstock to produce blendstocks of various grades of gasoline products and petrochemical naphtha. The design feed is the overhead C_5 and C_6 from Deisohexanizer in SGRU. The design capacity is 15250 BPSD with a minimum operating rate for this unit to be 7625 BPSD.

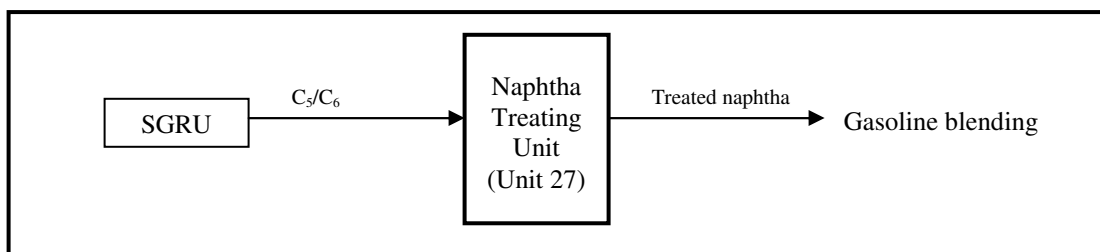


Figure 7.20: Block diagram of Unit 27

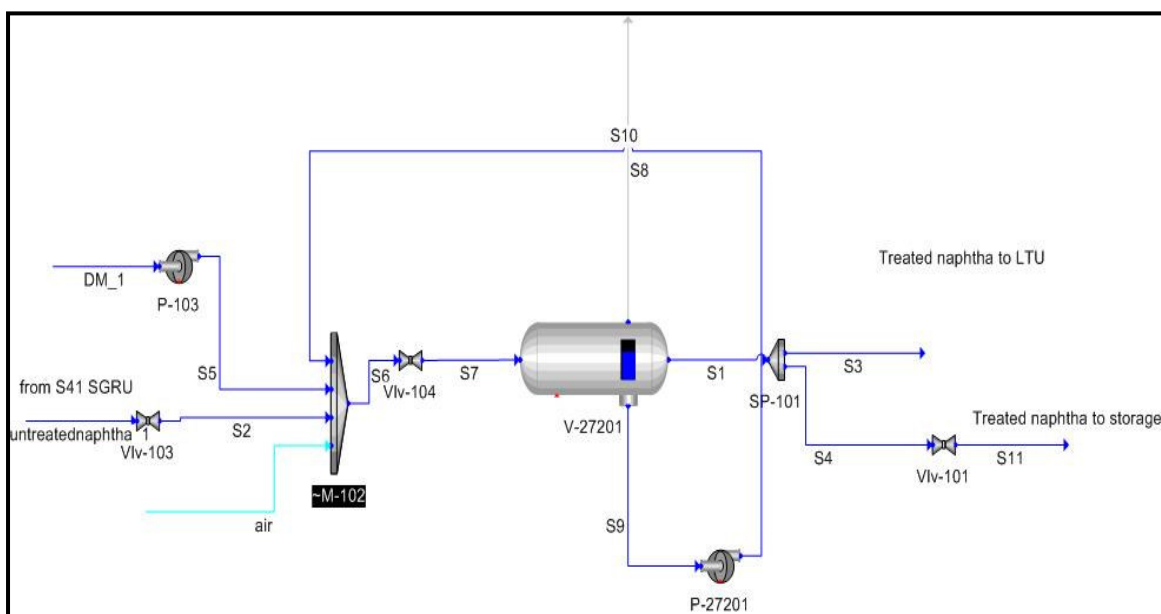


Figure 7.21: Flow diagram of Unit 27

Summary of sub-units within the NTU

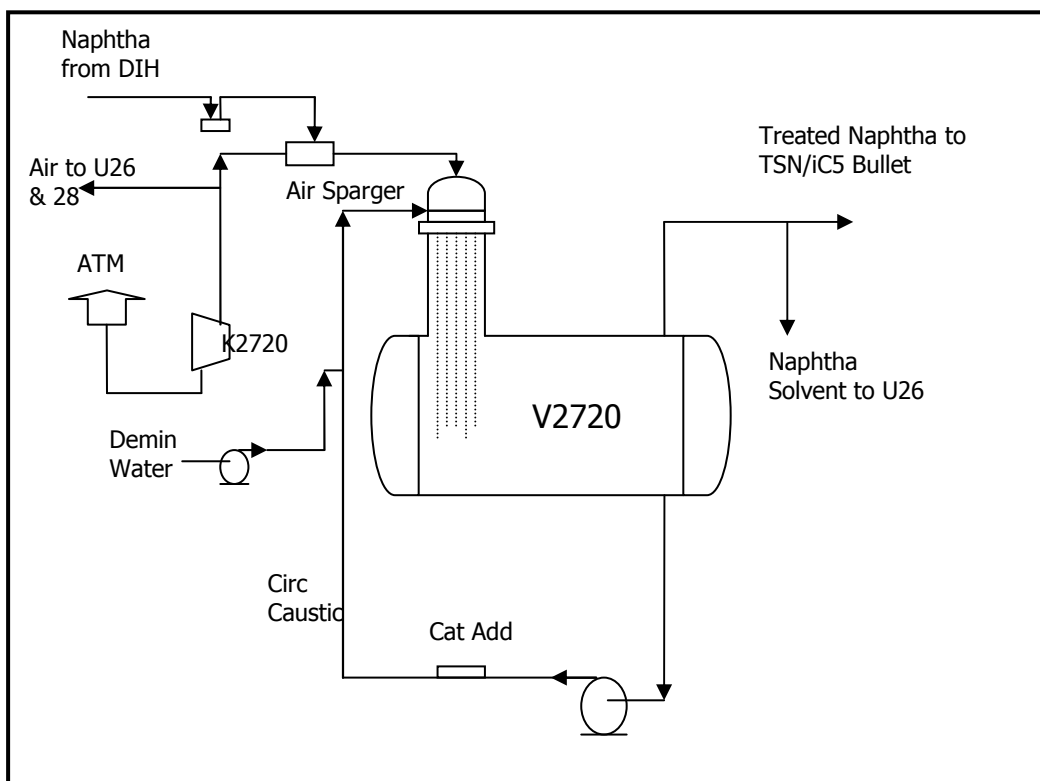


Figure 7.22: Flow diagram of Unit 27

This unit consists of a phase separator equipped with fiber-film contactor. The process oxidizes the mercaptans to disulphides via contacting the naphtha feed with caustic while concurrently flowing down the fibre film contactor in the presence of oxidation air and catalyst. Air is injected with incoming feed naphtha. The mixture then enters the top of Separator shroud and gets contact with circulating caustic and catalyst in the fiber-film contactor. R-SH and traces of H_2S in naphtha is extracted and removed from naphtha stream in the fiber-film. NaSR is oxidized to reproduce caustic while Na_2S is oxidized to form $Na_2S_2O_3$ which ends up in caustic solution. Reactions occur in the fibre-film. Make up demineralised water is provided to the process to prevent caustic dehydration and to maintain water-saturated naphtha in the unit.

Kerosene Treating Unit (Unit 28-1)

Kerosene Treating Unit is designed to remove H_2S and naphthenic acid and reduce mercaptan of the PSR-2 CDU kerosene and produce blendstocks for DPK and diesel. The feed is straight run kerosene from CDU-2 on processing Iranian Light Crude. The feed capacity is 13800 BPSD.

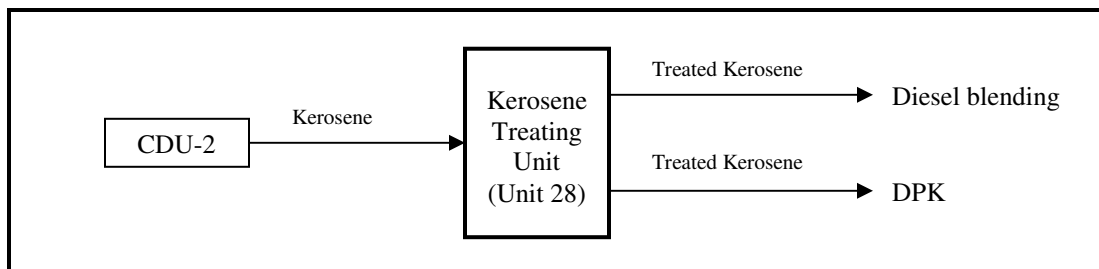


Figure 7.23: Block Diagram of Unit 28

Summary of sub-units within the KTU Unit

MERICAT II Section

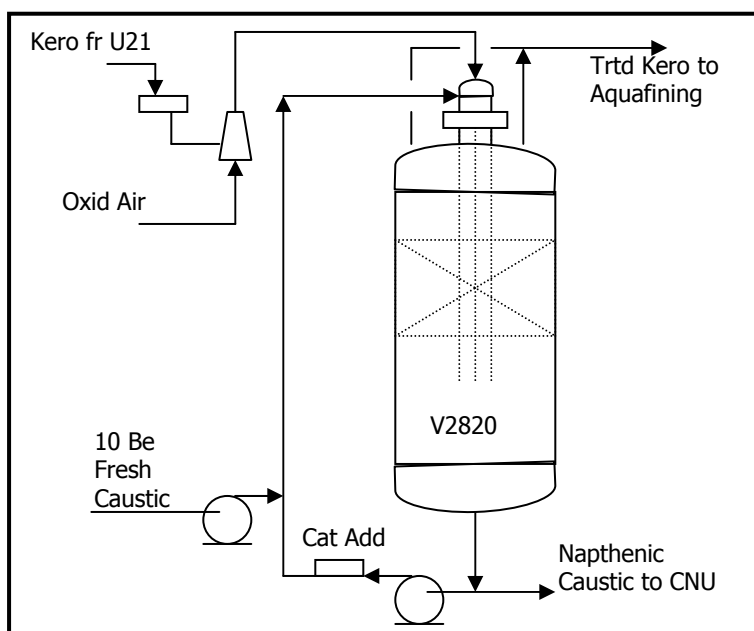


Figure 7.24: Flow diagram of Mericat II Section

This section consists of a vertical Phase Separator equipped with fiber-film and carbon bed that has been impregnated with catalyst. Caustic is used to extract naphthenic acids and oxidise R-SH to disulfides via kerosene feed/caustic contacting. Air is injected to incoming feed. Mixture enters top of the separator shroud and get contact with circulating caustic and catalyst in the fiber-film contactor. Caustic catalyst flow downward adhering to the fibers until it enter aqueous phase. Kerosene-air mixture concurrently flow and disengages upon entering hydrocarbon phase. Naphthenic acid is extracted from kerosene in the fibre-film. NaOOC-R is produced and ends up in spent caustic stream. Mercaptan is extracted in the fibre-film and carbon bed. NaSR is oxidized to reproduce caustic while DSO ends up in kerosene stream.

Carbon Bed Causticizing

Its purpose is to alkalinize the carbon bed so that it becomes an active oxidizing environment to promote R-SH removal.

Carbon Bed Hot Water Washing

When product specs cannot be attained, adjustments of the operating variables and recausticizing indicate that the catalyst is less active, the bed can be hot water washed to remove impurities which may have plugged off the carbon pores and render it inactive. The washing may also be effective in reducing the catalyst bed pressure drop which should not exceed $0.7 \text{ kg/cm}^2\text{g}$. The hot water washing is done by mixing LP steam and cold condensate into a line.

Catalyst Reimpregnation of Carbon Bed

Catalyst reimpregnation is necessary after hot wash if the catalyst activity starts to reduce. Acidification with glacial acetic acid is required before the reimpregnation.

AQUAFINING Section

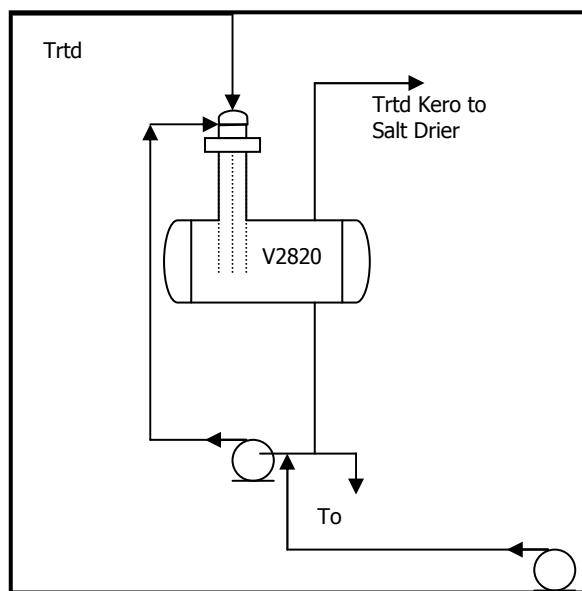


Figure 7.25: Flow diagram of the Aquafining Section

It is designed to wash out entrained caustic in kerosene. Kerosene is introduced into top of Separator shroud and contact with circulated demineralized. Demin water flows downward adhering to the fibers and enters aqueous phase. Kerosene concurrently flow and disengages upon entering hydrocarbon phase. Caustic is washed from the kerosene and transferred to aqueous stream. Fresh Demin water is continuously introduced while spent water is drawn out. pH of spent water is controlled at 9 – 11 and final spending targeted to 0.05 wt% of total alkalinity or 0.0288wt% of Na+ in effluent water.

Salt Drier Section

This section helps to remove traces of water in kerosene from Aquafining and extends the life of the downstream clay filter which would otherwise absorb the water and become spent. Removal of water will dry the kerosene below the saturation. Free water dissolves the salt to form a brine solution. The brine is drained to sewer periodically.

Clay Filter Section

Kerosene exiting Salt Drier is then flows downward through a clay-filled vessel to polish the kerosene from solids, moisture, emulsion and surfactants. It also ensures that water separation specification is met.

Caustic Neutralization Unit (Unit 28-2)

Caustic Neutralization Unit is designed to neutralize spent caustic from other treating units (U25, U26, U27, U28) using H_2SO_4 .

Summary of sub-units within the CNU

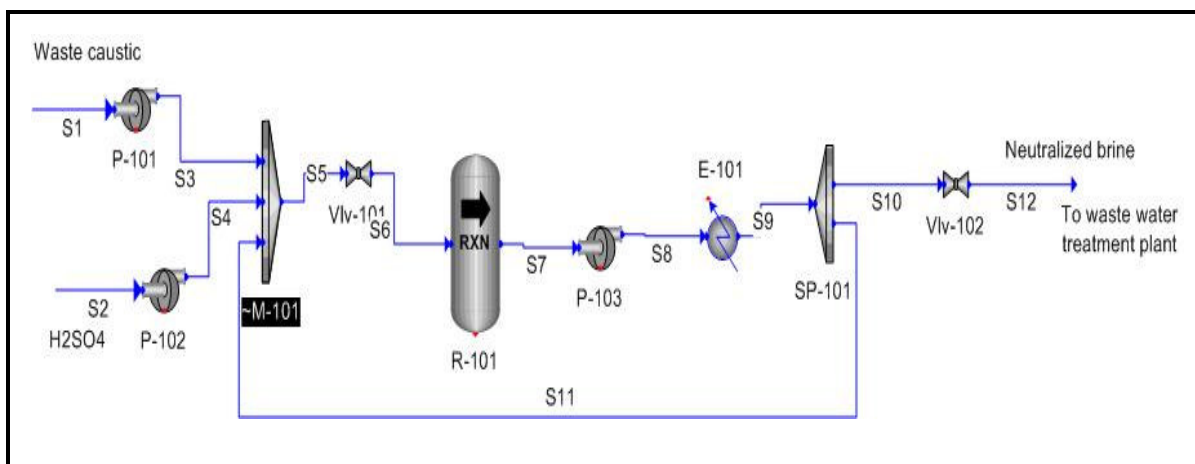


Figure 7.26: Flow diagram of CNU

Mericon Section

The spent caustic used from various units is collected in a caustic tank. It is then fed to a static mixer for neutralization using H_2SO_4 . The brine is sent to a phase separator and then partially discharged to sewer. Brine pH is controlled at 7.

Distillate Hydrotreater Unit (Unit 29)

Distillate Hydrotreater Unit is designed to process 35000 BPSD of feedstock which consists of light gas oil (LGO), heavy gas oil (HGO), light vacuum gas oil (LVGO) and coker distillate. The function of this unit is to reduce the sulfur content of the distillate feedstocks down to less than 0.04%, while meeting the color specification of 1.0 max by ASTM D-1500 to produce a suitable blend component for various grade of diesel product. Unit feed is tailored by use of feed tankage to suit production of domestic diesels. Recycle gas will be scrubbed for H₂S removal using MDEA to provide a H₂S content in the treated gas to be less than 50 ppm vol.

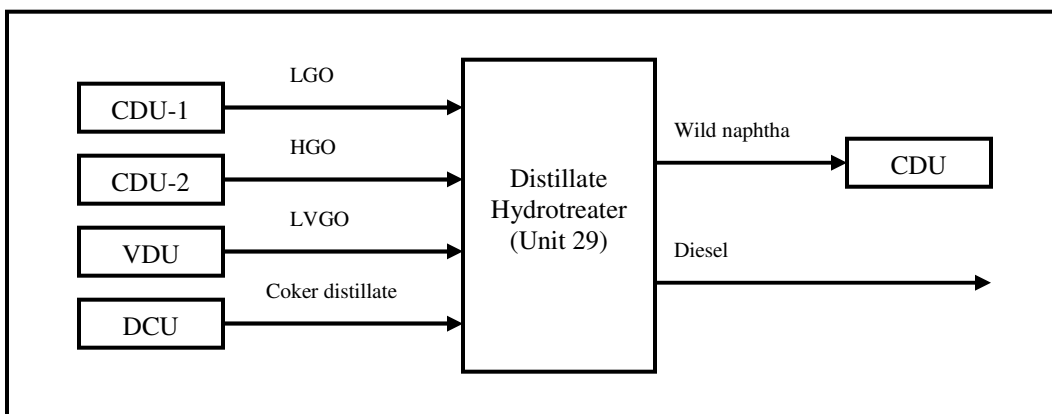


Figure 7.27: Block diagram of Unit 29

Summary of sub-units within the DHT

Hydrotreater

Hot feedstock enters the DHT and is collected in the feed surge drum. Feed is drawn from the feed drum and heated by the reactor effluent in the exchanger (also the feed furnace). Before entering the feed furnace, the feed is split into 4 passes where recycle gas being added to each pass. The heated distillate is then fed to the reactor. The reactor effluent leaves the bottom reactor and is used in a number of heat exchangers to heat up the untreated distillate feed and the recycle gas stream. It then enters the High Pressure

Separator (HPS) where the distillate product and the recycle gas are separated. The distillate product is sent to the Low Pressure Separator (LPS) before being fed to the distillate stripper. Flash gas from the LPS is sent to the DCU. The recycle gas from the HPS is treated by amine absorber to reduce its H₂S content. A purge stream taken from the recycle gas is then re-compressed by the recycle gas compressor. The compressor discharge is split into 2 streams. The first stream is used as quench gas in the reactor while the second is combined with the makeup hydrogen and added to the untreated distillate feed at the inlet to the furnace.

Distillate stripper

The feed to the stripper is drawn from the LPS. The feed is preheated by stripper bottoms and reactor effluent before entering stripper. Superheated steam is injected into the base of the stripper to strip sour gas and light ends to meet the hydrotreated distillate product flash point specification. The overhead product and the stripping steam is condensed in the condenser and cooled by the trim cooler, and being collected in the accumulator. Stripping offgas is sent to the SGRU. Part of the wild naphtha from the accumulator is returned to the stripper as reflux. The remaining naphtha is sent to the CDU. When CDU is shutdown, the naphtha is routed to the DCU. Stripping distillate product is withdrawn from the base of the stripper and cooled by the stripper feed and air cooler. The cooled product is then directed under temperature control to a 2-stage vacuum drying system.

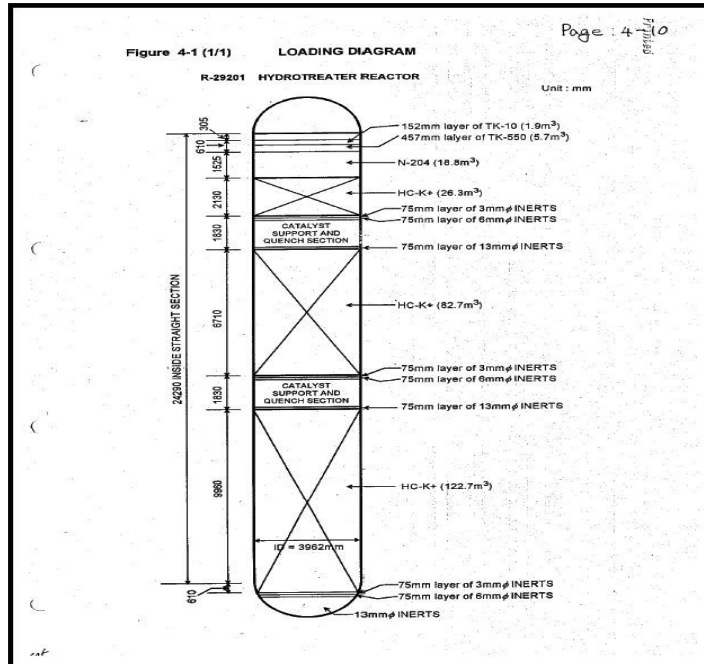


Figure 7.28 : DHT Reactor Diagram

Hydrocracker (Unit 30)

Hydrocracker is designed to produce high quality distillate products. It converts vacuum gas oil (MVGO and HVGO) to premium diesel, heavy diesel, kerosene and naphtha. Current MDR is 38000 BPSD. This unit consists of two stages of reactor section, high pressure section, low pressure section and a common fractionation section. The continuous make up hydrogen gas is provided from Unit 31 (HCDS). Recycle gas purge facility is provided, but the unit is normally able to operate without purge. Recycle gas is scrubbed for H_2S removal to be less than 500ppm.

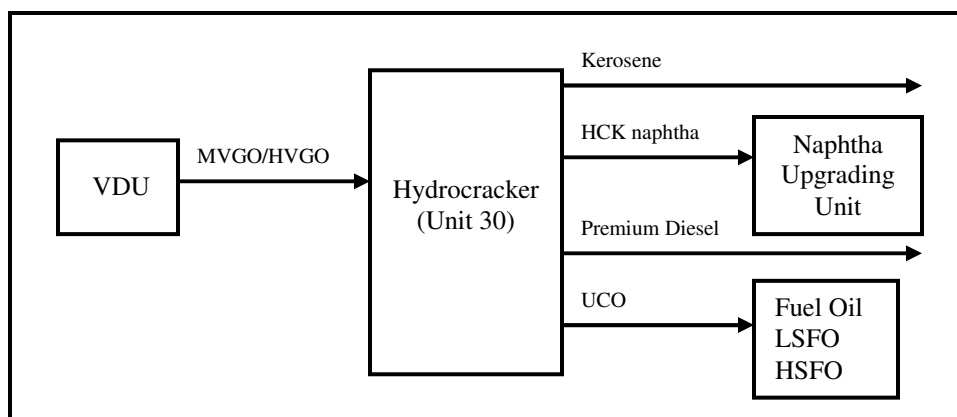


Figure 7.29: Block diagram of Unit 30

Summary of sub-units within the HCK

Reactor Section

Reactor fresh feed (mixture of VGO from VDU and tankage) is passed through a feed filter after routing through a series of heat exchangers and before entering the Filtered Surge Drum. It is then pumped up to reactor pressure by first stage feed pump. The oil feed is then mixed with hydrogen recycle gas and further heated by the first Stage Reactor Feed/ Effluent exchanger and Reactor Feed Furnace.

The feed enters Reactor 1 where it is denitrified, desulfurised and partially converted into products. Intermediate quench zones are provided between beds in order to prevent

runaway reaction. The reactor effluents from the first stage are cooled in a number of exchangers and the cooled reactor effluent is sent to HPS, where the rich gas is separated and recycled back to the reactors. The bottom liquid from high pressure section is sent to LPS into products. Fractionator bottom is recycled back to Reactor 2 as a recycle oil feed, where it is cracked with high selectivity for mid-distillated, and the effluents from both reactors are combined through HPS and LPS before being sent to fractionator for further separation.

Fractionator Section

Feed from LPS enters the stripper on level control where butane and lighter gases are being removed from overhead and sent to gas recovery section; and water present is removed.

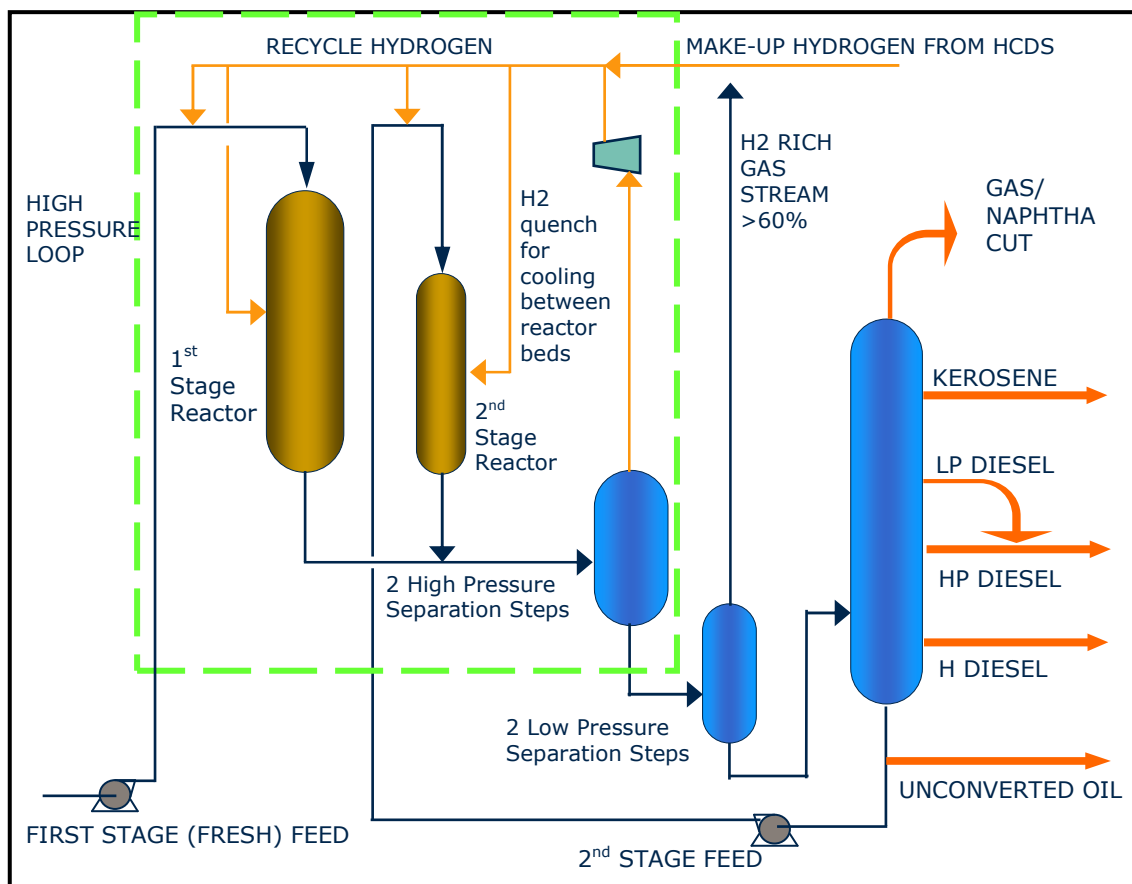


Figure 7.30: Flow diagram of the fractionator

Hydrogen Collection and Distribution System (Unit 31)

Hydrogen Collection and Distribution System is designed to collect low purity hydrogen off-gas from the producing units, and to purify, compress and distribute the product hydrogen at the quality and conditions required by hydro-processing units.

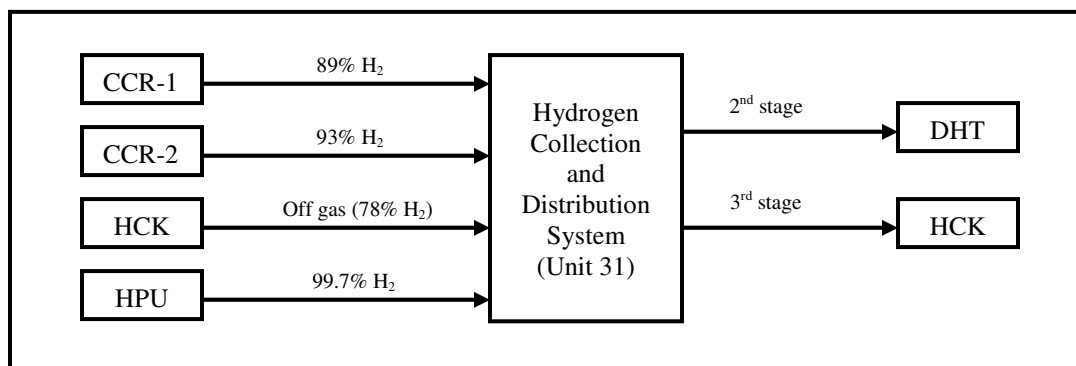


Figure 7.31: Block diagram of Unit 31

Summary of sub-units within the HCDS

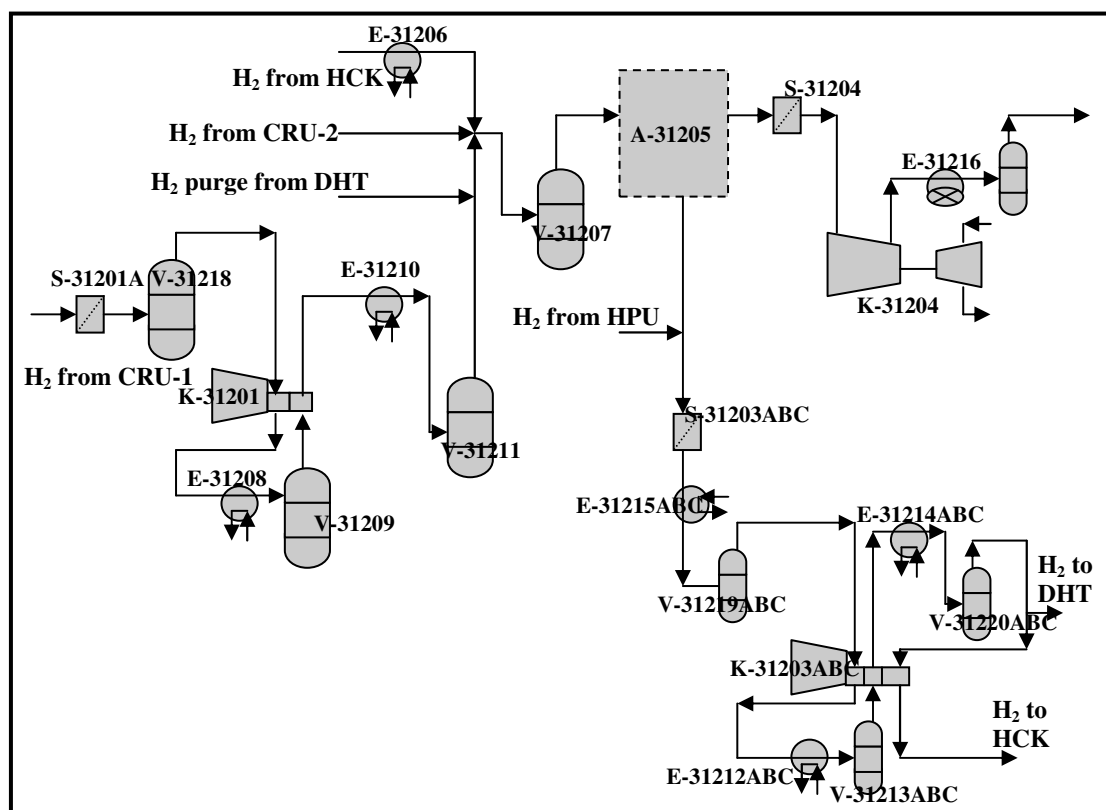


Figure 7.32: Flow diagram of Unit 31

H₂ rich gas from CCR-1 HPS is passed to the 2-stage LP H₂ Compressors, where it is compressed in two of the three machines. The HPS Pressure is controlled by adjusting the H₂ flow to the fuel gas header to maintain the system pressure and the header pressure for the H₂ to NHT-1. The purpose of HP compressors is to pressurize H₂ from PSR-1 from 2.4 bar to 25 bar. When the H₂ flow exceeds the required flow rate for PSR-2 or LP compressors are not operating, it will open the valve to vent the H₂ flow to flare header. In contrast, PSR-1 has to bleed a certain amount of H₂ to the fuel gas header to keep the quality. The compressed gas mixes with the HCK off-gas, CCR-2 H₂ gas and DHT H₂ gas. The combined stream then passes to PSA unit 31 for purification. The purified H₂ is then cooled and mixed with the H₂ from HPU before being compressed in the three stages by HP H₂ Compressors to supply make-up H₂ to DHT and HCK. To maintain suction pressure to be constant, a small amount of H₂ is normally vented to the fuel gas header. This H₂ venting is also the buffer to absorb the fluctuating difference between H₂ supply and consumption in DHT and HCK. The slipstream of H₂ from 2nd stage discharge passes to DHT unit. The discharge pressure is maintained by a split range of pressure controller which acts on DHT feed H₂ flow controller and a recycle stream back to the 1st stage compressor suction. H₂ from the 3rd stage of the compressor is then passed to HCK.

Pressure Swing Absorption

The H₂ gas is passes here for purification. Impurities such as CO, CO₂, CH₄ and C₂ are removed. Impurities are adsorbed at high pressure and then desorbed at low pressure. Internally, it is a semi-batch process using multiple absorbers providing constant high purity H₂ and off gas from constant feed flow. There are 10 adsorbent beds operating on an alternating cycle of adsorption and regeneration to provide a constant flow of purified hydrogen. The product leaves the system close to the feed pressure. Off gas is available at low pressure as fuel gas.

Delayed Coker Unit (Unit 32)

Delayed Coker Unit is designed to process 21000 BPSD vacuum residues (VR) from VDU in order to produce gas oil, distillate, naphtha, gases and a by-product of green coke and also to process non-biological refinery sludge. It aims to maximize liquid yield and minimize coke yield; and to minimize noise, oil and water pollution.

Delayed Coker is a low pressure thermal cracking process; which reduces the heaviest and least desirable element of crude bottoms to lighter distillate products and converts the remainder of the charge to marketable petroleum coke. This unit is important as the demand for gasoline and distillate products are increasing while the demand for residual fuel oil is decreasing.

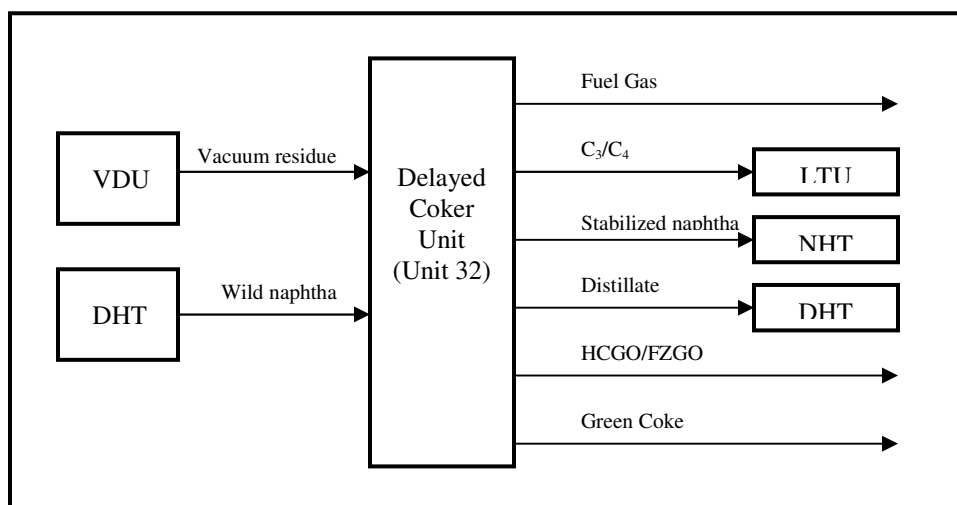


Figure 7.33: Block diagram of Unit 32

Summary of sub-units within the DCU

DCU consists of 4 packages:

- Fractionation, heater and coke drum section
- Closed blowdown section (CBS)
- Light Ends Section
- Green coke handling section

Fractionation, Heater and Coke Drum Section

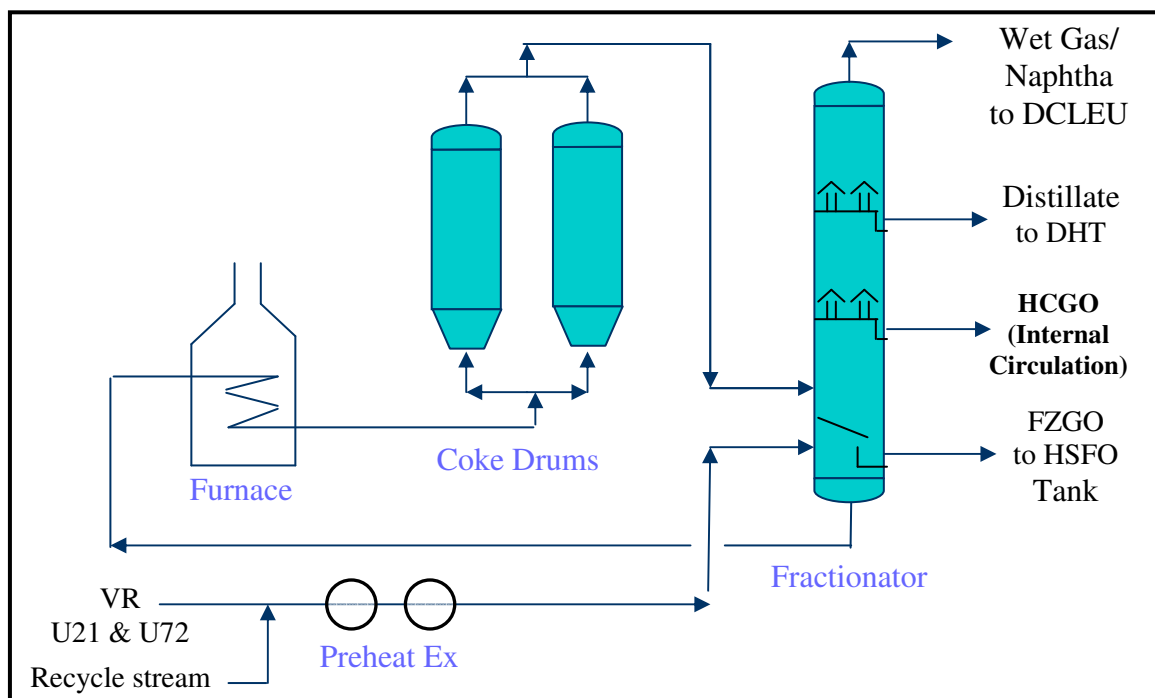


Figure 7.34: Flow diagram of Fractionation, heater and coke drum section

Residue Feed Control

Fresh feed, recycle from main fractionator and 20% heavy coker gasoil (HCGO) recycle based on fresh feed are charged to the heater via 4 tube passes. HP steam is injected into each pass to maintain a minimum linear velocity of combined steam and oil.

Coker Main Fractionator

The total feed is split into 2 streams and routed through separate preheat exchangers. One stream is preheated with HCGO and the other with flash zone gasoil (FZGO). Each heater outlet temperature is controlled by adjusting the fuel gas flow to each section of the heater. A higher temperature will reset the flow controller to a higher flow rate. The fractionator has 2 pumparound circuits and 2 total draw trays.

Furnace

A low fuel gas header pressure will cause a shutdown of the furnace fuel gas system.

Coke Drum

The combine heater effluent enters one of two coke drum, where the cracking reaction takes place. 4 nuclear level devices monitor the coke level, one for each section of the coke drum. The vapor flow is quenched with HCGO to minimize the cracking reaction hence to prevent coke laydown in the overhead line. Interlocks are provided to prevent closing of overhead to Blowdown Ball Valves, from being closed during a coke drum water quench. The quenched vapor enters the flash zone section of main fractionator where the vapor contacted with wash liquid to condense heavier hydrocarbons.

Closed Blowdown Section (CBS)

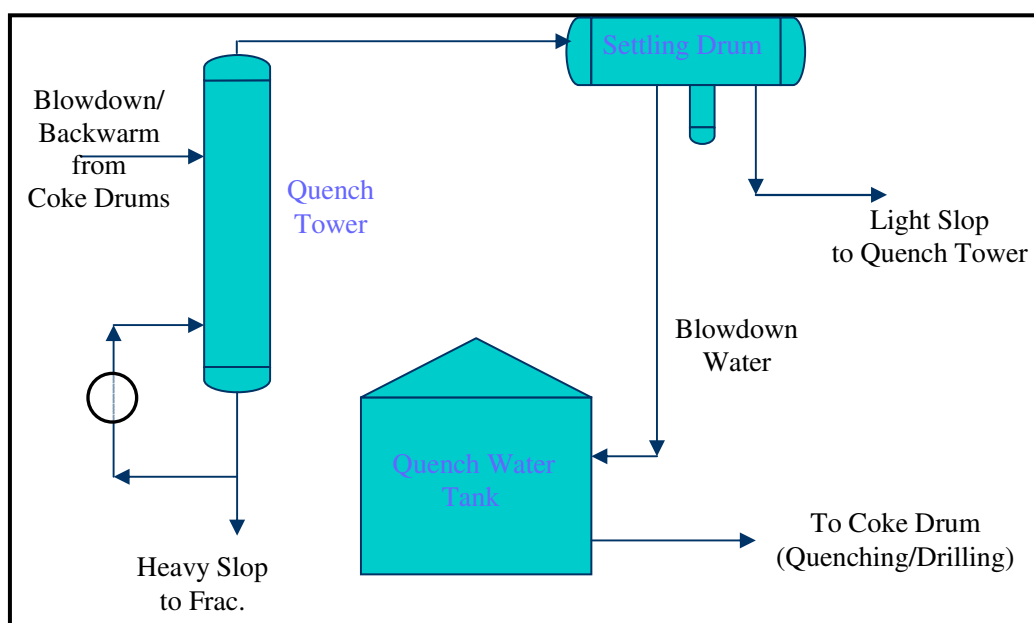


Figure 7.35: Flow diagram of the Closed Blowdown Section

CBS is specifically designed to reduce pollution due to coke drum switches. It condenses and collects steam and hydrocarbon vapor generated during steamout/backwarming and water quenching of the coke drums. The condensable hydrocarbons are recovered and charged back into coking section, while the non-condensable materials flow to fractionator overhead accumulator or flare. It also recovers the heavy condensable

hydrocarbon portion from emergency release of hot hydrocarbon from the coke drums, as well as to cool the drained water from the coke drums.

Light Ends Section

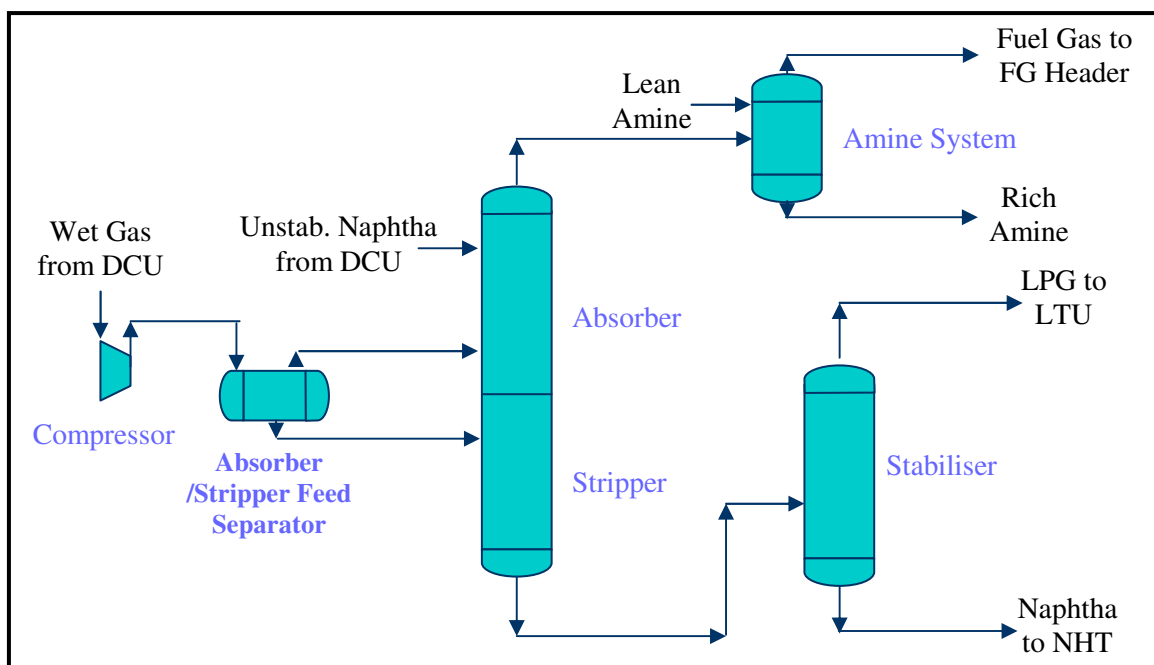


Figure 7.36: Flow diagram of the DCLEU

This section fractionates the light end streams from DCU. It is designed to fractionate unstabilized naphtha and lighter streams from DCU to produce refinery fuel gas, C₃/C₄ stream and stabilized naphtha. The heat integration of the light end is incorporated with DCU, thus shutting down the DCU will shutdown DCLEU as well.

Absorber/Stripper

The main fractionator wet gas and Pressure Swing Adsorber (PSA) waste gas are fed to compressor suction drum, and then cooled by fan cooler before entering the stripper feed separator. The gas stream from the separator and the overhead naphtha distillate from main fractionator are then sent to the absorber where it is contacted with lean oil from the bottom of stabilizer. The overhead gas from absorber is sent to Amine Contactor where H₂S is removed by lean amine. The sweet gas is then sent to the fuel gas header. The rich

oil and naphtha from bottom of Absorber is sent back to the feed separator. Liquid from feed separator is fed to the stripper. After being stripped of the light gases, it is sent from the bottom of the Stripper to Stabilizer.

Pumparound

Temperature is controlled by manipulation of amount of pumparound to the steam generator, in dual split. The pumparound is used to reboil the light ends stripper. Residual heat is removed in the trim cooler. The return temperature is controlled by adjusting the fan speed.

Green coke handling section

The green coke/sludge handling section can be done through 2 steps, the Waste Addition Step During Coke Drum Quenching; and Sludge Dewatering in CBS. It recycles all non-bio solid sludge produced by the refinery and converts into coke, slop oil and water.

Hydrogen Production Unit (Unit 39)

Hydrogen Production Unit is designed to produce high purity of H_2 from the natural gas (from supply pipeline through feed gas compressor) and supplies H_2 to HCDS. However, it only acts as a backup to the H_2 shortfalls between supply and consumption. It is coupled with PSA to reach the required hydrogen specification.

Summary of sub-units within the HPU

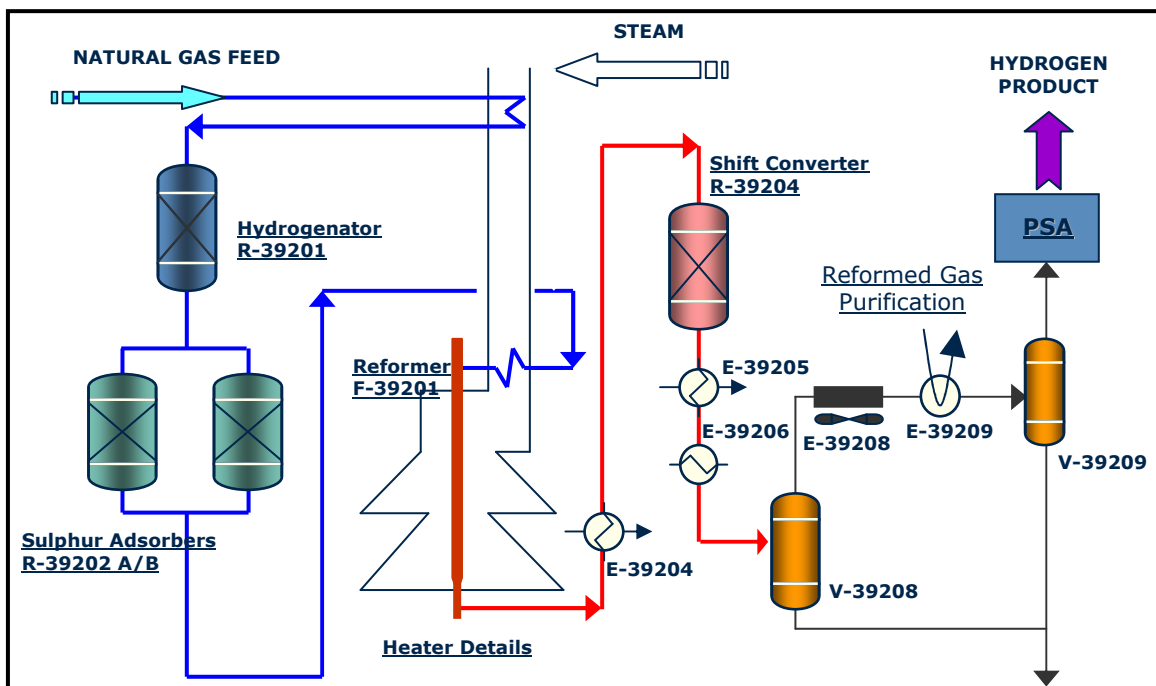


Figure 7.37: Flow diagram of Unit 32

The process involved 5 main principals mentioned below:

- Sulphur Removal
- Natural Gas Steam Reforming
- High Temperature Shift Conversion
- Reformed Gas Purification by PSA
- Steam Generation

The natural gas is mixed with the purged H_2 in the feed surge drum before being compressed by the feed compressors. The drum pressure is maintained by adjusting the gas flow. The purge H_2 comes from the purified H_2 header. The feed compressor discharge pressure is maintained by adjusting the spillback flow. The compressed natural gas is then heated in the Reformer heater and goes to the Hydrogenator and sulfur absorber for the natural gas treating. The treated natural gas mixes with the steam from the Steam Generator. The mixture goes through the heater for the reforming reaction. After reforming, the effluent is cooled down from about $750^{\circ}C$ to $350^{\circ}C$ before going through the Shift Converter to convert CO and steam in the Reformer effluent to hydrogen and CO_2 . Raising the reformer effluent temperature will reduce the methane slip. Shift converter effluent is further cooled down from $390^{\circ}C$ to $36^{\circ}C$ and knockout the condensate before going to the PSA for purification from the overhead. Condensate from bottoms goes to Deaerator systems to strip off any CO_2 or dissolved O_2 present.

Hydrogenation

Takes place in Hydrogenator

Hydrogenates any un-reactive organic sulfur compound to H_2S

Catalyst: Co-Mo

De-sulfurisation

H_2S removal

Vessels are filled with ZnO as catalyst

Catalyst is loaded in 2 vessels connected in series

NG Steam Reforming

Major reaction step to produce H_2

Reaction is endothermic

Catalyst: Calcium Aluminate support impregnated with (Nickel Oxide) NiO

Heat is supplied to promote the reaction since the reaction is endothermic

Steam added in excess to push the equilibrium to the desired direction

Recommended steam to hydrocarbon ratio is 3.0

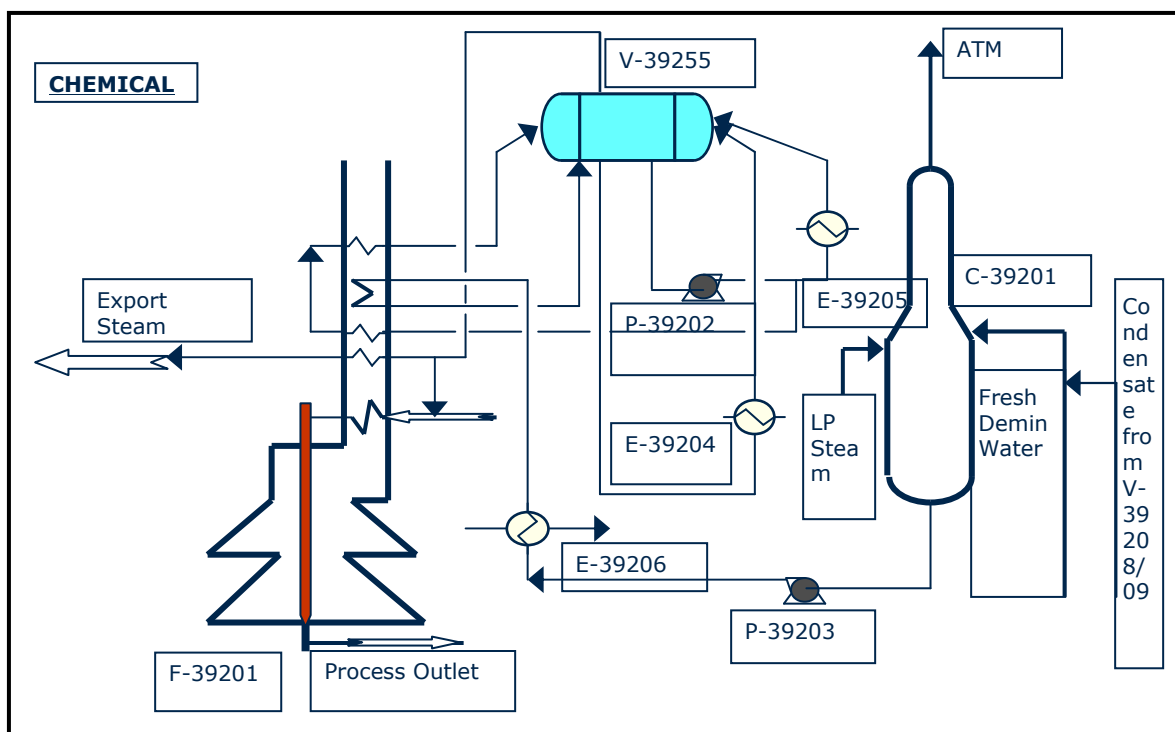


Figure 7.38: Flow diagram of the NG Steam Reforming

Steam Reformer

Single cell, terraced wall, natural draft reformer

Number of tubes filled with catalyst: 54; Number of burners: 36

Radiant section: steam reforming reaction (48% of heat supplied)

Convection section: steam generation, reformer feed preheating, steam superheating, hydrocarbon feed preheat

Design basis: uniform heat flux at all tubes (hence the terraced wall)

Design duty is 31.9 MW supplied by burning fuel gas and PSA tail gas

Reformed Gas Purification

HTSC effluent is removed of water by flash separation

Dry gas is purified to remove CO, CO₂ and CH₄ to deliver high purity H₂ via PSA

Pressure Swing Adsorber

To purify the hydrogen product.

The purity increase to 99.7 volume % as all impurities were removed to tail gas stream.

8.0 UTILITIES

Introduction

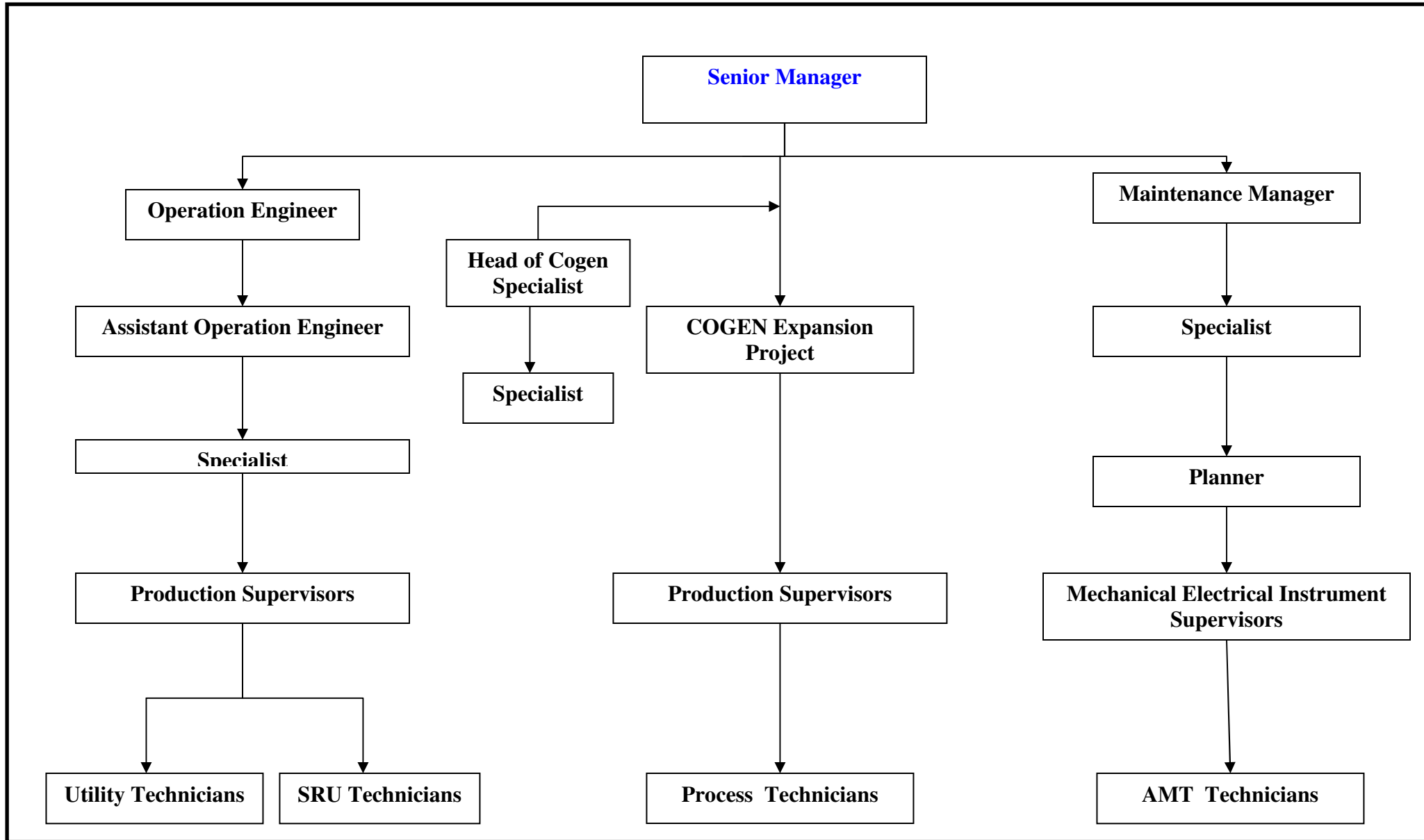
The Utilities department in the PETRONAS Penapisan Melaka Sdn .Bhd is incharge of supplying common facilities and services to the units in the refinery to enable the units to function.

Units that are available in area 5A are as follows:

- I. Unit 51- Steam, Condensate and Boiler Feed Water (BFW) supply
- II. Unit 52-Water supply system
- III. Unit 53-Cooling Water supply system
- IV. Unit 54- Air unit
- V. Unit 55-Nitrogen System
- VI. Unit 56-Chemical Unit
- VII. Unit 58-Fuel Unit
- VIII. Unit 64-Flare Unit
- IX. Unit 65-Fire Unit

The Utility and Cogen department is incharge of maintaining and operating the Utilities services and facilities in the refinery. The people who are involved in this section can be represented in the organizational chart in the following page.

UTILITIES & COGEN ORGANIZATIONAL CHART



Summary of Area 5A

Area 5A utilities provide some common facilities and services to the units in the plant.

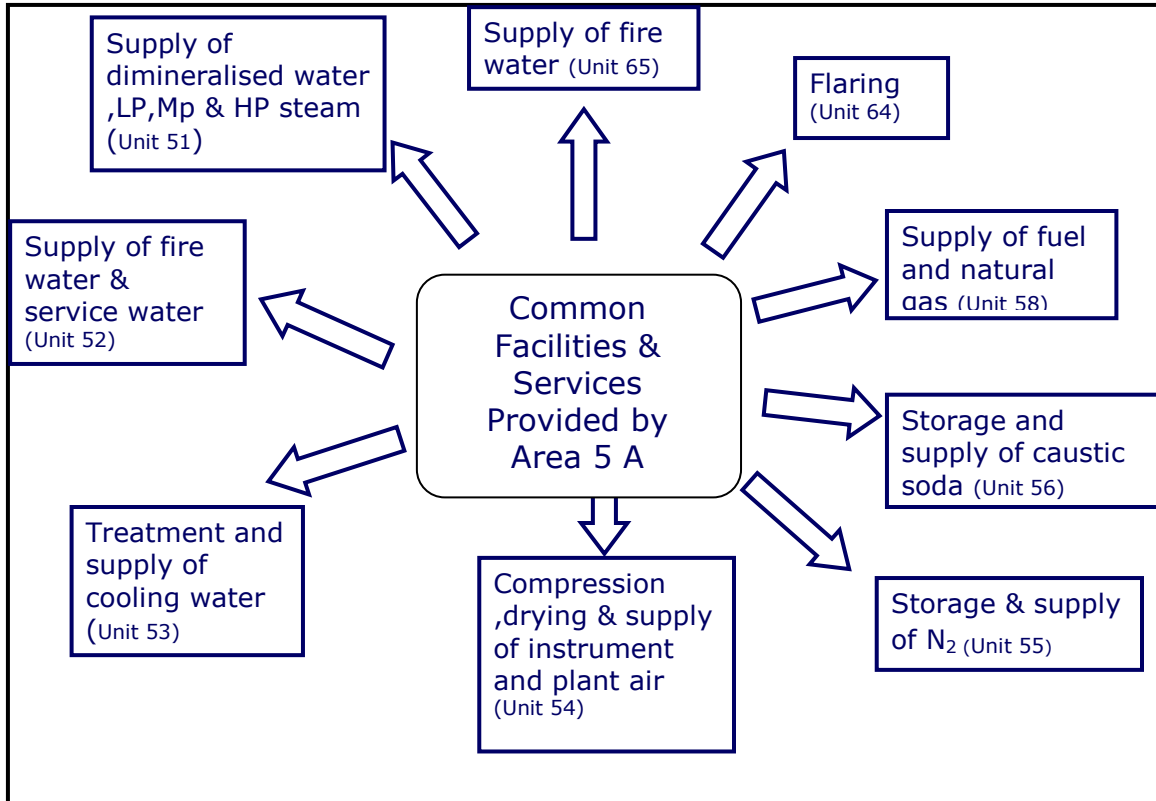


Figure 8.1: Summary of Area 5A

Description of Unit 51 (Steam, Condensate and BFW System)

1. The main purpose of this unit is to :
 - i. To produce high pressure steam at 44 kg/cm^2 . This high pressure steam is mainly used as motive steam to drive turbines and compressors and also for process and cleaning purposes.
 - ii. To cater for the whole refinery's steam requirement. Normal steam requirement is 220 tonne/hr.
2. This system is composed of the following subsystems:
 - i. Demineralized Water System
 - ii. BFW System
 - iii. Steam System
 - iv. Condensate System
3. The major equipments which are involved in this unit are :
 - i. Demineralisation trains (2 x 70 m³/hr, 1 x 140 m³/hr)
 - ii. 5 High Pressure Boilers (115 t/h at MCR, 44.6kg/cm²)
 - iii. 3 Deaeration trains (1 x 55 m³, 2 x 108 m³/hr)
 - iv. HP and MP BFW Pumps
 - v. Condensate Filtration Package (1 x 332.2 t/hr)

4. The overall process which is occurring in this unit can be described through the flow diagram below.

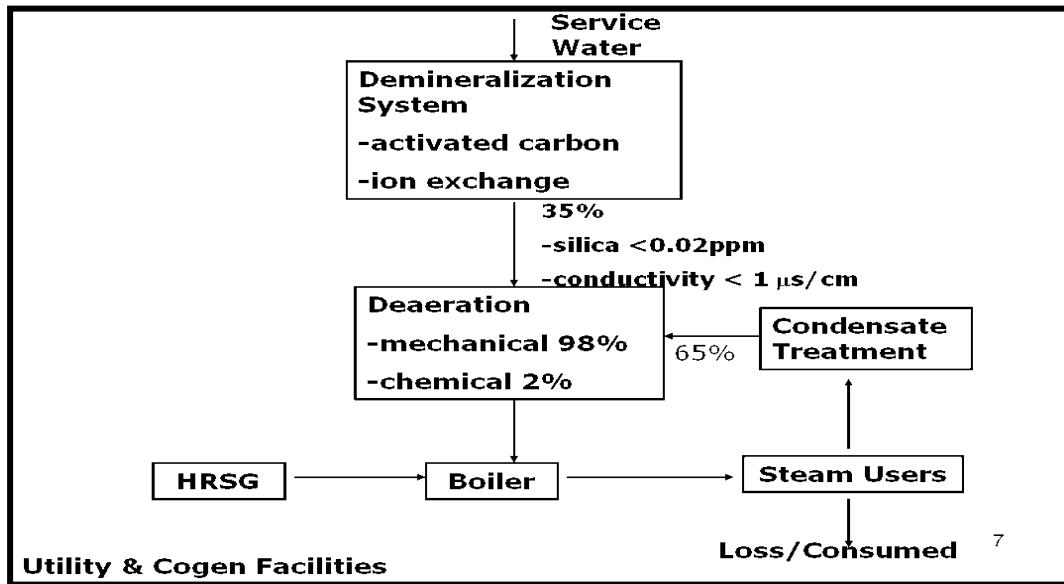


Figure 8.2: Flow diagram of Unit 51

5. Demineralized Water System

The Demineralized water system is designed based on the requirement of water quality for the boiler feed water at PSR-1 and at PSR-2 and the raw water quality.

The steam conditions:

	PSR-1	PSR-2
Pressure (kg/cm ² G)	12.0	44.6
Temperature (°C)	265	460

Table 8.1

The water quality at the outlet of the demineralized water system is :

Properties	Amount
Conductivity at 25°C	less than 1.0 $\mu\text{S/cm}$
Total hardness (CaCO_3)	less than 0 ppm
Silica (SiO_2)	less than 0.02 ppm
Iron (Fe)	less than 0.1 ppm

Table 8.2

6. BFW System

- a) To prepare the high quality BFW for steam production the following process are employed:

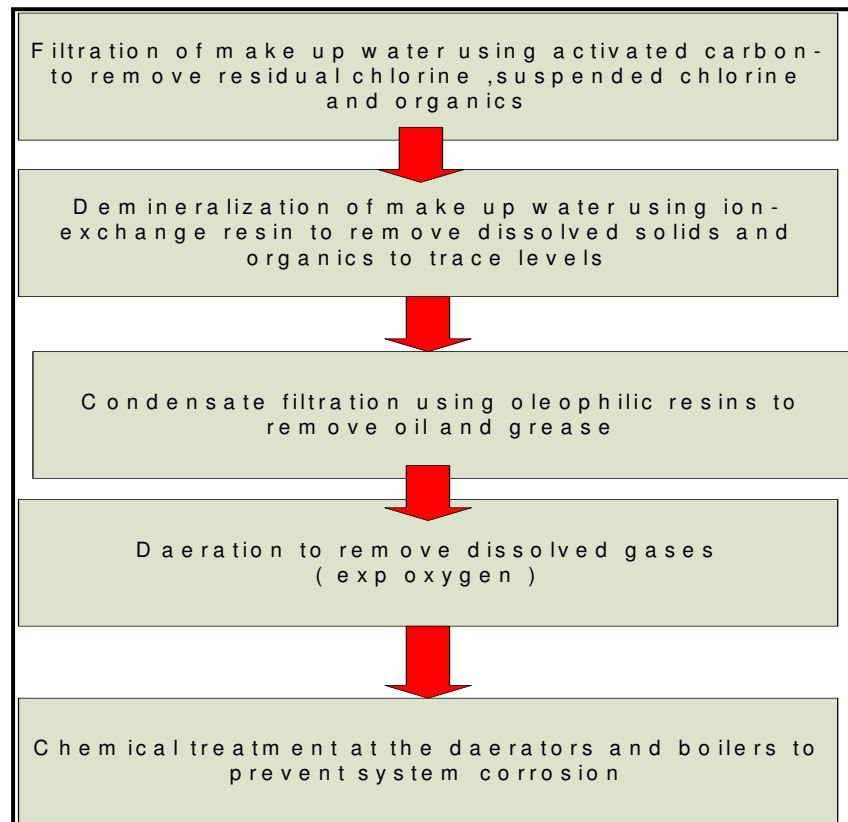


Figure 8.3: Flow diagram of Producing BFW

b) The constitutions of the BFW system are :

v. Dearator	145 ton/hr
vi. Storage Capacity	55m ³
vii. MP BFW pump	140m ³ /hr
viii. HP BFW pump	132m ³ /hr
ix. Chemical injection unit	

c) The water quality produced by the BFW system is as follows:

Properties	Amount
Dissolved O ₂ at the deaerator outlet	less than 0.007ppm
pH	8.0 to 9.5
Conductivity at 25°C	less than 4.0 µS/cm
Oil and organic matter	Traces
Total hardness (CaCO ₃)	less than 1.0 ppm
Silica (SiO ₂)	less than 0.02 ppm
Iron (Fe)	less than 0.1 ppm
Copper (Cu)	Less than 0.005 ppm

Table 8.3

7. Steam System

a. The steam conditions produced by the steam system is as follows:

Steam Conditions	Pressure (kg/cm ² G)	Temperature (°C)
Medium Pressure (MP)	10.5	260
Low Pressure(LP)	2.0	133
Low Low Pressure (LLP)	1.05	121
LP condensate	2.1	Saturated temperature
High Pressure (HP)	44.6	465

Table 8.4

8. Process Description

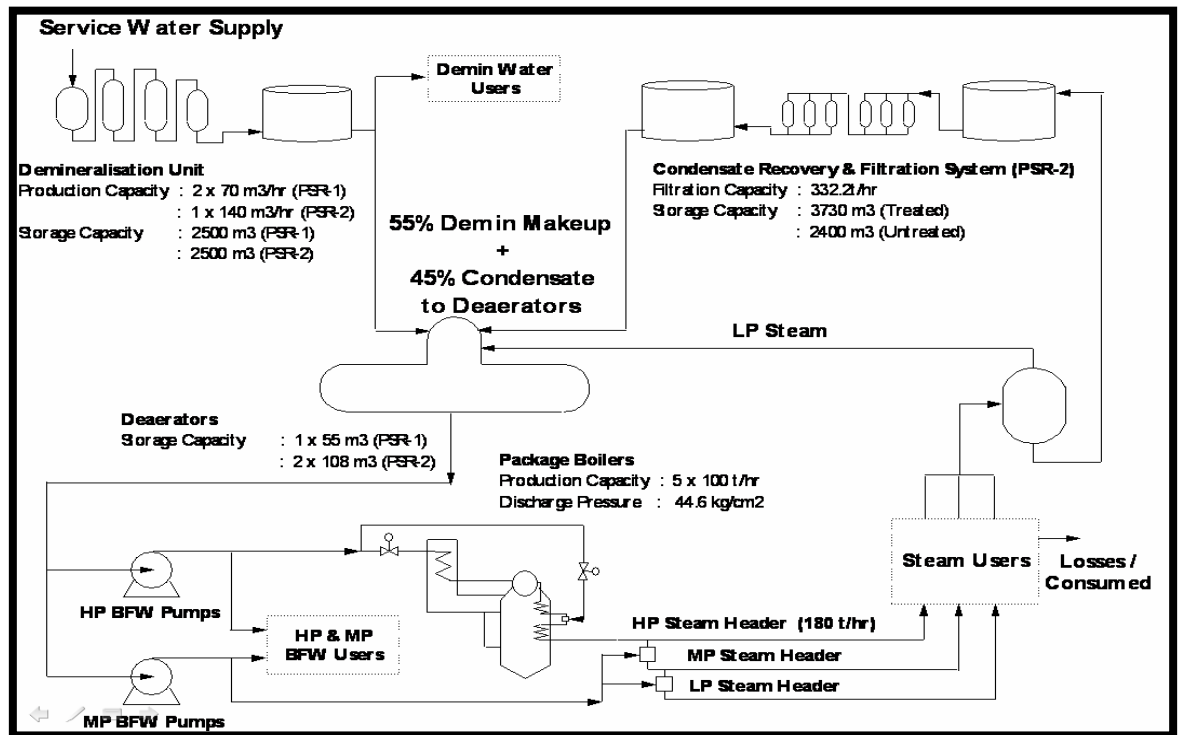


Figure 8.4: Process Description of Unit 51

a) Demineralized Water System

The service water first passes through the Activated Carbon Filters whereby residual chlorine, suspended chlorine and organic materials are removed. The clarified water then enters the Cation Exchangers where all cations (eg Ca^{2+} , Mg^{2+}) are removed and the anion exchangers where all anions (eg SO_4^{2-} , CO_3^{2-}) are removed by adsorption. Then the water is passed through a mixed bed column where final purification of the water takes place making it suitable for usage in boilers under MP and HP mode.

The treated water is sent to the Demineralized Water Tank where it is pumped out to the header. The demineralized water quality is maintained essentially by controlling the quality of the Activated Carbon Filter, Cation, Anion and Mixed

bed exchangers. These bed are regenerated in an automatic sequence . Extra high conductivity will activate an automatic train change over and initiate regeneration (with 98% sulphuric acid and 50° Be Caustic after dilution with ejector water) of the “exhausted “ beds .

The Demineralized water is then used as make up to Dearator, as dilution water for chemicals (Acids , Caustic) and as backwash water for Cation , Anion and Mixed bed exchangers.

b) Deaerator

The Demineralized water make up to the Deaerator is very critical since it maintains the Deaerator liquid level. Lack of level in the Deaerator starves the BFW pumps of suction and thereby affects the pump as well as the boiler. This can lead to failure of the superheater and economizer tubes due to overheat.

The other source of supply to the Deaerator is the condensate return from the plants through the Condensate Flash Drum. The Deaerator pressure is controlled by the LP steam flow to Deaerator top section.

Steam stripping ensures the dissolved oxygen content of the BFW to 7ppb in the Deaerator outlet. Oxygen scavenger and Morpholine are dosed into the Deaerator and mixed with returned condensate and fresh Demineralized Water.

c) Steam System (Boiler)

The BFW is sent to the Boiler and economizer coils in the boiler flue gas duct. The preheated BFW is sent to the boiler steam drum which is part of the boiler package for HP/MO Steam generation. Continuous and intermittent blowdown are adopted to maintain the quality of the BFW.

The steam from the boiler is superheated and goes for distribution.

d) Condensate System

MP condensate generated from heat exchangers in the CDU unit is gathered in the MP condensate flash drum and is separated into LP steam and LP condensate.

Recovered LP steam is discharged to LP steam header and recovered LP condensate is discharged to the LP condensate header through level control valve and finally sent to the Condensate Flash Drum (V-5102).

LP condensate generated from the heat exchangers is sent to the Condensate Flash Drum.

Flashed LLP steam generated in V-5102 and remaining condensate is transferred to the Deaerator using condensate pumps.

A hydrocarbon detector is installed on the discharge line of the condensate pump and the contaminated condensate is sent to covered sump.

Description of Unit 52 (Water Supply System)

1. This system consists of fire water storage, city water (utility water) system and potable water system. City water is used as raw water.
2. The purpose of this system is as follows:
 - a. To receive city water from Lembaga Air Melaka (LAM) reservoir.
 - b. To store city water for fire water , utility water , service water and potable water.
 - c. To transfer the water from the tank to the user using pump or by static head.
3. Major Equipments that are involved in this system:
 - i. Utility and portable water pumps
 - ii. City, fire and portable water tanks

4. System Capability:

Storage capacity:	City water	2 x 4,085 m ³
	Firewater	2 x 10,723 m
	Potable water	1 x 5 m ³

5. System Capacity:

	Capacity Under Normal Condition (m ³ /hr)
Make up water for cooling	45.7
Feed water for boiler make up water treating unit	73.5
Desalter's feed	16.7
Other consumptions	6
Total Requirement	141.9

Table 8.5

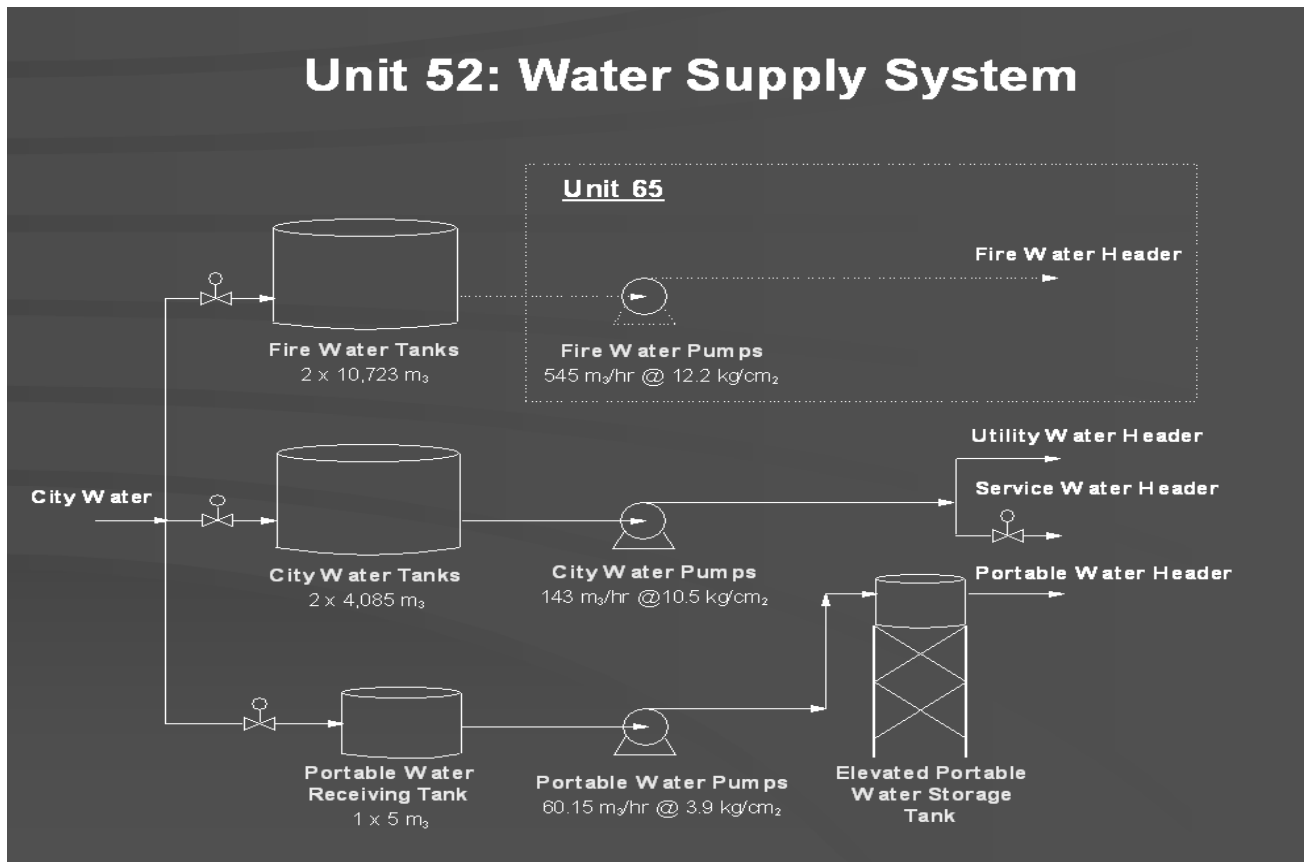


Figure 8.5: Process Flow Diagram of Water Supply System

6. Fire Water Tanks

- a. The fire water tanks are fixed roof tanks of 27,700mm diameter and 18,300mm height. The storage capacity of each tank is 10,723 m³. The level in the tank is always kept under level control.
- b. A fail open level control valve is provided in the inlet of each tank which opens at low level and closes at high level in the tank.
- c. Each tank is provided with the local level indicator and level alarm high-high and low-low which goes to DCS panel. Tanks are also provided with overflow and drain connections.
- d. A 12'' outlet line from each tank is routed to the suction header of utility water pumps. One utility water pump is adequate to meet the utility water requirement. Two pumps are required for maximum demand operation.

7. City Water Tanks

- a. Fixed roof type , two nos city water tanks are provided to store service water and utility water requirements of the refinery. The storage capacity of each tank is 4085m^3 .

8. Potable Water System :

- a. Potable water system consists of:
 - i. Potable water receiving tanks
 - ii. Potable water pumps
 - iii. Elevated potable water storage tank
- b. The potable water receiving tanks receives water through a 6''line tapped from city water line under level control. The water is then transferred to an elevated potable water storage tank with the help of motor driven centrifugal pumps.
- c. The pump starts automatically by low level switch provided on the elevated tank and stops by high level switch. The level alarm high high is also provided on the tank which is indicated on the DCS panel .This helps the operators to switch off the pump if it does not stops at high level. The tank is fitted with level indicator at grade over flow line and drain connections.
- d. The potable water in the elevated tank is supplied to the building, safety shower and ship loading area.

Description of Unit 53 (Cooling Water System)

1. This system consists of a two cell cooling tower, three cooling water circulating pumps, chemical treatment facilities, two cooling water filters, make-up facility of utility water, and supply and return headers of cooling water.
2. The purpose of this system is as follows:
 - a. To transfer the cooling water to cool heat exchangers , pumps, turbines and compressors
 - b. To recover the used cooling water from heat-exchangers, pumps, turbines, compressors, etc.
 - c. To cool the recovered cooling water.
 - d. To maintain the quality of circulating cooling water.
3. Major Equipments:
 - a. Two separate system for PSR-1 (open loop) and PSR-2 (closed loop and once through system)
 - b. PSR-1 : Cooling tower, FD Fan, Cooling water pumps, Chemical injection skid
 - c. PSR-2 : Cooling water tank, cooling water pumps, sea water intake, sea water pumps, titanium plate heat exchanger.
4. Method of Heat Removal and Chemical Treatment
(PSR 1 open loop system):
 - a. Open evaporation cooling system where heat transferred from process units are removed in a cooling tower.
 - b. Continuous phosphate –based chemical treatment to control deposition, corrosion, microbiological fouling in heat exchanging equipment.
 - c. Sodium hypochlorite as biocide, acid to control high pH, side-stream filter to control turbidity

5. Process Description and Controls

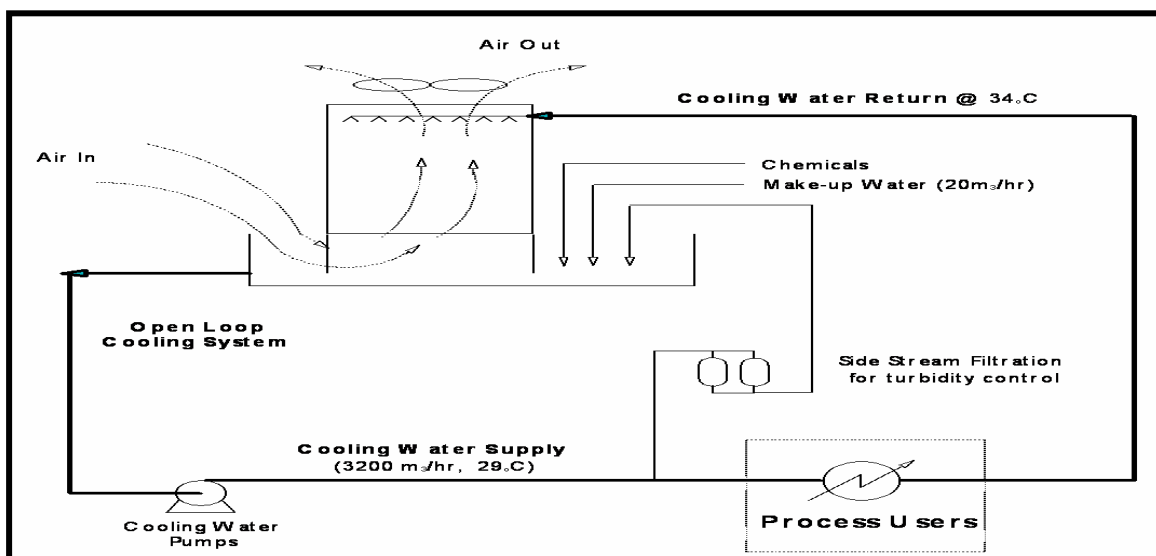


Figure 8.6: Process Flow Diagram of PSR-I Cooling Water System

- Hot water returned from the plant is cooled by the induced draft counter flow cooling tower. The water is cooled from 34 °C to 29° C .Cooling water return headers delivers hot water to these cells where it is distributed on either sides of each cell.
- Water falls across the path of airflow and gets cooled. Cold water is collected in the cooling tower basin. The water from the basin flows into the suction sump of cooling water circulating pumps.
- The cooled water from the cooling water sump is pumped by the cooling water circulating pumps to the various users. Cooling water is pumped by three centrifugal pumps each of 1460 m³/hr rated capacity at a pressure of 5.0kg/cm²g.
- The discharge line of the cooling water pumps are connected to the 24'' supply headers which supply the cooling water to main users. A 6'' tapping has been taken from the 24'' supply header for cooling water filters. A 6'' return line from the filter is connected to the cooling tower sump top.

- e) A hydrocarbon detector is provided at the return header of cooling water to detect the presence of light hydrocarbon in case of leakage in the heat exchangers in the plant.
- f) To supplement the loss of water due to blow down, evaporation and drift losses, fresh raw water is added to the cooling water sump as make-up.

6. Cooling Water Treatment

For efficient heat transfer and longer life of heat exchanger equipment, it is essential to treat the cooling water. The cooling water treatment comprises:

a) Treatment for pH control

This is done by acid injection of Sulfuric Acid (H_2SO_4) to lower the alkalinity of water and prevent deposits in the circulating water system. The rate of acid dosing depends on the pH value of the circulating water to be controlled. For practical purpose a pH value of 7.5 to 8.0 is maintained.

b) Treatment for prevention of Corrosion

Corrosion inhibitor is continuously added to the cooling water to prevent metal surfaces from being corroded by forming an inert anticorrosive film on the metallic surfaces coming in contact with the cooling water.

c) Treatment for prevention of Bio-Fouling

Sodium hypochloride (Biocide) injection is provided to prevent the growth of algae, slime and bacteria. Chlorination is done by mean of reciprocating pumps. Shock dosing needs to be done twice a day. The level of free residual chlorine

needs to be maintained below 0.3 to 0.6 ppm at the return line of the cooling water.

d) Control of Total Dissolved Solids

Make up of cooling water is from city water which contains certain amount of dissolved solids. The treatment chemicals are added to this which also contributes to the addition of dissolved solids. During operation the concentration of dissolved solids tends to build up due to continuous evaporation of pure water at the cooling tower. TDS is normally controlled between 500-800 ppm. This is obtained by certain amount of blow down achieved by the help of the 4" blow down pipe line tapped from 24" supply header. The blow down water is then sent to the Effluent Treating Systems (ETS).

e) Control of suspended solids by the cooling water filters.

The water in the cooling tower and the distribution system contains suspended solids. These solids come from the process system where the water is used for cooling, or is blown down into the cooling tower from the surrounding environment.

High level of suspended solids in the cooling water can be detrimental to the system due to:

- i. Blockage of the pipework with solids.
- ii. Coating of the heat exchangers, thus lowering the cooling efficiency.
- iii. Erosion due to abrasion.

The cooling water filters are utilized to maintain suspended solids in the cooling tower systems from 20-10 ppm.

7. Method of Heat Removal and Chemical Treatment

(PSR 2 closed loop and once through system) :

- a) Closed recirculating cooling system where heat transferred from process units is removed via heat exchange with once-through sea water system with a series of titanium plate heat exchangers
- b) Nitrate-based treatment program
- c) Electrolytic chlorinator is used to extract chlorine from sea water for disinfection

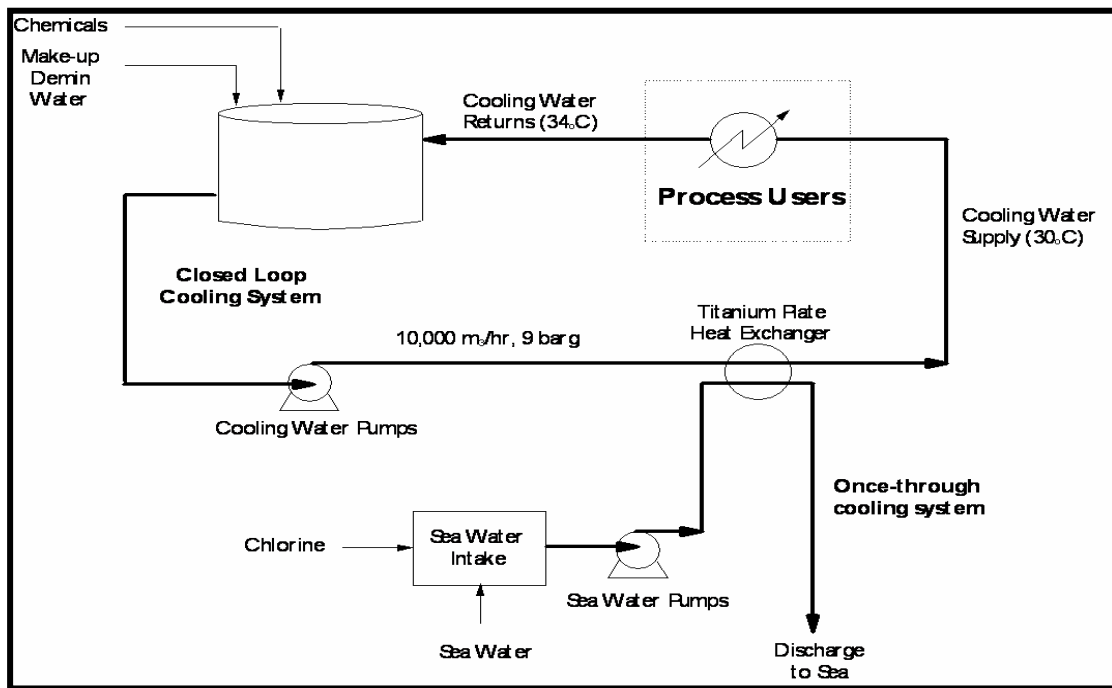


Figure 8.7: Process Flow Diagram of PSR-2 Cooling Water System

Description of Unit 54 (Instrument and Plant Air System)

1. Objective :
 - i. To compress and dry air for the following purposes :
 - a. to provide instrument air for pneumatic control (total capacity of 4,892 Nm³/hr)
 - b. to provide plant air for general cleaning, blowing and operate pneumatic tools
 - c. To chemically treat and maintain the desired quality of the cooling water to supply to process units.
 - ii. To maintain header pressure at the pressure of 7 kg/cm²
2. Major equipments used comprises of :
 - a. 5 compressor (2 for PSR-1 and 3 for PSR-2)
 - b. 3 trains of filters and dryers (1 for PSR-1 and 2 for PSR-2)
 - c. Instrument Air Receiving Tank I (I for each PSR-1 and PSR-2)

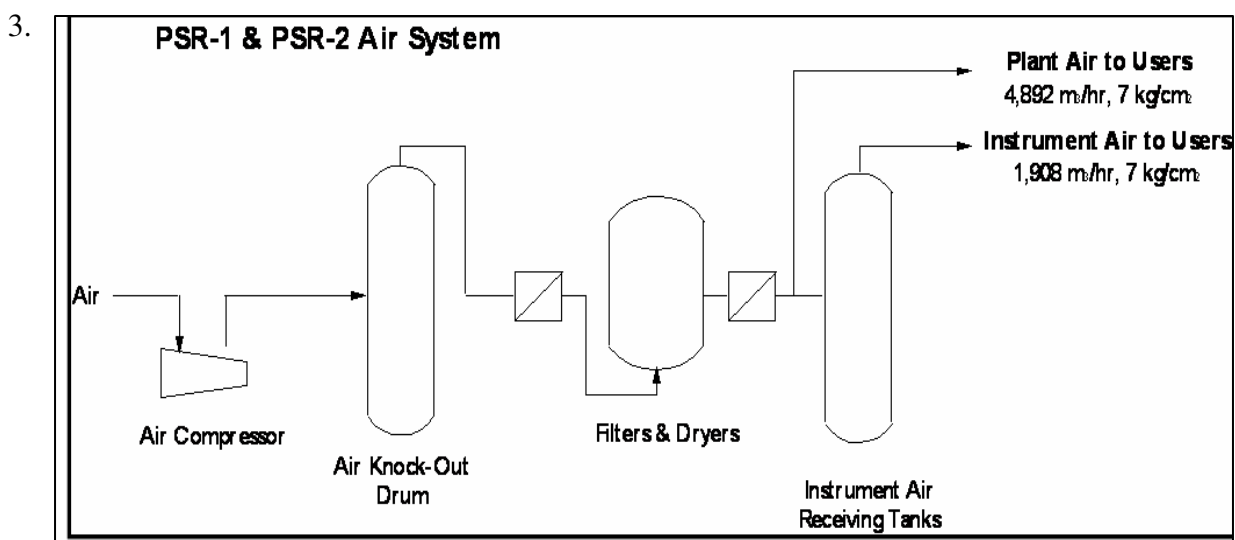


Figure 8.8: Process Flow Diagram of Plant Air System

Two air compressors each of capacity $3380 \text{ Nm}^3/\text{hr}$ with a discharge pressure of $9.0 \text{ Kg/cm}^2\text{g}$ are provided. These are oil free (non seal oil) type centrifugal compressors complete with intercoolers, after coolers and associated piping and instruments. One of the compressors is steam turbine driven and the other is electric motor driven. Each of the compressors has 100% design capacity.

The air from the compressor flows to the Air Knock-Out Drum. The knock-out drum has a capacity of one minute holding time of total instrument and plant air requirement. Air from the receiver goes to the instrument air dryers. Upstream from the dryers a 4" tapping is provided which supplies plant air to various consumers.

Two dryers each of capacity $2800 \text{ Nm}^3/\text{hr}$ have been provided. These are twin bed, split flow closed circuit type dryers. Dried air free from any solid particles out the dryers is led to the instrument air receiver through a 4" header.

A 4" line tapped upstream of the Instrument Air Receiver V-5402 from the main 4" line supplies plant air to various consumers.

Description of Unit 55 (NITROGEN SYSTEM)

1. The main function of this unit is to store , evaporate and supply nitrogen for :
 - a. CCR Unit Usage : 247 Nm³/h (PSR-1) and 378 Nm³/hr (PSR-2)
 - b. General Usage : 1 x 2868 Nm³/hr, 1 x 1000 Nm³/hr (PSR-1)
: 2 x 3250 Nm³/hr
2. 6 nitrogen packages are provided. 3 for PSR 1 and 3 for PSR2.
3. Nitrogen supplied by tankers is stored in pressurized vessel .This Nitrogen is than supplied to the vaporizer, whereby the Nitrogen is evaporated via heat exchange with LP steam. The Nitrogen is then supplied to the CCR unit or used for general purposes.

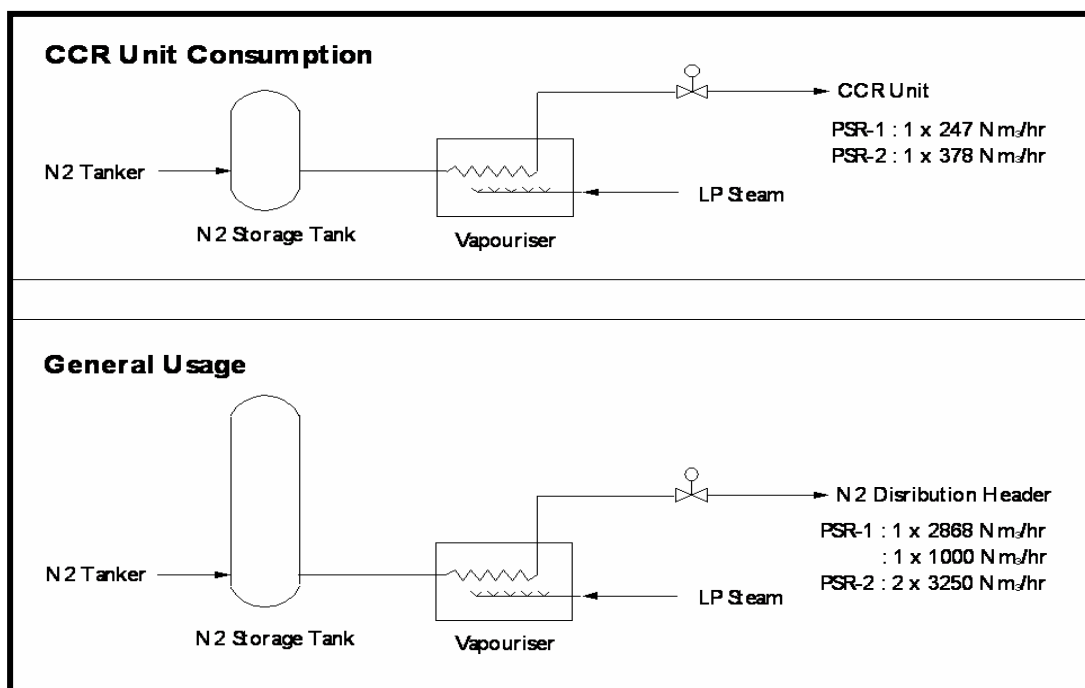


Figure 8.9: Process Flow Diagram of Nitrogen System

Description of Unit 56 (Chemical System)

1. The main purpose of this unit is to store and to supply dilute NaOH to users. Caustic is used for process treatment, resin regeneration, effluent treatment and for contact points cleaning.
2. Major equipments involved are two 50°Be (Baume) tanks, one 20°Be tank and one 10°Be tank and transfer pumps.
3. 50°Be of NaOH is received by tank trunks and it is then pumped to 10°Be storage tank whereby the NaOH is diluted to 10°Be using cold condensate. The diluted NaOH (10°Be) is then supplied to the PSR-1 users.
4. As for PSR-2 , 50°Be of NaOH is pumped to 20°Be storage tank whereby the NaOH is diluted to 20°Be using cold condensate. The diluted NaOH (20°Be) is then supplied to the PSR-2 users.

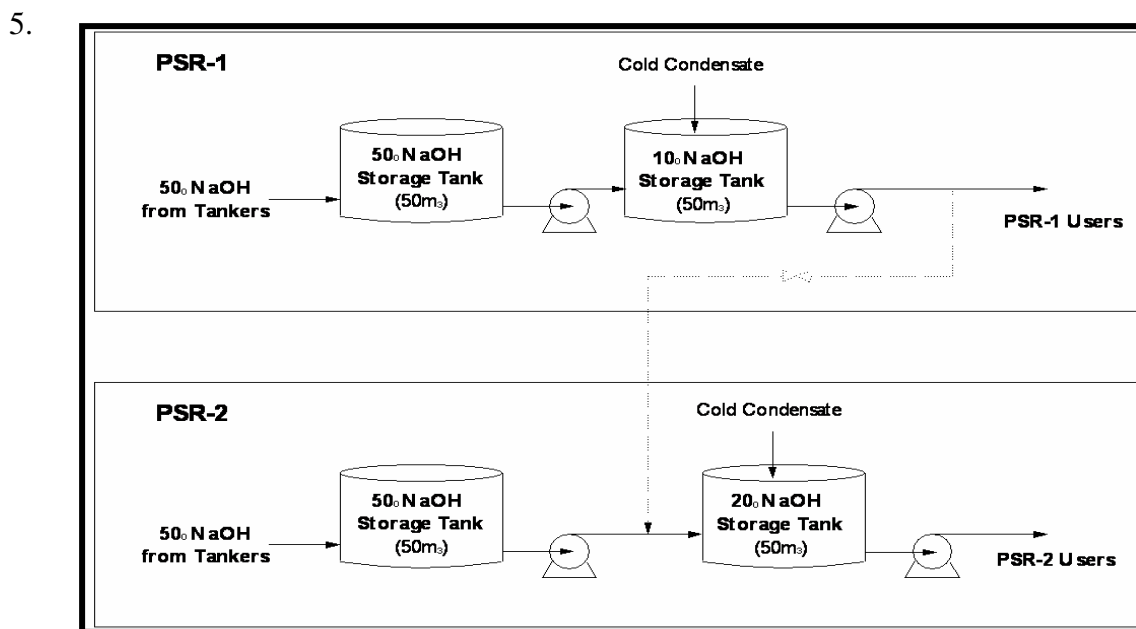


Figure 8.10: Process Flow Diagram of Chemical Unit

Description of Unit 58 (Fuel System)

1. The fuel system is provided for the operation of the furnaces and the boilers, purging of flare header and blanketing of vessel.
2. The system is composed of the fuel gas system and the fuel oil system. Diesel or Kerosene is used as fuel oil and refinery fuel gas is used as the main source of fuel gas. Natural gas and LPG are used as back-up fuels.
3. The major equipments involved in this system are :
 - a. LPG Vaporizers
 - b. LPG Superheater and Fuel Gas Drum
 - c. Fuel Oil Day Tank and Circulation Pumps
4. Fuel oil system

Fuel oil day tank of 960m³ is provided for storage of diesel oil or kerosene. Two fuel oil circulating pumps which is motor driven centrifugal type are provided to supply diesel or kerosene to the boiler.

If the pressure of fuel gas header becomes low, the supplement fuel gas needs to be supplied. After the natural gas becomes available, natural gas becomes the first priority and the LPG has the second. High low pressure alarm is also provided on fuel gas header which indicates on the DCS panel.

5. Fuel gas system

Natural gas is to be used as the first back up fuel after it becomes available when supply pressure in fuel gas header is below the set pressure. Low pressure signal indicated by 58-PRCA-004 opens the control valve on natural gas line.

Natural gas is received from natural gas main pipeline from outside of the refinery. The temperature, pressure, flow and density recorders are installed at the natural gas metering station and monitored on the DCS panel.

6. Off-Spec LPG

Off-spec LPG produced in the gas concentration unit (unit-14) can be introduced to the fuel gas system and used along with the refinery fuel gas whenever off-spec LPG is generated. The excess fuel gas is discharged to the flare line through the control valve. The switching over operation for use of off-spec LPG is done manually in Unit-14.

7. Fuel and Natural Gas System

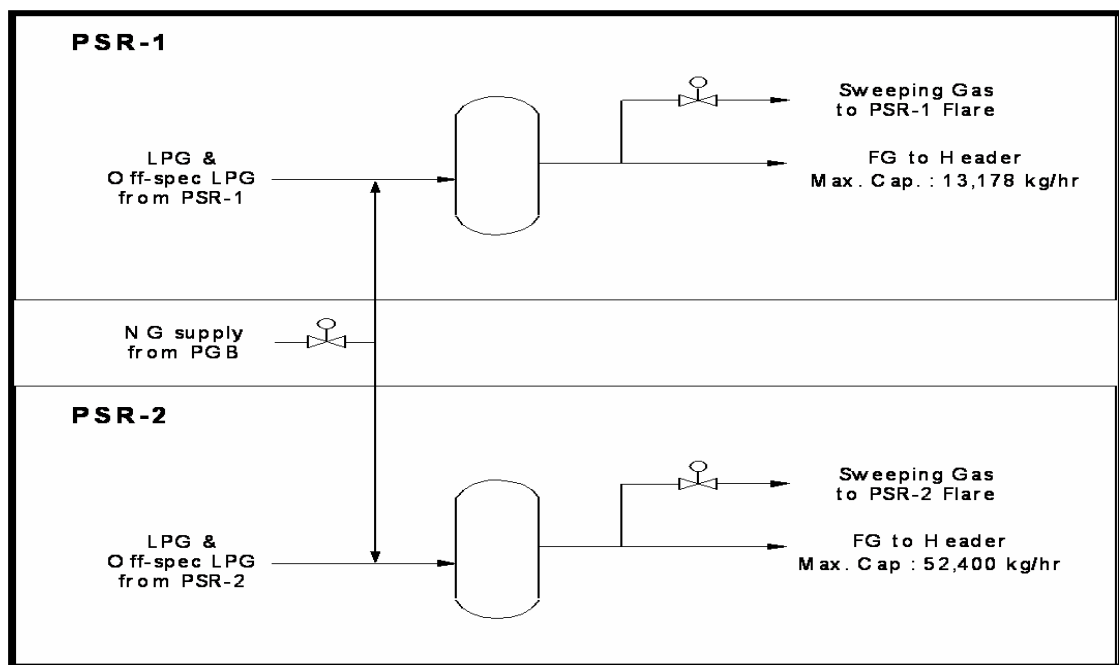


Figure 8.11: Flow Diagram of Natural Gas System

8. Fuel oil System

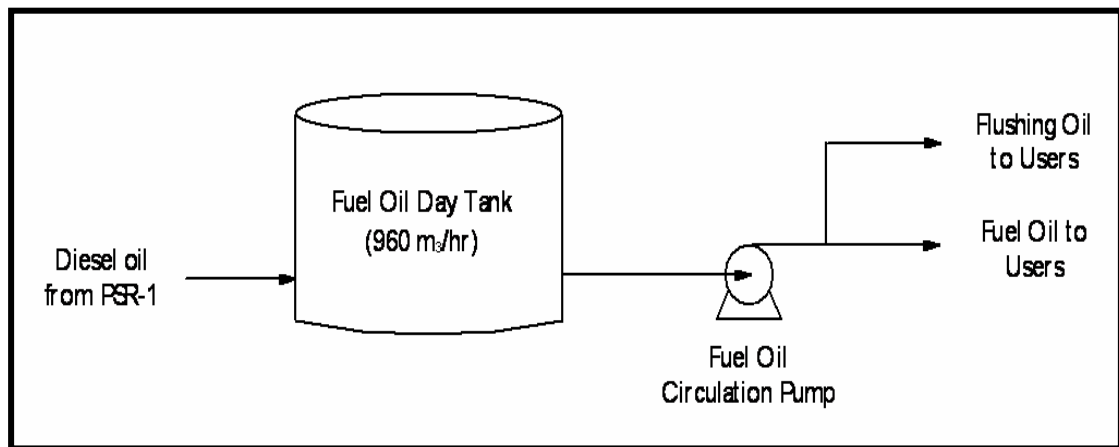


Figure 8.12: Process Flow Diagram of Fuel Oil System

Description of Unit 64 (Flare System)

1. To handle normal gas release, emergency gas and gas liquid release from the refinery operations.
2. Major equipments involved are :
 - a. Sour flare knock- out drum
 - b. Main flare knock-out drum
 - c. Flare seal drum
 - d. Main flare
 - e. Sour flare
3. The flare gas shall be categorized as :
 - a. Sour flare -contains more than 30vol% H₂S
 - b. Main flare - contains flammable or toxic material
 - c. Atmospheric flare - cooling water, steam, inert gas, air
4. The sour flare gas is steam traced to avoid condensation of H₂S.
5. Capacity:

PSR1 Main Flare	725	t/h
PSR1 Sour Flare	101.1	t/h
PSR2 Main Flare	773	t/h
PSR2 Sour Flare	77.6	t/h
6. Flare gas is first sent to the knockout drums to remove liquids. Liquid from the main flare knockout drums is sent to the Light Slop Tank (Unit 77).Liquid from the sour flare knockout drums is sent to the Degassing Drum and Sour Water Stripping Unit (Unit 34).

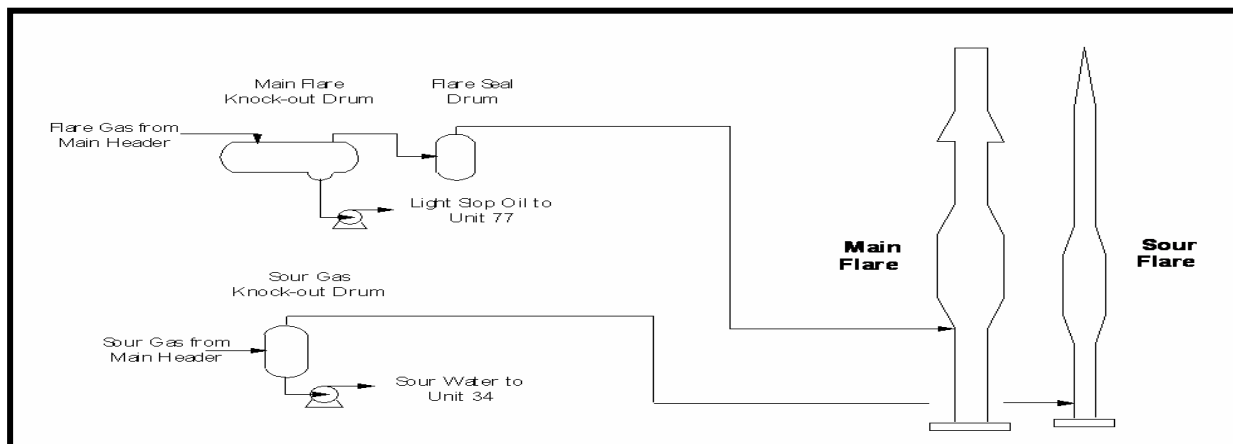


Figure 8.13: Process Flow Diagram of Flare Unit

Description of Unit 65 (Fire System)

1. To provide and maintain fire water facility to refinery users during emergency.
2. Major equipments involved are :
 - a. Fire water pumps
 - b. Fire water distribution system
 - c. Foam extinguishing system
 - d. Fire and gas detection system,
 - e. Portable and mobile fire fighting equipment (PSR-1 & 2)
3. Fire water storage: Tanks → 2x 10,723 m³
Pond #2 → 10,000
4. Methods of operations :
 - a. City water received from PAM is stored and maintain at desired level in Fire water tanks
 - b. Jockey pumps (2 x 56.8 m³/h) will maintain the water header pressure above 12.5 kgf/cm².
 - c. The firewater pumps (4 x 545 m³/h) will operate based on a preset sequence once the header pressure dropped below 11.8 kgf/cm².

9.0 Overview of COGENERATION Plant

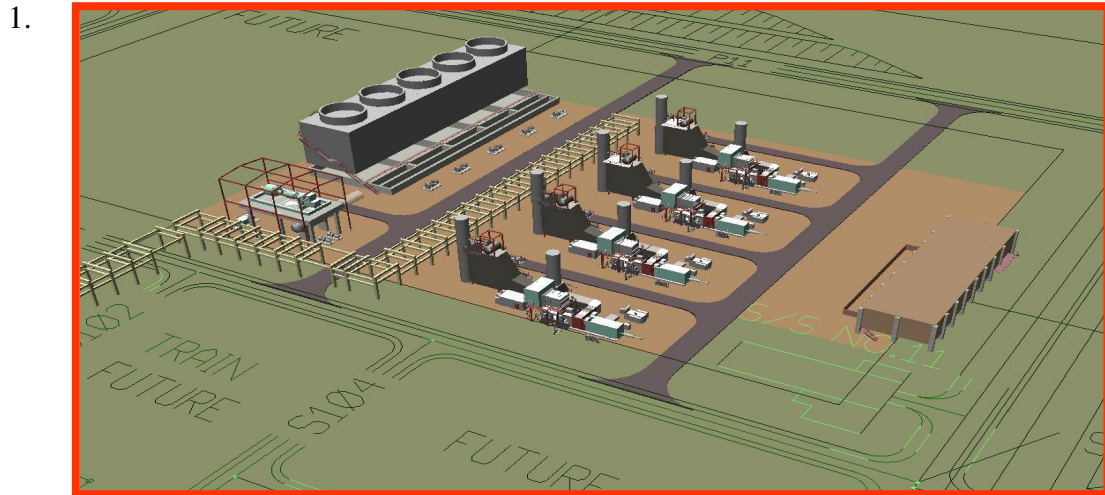


Figure 9.1: Layout of COGEN Plant

2. The Purpose of Cogeneration Plant :
- a. To reduce operating costs by generating power in-house and generating steam using waste heat from the GTGs (Gas Turbine Generators)
 - b. To provide a reliable generation and distribution of power to PSR-1 & PSR-2.
 - c. To meet the refinery power demand (70 MWe) without the need to import power from TNB under normal operating conditions.
 - d. To be capable of operating in “Island Mode”, whilst maintaining the capability to import power from TNB during maintenance and upset conditions.
 - e. To provide a reliable supply of steam to PSR-1 and PSR-2 including sufficient steam for safe shutdown in the event of total power failure

3. Power Generation Cogeneration is defined as the simultaneous generation of heat and power both of which are used. The Cogeneration Plant comprises four Gas Turbines (GTs), a Steam Turbine (ST), four Heat Recovery Steam Generators and a condensate recovery facility. The GTs and ST each drive an AC electrical generator. The combined driver / generator units are referred to as a GTG and an STG as appropriate. The GTGs and the STG will be of similar capacity.

a) The Gas Turbine engine comprises three main sections:

- Compressor- ambient air is compressed to high pressure
- Combustion -fuel is introduced into the high pressure air and combusted producing high pressure, high velocity gas.
- Turbine -high pressure, high velocity hot gas passes over and acts upon the turbine blades forcing them to rotate at high speed). The rotating shaft will drive the AC generator via a gearbox. The exhaust gas is hot and rich in oxygen and will be ducted to the inlet of the HRSG where it will be used to generate HP steam.

b) The HRSG comprises of:

- Economizer
- steam drum
- evaporator superheater
- de-superheater
- Stack.

c) High pressure steam production is increased by the installation of additional burners at the inlet section of the HRSG to enable supplementary

firing. This will enable steam production to be maintained when the GTG is operating below 100% load.

- d) The STG takes in HP steam from the HRSGs as well as surplus MP and LP steam from the refinery. The MP and LP steam flows to the turbine are dependent on the refinery conditions and, under normal conditions; the STG will operate at turndown with a reduced flow of HP steam. The STG is designed to ramp up to give additional power in the event of a GTG trip.
- e) The Steam Turbine comprises of:
 - i. HP section
 - ii. MP section
 - iii. LP section

The wet exhaust steam at exit from the LP section is condensed in a water cooled condenser. The condenser operates under vacuum. Non-condensibles are removed from the condenser by the vacuum system.

The clean condensate from the STG is pumped to the refinery Clean Condensate Tank for use as make-up to the refinery BFW system. The vacuum condensate can be contaminated with cooling water leaking from the condenser tubes. The conductivity of the condensate will be monitored continuously. In the event of contamination, the operators will return the condensate to the Contaminated Condensate Tank upstream of the Condensate Filtration Unit. If the contamination exceeds the maximum allowable loading on the mixed beds within the Condensate Filtration Package, then the condensate will be dumped to drain.

- 4. Cogen plant is owned 100% by PPMSB so it sells HP steam, electricity to refinery and buys BFW and excess MP and LP from refinery.

5. Plant overview.

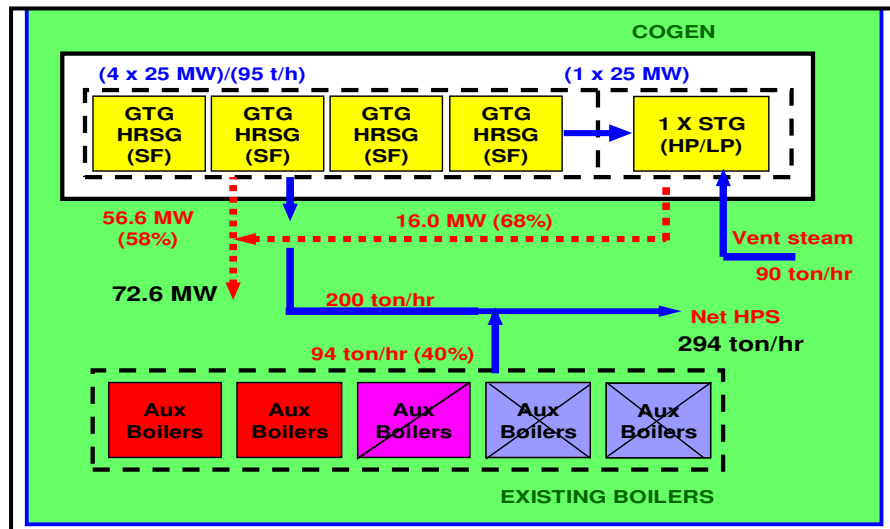


Figure 9.2: Overview of COGEN Plant

6. TNB Parallel and Island Operation

The system has been designed to allow both parallel and island operation. While island operation is preferable on commercial grounds, running parallel with one feeder firm standby will increase system availability. The other three feeders will be on non-firm.

10. Bibliography

Information for this document was sourced from:

- a. Operating Manuals
- b. Refinery Management System (RMS)
- c. Presentation Slides of Skill Group 16.1

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